```
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"type":"Ticket",
"attributes":{
"subject":"Problem with Ticket",
"description":"Hello,\r\n\r\nI have a problem with my ticket.\r\n\r\nCan you please check it?\r\n\r\nThanks\r\n\r\n\r\n
```

Mr. Gabriel Sanguinetti
Director de Ingeniería de Proyectos Petrolíferos

Mr. Juan Cabral
Mr. Gabriel Sanguinetti
Mr. Ernesto Lopez
Mr. Ernesto Lopez
Mr. Gabriel Sanguinetti
Mr. Juan Cabral
Mr. Gabriel Sanguinetti
Mr. Ernesto Lopez

```
{
  "_id": "68a8d7f7b7b0ec",
  "_rev": "1-d7d0d0d0d0d0d0d0",
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    },
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      "stub": true
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      "length": 1,
      "stub": true
    },
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    },
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      "stub": true
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      "stub": true
    },
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      "stub": true
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      "length": 1,
      "stub": true
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      "length": 1,
      "stub": true
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      "stub": true
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      "stub": true
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      "stub": true
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      "stub": true
    },
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      "stub": true
    },
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      "stub": true
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      "stub": true
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      "length": 1,
      "stub": true
    },
    "90.mp3": {
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      "digest": "md5-ZGRlMjAyMjAyMjAyMjAyMg==",
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      "stub": true
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}
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"_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_local_seq":"0000-00-00T00:00:00.000+00:00",
"_revisions":{"start":1,"ids":["00000000000000000000000000000000"]},
"_revs_info":[{"rev":"1-00000000000000000000000000000000","status":"available"}],
"_rev_tree":[["1-00000000000000000000000000000000",{"status":"available","children":{}}]],
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"_local_seq_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_attachment_stubs":{"0000-00-00":{"content_type":"application/octet-stream","digest":"md5-00000000000000000000000000000000","length":0,"revpos":1,"stub":true}},
"_attachment_data":{"0000-00-00":"AAAA
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```
BEGIN:VCARD
VERSION:2.1
N:;Ivan Ivanovich;;;
FN:Ivan Ivanovich
ORG:Ivan Ivanovich Corporation;Marketing Department
TITLE:Director
TEL;WORK;VOICE:(495)
TEL;CELL;VOICE:(495)
EMAIL;PREF;INTERNET:
END:VCARD
BEGIN:VCARD
VERSION:2.1
N:Ivanov;Ivan;;;
FN:Ivan Ivanov
ORG:Ivanov Corporation;Marketing Department
TITLE:Director
TEL;WORK;VOICE:(495)
TEL;CELL;VOICE:(495)
EMAIL;PREF;INTERNET:
```

Generic Medical Consent

**Author: Dr. Ashish Gupta, Dr. Ankur Gupta, Dr. Ankur Gupta, Dr. Ankur

```
<!-- ~~~
```

```
<div class="page">

<div>
<span id="1" />
<span id="2" />
<span id="3" />
<span id="4" />
<span id="5" />
<span id="6" />
<span id="7" />
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<span id="10" />
<span id="11" />
<span id="12" />
<span id="13" />
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```
<?xml version="1.0"?>

<!DOCTYPE
  PLOT_SPECIFICATION SYSTEM "http://www.fr.com/fr/fr.dtd">

<PLOT_SPECIFICATION
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  VERSION "1.0"
  DATE "Wed Jan 13 00:00:00 2010"
  DEVICE "unknown"
  TERMINAL "unknown"
  OUTPUT "/tmp/tmpkbcb8Ia_.png"
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<?xml version="1.0"?>

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<PLOT_TEXT
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  TEXT "Gnuplot"
  FONT "Helvetica"
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  ROTATION "0.000000"
  ALIGNMENT "CENTER"
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<?xml version="1.0"?>

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  ALIGNMENT "CENTER"
  COLOR "0.000000,0.000000,0.000000"
>
```

Gnuplot is a command-line driven interactive function plotting program. Originally intended as a means to stimulate students' interest in mathematics, it has been used in scientific fields for over a decade. Gnuplot is capable of plotting 2D functions and data in various styles, and 3D functions and data in the form of surfaces and contour maps. Gnuplot can output to a variety of formats, including PostScript, PNG, and SVG.

Gnuplot is a portable command-line driven interactive data and function plotting utility for UNIX, IBM OS/2, MS Windows, DOS, and many other platforms. It was originally intended for students to explore mathematical functions and data in a friendly way, but has been used for many scientific and engineering applications since then. Gnuplot supports many types of plots in either 2D or 3D. It can plot functions and data read from ASCII files. Gnuplot can output to a variety of formats, including PNG, PostScript, and SVG.

```
<?xml version="1.0"?>
```

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<div>
<div>
<div>
```

Ad Clash Report

Ad Clash Report

Ad Clash Report

{
"id": "1707.07187",
"title": "Adversarial Examples for Semantic Segmentation and Object Detection",
"abstract": "Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.",
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"subjects": "Computer Vision and Pattern Recognition (cs.CV); Machine Learning (cs.LG); Machine Learning (stat.ML)",
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"licenses": "Creative Commons Attribution Non-Commercial Share Alike 4.0"
}

Adversarial Examples for Semantic Segmentation and Object Detection

Cihang Xie[1], Jianyu Wang[2], Zhishuai Zhang[1], Yuyin Zhou[1], Lingxi Xie[1], Alan Yuille[1]
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{cihang.xie, zhishuai.zhang, yuyin.zhou, lingxi.xie, alan.yuille@jhu.edu, jianyuwang@hust.edu.cn}

Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.

Intro

Cross-Sectional Study of Dietary Patterns and Anemia Among Adolescent Girls in Rural Tribal Hamlets of Gadchiroli, Maharashtra

Dr. Ananya Bhattacharjee, Dr. Sudhir Kumar Tripathi, Dr. Meenakshi Rao, Dr. Vikram Singh Rathore, Dr. Nandini Sen, Dr. Ashok Kumar Jain, Dr. Sunita Devi Yadav, Dr. Rajesh Kumar Mishra, Dr. Pooja Sharma, Dr. Anil Kumar Verma, Dr. Sanjay Kumar Gupta, Dr. Ramesh Chandra Pandey, Dr. Kavita Singh, Dr. Mohan Lal Sharma, Dr. Geeta Devi Kushwaha, Dr. Vinod Kumar Singh, Dr. Ashok Kumar Mishra

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Jan-2021 Impact Factor: 4.101 (SJIF) Peer Reviewed Journal International Advanced Research in Engineering and Technology
Page 1

Design of Electrical Generator operated on Low speed Wind Mill for Village Electrification

Prof. M. Ashok Kumar, Prof. Ch. Ramesh, Prof. Ch. Ramesh, Prof. Ch. Ramesh

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NO. 7.5.2.7 **MAGNIFICENCE**

Sub-section 7.5.2.7 of Section 7.5 Magnificence

Sub

Sub-section 7.5.2.7 of Section 7.5 Magnificence

Section

Section 7.5 Magnificence of Chapter 7 Sacrifice and Magnificence

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Chapter 7 Sacrifice and Magnificence of Book 7.5 Sacrifice and Magnificence

Book

Book 7.5 Sacrifice and Magnificence of Volume 7 Sacrifice and Magnificence

Volume

Volume 7 Sacrifice and Magnificence of Part 7 Sacrifice and Magnificence

Part

Part 7 Sacrifice and Magnificence of Section 7.5 Sacrifice and Magnificence

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Sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-sub-section 7.5.2.7 Magnificence of Sub-section 7.5.2 Sacrifice and Magnificence

**Sub-

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Year VIII
Number 4
http://www.kvasy.sk
Print version: 1336-8672

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Release Date: [EBook #11]
Last Updated: [July 14, 2011]
Language: [English]
Character set encoding: [UTF-8]
Produced by: [Arthur DiBianca and David Widger]
Source: [Gutenberg]

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d615 | Page 5 Conclusion The Ultrasonic Assisted Extraction (UAE) of Ginger Oleoresin from Ginger Rhizome Powder using Ethanol as solvent and optimized UAE. This study concluded following points.
1. Ginger Oleoresin is a valuable natural product.
2.

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Verify Layer3 Subinterface Configuration on the CLI
Related Topics

Sub-Interface - Layer3 Subinterface

Palo Alto Networks firewalls support Layer 3 routing on subinterfaces to accommodate multiple virtual routers and security zones on a single physical interface. The ability to define subinterfaces is useful when you want to route traffic for multiple virtual routers or security zones through a single physical interface. Each subinterface can be assigned to a different virtual router and security zone. Each subinterface can be configured with its own IP address, static routes, RIP, OSPF, BGP, or multicast routing protocol. Subinterfaces are also useful if you have a firewall that is connected to a switch that is configured for multiple VLANs. You can define a subinterface for each VLAN and assign traffic from each VLAN to a different virtual router and security zone.

Configure a Layer3 Subinterface on the GUI

Step 1. Define the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Define the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Verify Layer3 Subinterface Configuration on the GUI

Step 1. Verify the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Verify the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Configure a Layer3 Subinterface on the CLI

```
set interfaces ethernet <slot>/<port_number>
set interfaces ethernet <slot>/<port_number> mtu <mtu_size>
set interfaces ethernet <slot>/<port_number> link-speed <speed_value>
set interfaces ethernet <slot>/<port_number> link-duplex <duplex_value>
set interfaces ethernet <slot>/<port_number> link-state <state_value>
set interfaces ethernet <slot>/<port_number> layer3 ip <ipv4_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 <ipv6_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 enable-duplicate-address-detection yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement enable yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement hop-limit <hop_limit_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement reachable-time <reachable_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement retransmit-time <retransmit_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-server <dns_server_address_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-suffix <dns_suffix_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery neighbor <ipv6_address_value> <mac_address_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type prefix-only
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 
```

Regular Expressions Cheat Sheet

Regular Expressions Cheat Sheet - Quick Reference Guide for Pattern Matching

Basic Syntax

Pattern	Description
`.`	Any character except newline
`^`	Start of string/line
`$`	End of string/line
`\d`	Digit (0-9)
`\D`	Non-digit
`\w`	Word character (a-z, A-Z, 0-9, _)
`\W`	Non-word character
`\s`	Whitespace (space, tab, newline)
`\S`	Non-whitespace
`\b`	Word boundary
`\B`	Non-word boundary
`[abc]`	Character set: a, b, or c
`[^abc]`	Negated set: not a, b, or c
`[a-z]`	Range: a through z
`[a-z0-9]`	Multiple ranges: a-z or 0-9
`\n`	Newline
`\t`	Tab
`\r`	Carriage return
`\\`	Literal backslash

Quantifiers

Pattern	Description
`*`	0 or more
`+`	1 or more
`?`	0 or 1 (optional)
`{n}`	Exactly n times
`{n,}`	n or more times
`{n,m}`	Between n and m times
`*?`	Lazy 0 or more
`+?`	Lazy 1 or more
`??`	Lazy 0 or 1
`{n,m}?`	Lazy between n and m times

Groups and Alternation

Pattern	Description
`(abc)`	Capture group
`(?:abc)`	Non-capturing group
`(?P<name>abc)`	Named capture group
`\1`	Backreference to group 1
`\2`	Backreference to group 2
`\k<name>`	Backreference to named group
`a|b`	Alternation: a or b
`(?=abc)`	Positive lookahead: followed by abc
`(?!abc)`	Negative lookahead: not followed by abc
`(?<=abc)`	Positive lookbehind: preceded by abc
`(?<!abc)`	Negative lookbehind: not preceded by abc

Common Patterns

Pattern	Description
`[0-9]+`	Integer
`[0-9]*\.[0-9]+`	Decimal number
`[a-z]+`	Lowercase letters
`[A-Z]+`	Uppercase letters
`[a-zA-Z]+`	Letters
`[a-zA-Z0-9]+`	Alphanumeric
`[a-zA-Z0-9_]+`	Word characters
`[a-zA-Z0-9_-]+`	Slug
`[a-zA-Z0-9.-]+`	Domain name
`[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}`	Simple domain
`[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}`	Email address
`https?://[a-zA-Z0-9.-]+(:[0-9]+)?(/[a-zA-Z0-9._~:/?#[]@!$&'()*+,;=%-]*)?`	URL
`[0-9]{1,3}(\.[0-9]{1,3}){3}`	IP address
`[0-9]{4}-[0-9]{2}-[0-9]{2}`	Date (YYYY-MM-DD)
`[0-9]{2}:[0-9]{2}:[0-9]{2}`	Time (HH:MM:SS)
`[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}(Z|[+-][0-9]{2}:[0-9]{2})`	ISO 8601
`[0-9]{1,2}/[0-9]{1,2}/[0-9]{4}`	Date (MM/DD/YYYY)
`[0-9]{1,2}-[0-9]{1,2}-[0-9]{4}`	Date (MM-DD-YYYY)
`[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?(\s?[AP]M)?`	Time (12-hour)
`[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?`	Time (24-hour)
`[0-9]{1,2}(st|nd|rd|th)`	Ordinal
`[0-9]{1,2}(st|nd|rd|th)\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)`	Ordinal date
`[0-9]{1,2}\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}`	Date (DD Mon YYYY)
`[0-9]{1,2}\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)`	Date (DD Mon)
`(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{1,2}`	Date (Mon DD)
`(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}`	Date (Mon YYYY)
`[0-9]{1,2}(st|nd|rd|th)\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}`	Date (DD Mon YYYY)
`[0-9]{1,2}\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}\s?[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?(\s?[AP]M)?`	Date (DD Mon YYYY HH:MM:SS)
`[0-9]{1,2}(st|nd|rd|th)\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s?[0-9]{4}\s?[0-9]{1,2}:[0-9]{1,2}(:[0-9]{1,2})?(\s?[AP]M)?`	Date (DD Mon YYYY HH:MM:SS

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Accepted: 15 November 2017
Online First: 27 November 2017

```
<?xml version="1.0" encoding="UTF-8" ?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:content="http://purl.org/rss/1.0/modules/content
```

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Section 1.10.32 of "de Finibus Bonorum et Malorum", written by Cicero in 45 BC

"Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboratoriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat, quo voluptas nulla pariatur?"

1914 translation by H. Rackham

"But I must explain to you how all this mistaken idea of denouncing pleasure and praising pain was born and I will give you a complete account of the system, and expound the actual teachings of the great explorer of the truth, the master-builder of human happiness. No one rejects, dislikes, or avoids pleasure itself, because it is pleasure, but because those who do not know how to pursue pleasure rationally encounter consequences that are extremely painful. Nor again is there anyone who loves or pursues or desires to obtain pain of itself, because it is pain, but because occasionally circumstances occur in which toil and pain can procure him some great pleasure. To take a trivial example, which of us ever undertakes laborious physical exercise, except to obtain some advantage from it? But who has any right to find fault with a man who chooses to enjoy a pleasure that has no annoying consequences, or one who avoids a pain that produces no resultant pleasure?"

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |

```
<mx:Text id="mainMenuTitle"
  fontFamily="Arial" fontSize="24" fontWeight="bold"
  color="0x000000" />
<mx:Button id="newGameBtn" fx:Resolve="@Resolve(!$ctx.isGameInProgress)"
  fx:Bind="{$ctx.isGameInProgress ? 'Resume Game' : 'New Game'}" fx:Event="$ctx.startNewGame" />
<mx:Button id="loadGameBtn" fx:Resolve="@Resolve($ctx.hasSaveFiles)" />
<mx:Button id="optionsBtn" />
<mx:Button id="quitBtn" />
<mx:Text id="versionLabel" />
<mx:Text id="copyrightLabel" />
</mx:VBoxContent>
</mx:Panel>
<mx:Panel id="gameHUD" />
<mx:Panel id="pauseMenu" />
<mx:Panel id="gameOverScreen" />
<mx:Panel id="levelUpDialog" />
<mx:Panel id="inventoryPanel" />
<mx:Panel id="dialogueBox" />
<mx:Panel id="tutorialOverlay" />
<mx:Panel id="loadingScreen" />
<mx:Panel id="errorDialog" />
<mx:Panel id="confirmDialog" />
<mx:Panel id="creditsScreen" />
<mx:Panel id="mapPanel" />
<mx:Panel id="questLogPanel" />
<mx:Panel id="shopPanel" />
<mx:Panel id="craftingPanel" />
<mx:Panel id="settingsPanel" />
<mx:Panel id="saveGamePanel" />
<mx:Panel id="achievementsPanel" />
<mx:Panel id="statsPanel" />
<mx:Panel id="bestiaryPanel" />
<mx:Panel id="skillTreePanel" />
<mx:Panel id="partyPanel" />
<mx:Panel id="worldMapPanel" />
<mx:Panel id="minimapPanel" />
<mx:Panel id="chatPanel" />
<mx:Panel id="notificationPanel" />
<mx:Panel id="tooltipPanel" />
<mx:Panel id="contextMenuPanel" />
<mx:Panel id="modsPanel" />
<mx:Panel id="debugPanel" />
<mx:Panel id="cutscenePlayer" />
<mx:Panel id="bossHealthBar" />
<mx:Panel id="hotbarPanel" />
<mx:Panel id="quickSlotPanel" />
<mx:Panel id="mouseCursorOverlay" />
<mx:Panel id="screenFadeOverlay" />
<mx:Panel id="loadingTipsPanel" />
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<mx:Panel id="gameOverRetryPanel" />
<mx:Panel id="levelUpStatsPanel" />
<mx:Panel id="inventoryGridPanel" />
<mx:Panel id="dialogueChoicesPanel" />
<mx:Panel id="tutorialStepPanel" />
<mx:Panel id="loadingProgressBar" />
<mx:Panel id="errorStackTrace" />
<mx:Panel id="confirmDialogMessage" />
<mx:Panel id="creditsScrollPanel" />
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<mx:Panel id="questLogFilterPanel" />
<mx:Panel id="shopItemGrid" />
<mx:Panel id="craftingRecipeGrid" />
<mx:Panel id="settingsTabPanel" />
<mx:Panel id="saveGameSlotPanel" />
<mx:Panel id="achievementGrid" />
<mx:Panel id="statsCharacterPanel" />
<mx:Panel id="bestiaryEntryPanel" />
<mx:Panel id="skillTreeNodeGrid" />
<mx:Panel id="partyMemberPanel" />
<mx:Panel id="worldMapRegionPanel" />
<mx:Panel id="minimapZoomPanel" />
<mx:Panel id="chatInputPanel" />
<mx:Panel id="notificationStackPanel" />
<mx:Panel id="tooltipContentPanel" />
<mx:Panel id="contextMenuItemsPanel" />
<mx:Panel id="modsInstalledPanel" />
<mx:Panel id="debugConsolePanel" />
<mx:Panel id="cutsceneSubtitlePanel" />
<mx:Panel id="bossPhaseIndicatorPanel" />
<mx:Panel id="hotbarSlotPanel" />
<mx:Panel id="quickSlotCooldownPanel" />
<mx:Panel id="mouseCursorTooltipPanel" />
<mx:Panel id="screenFadeOverlayPanel" />
<mx:Panel id="loadingTipsRotationPanel" />
<mx:Panel id="pauseTimeIndicatorPanel
```

```
<div id="ad_leaderboard_wrapper_wrapper_wrapper
```

一种电缆 专利说明书

[19]中华人民共和国专利局
[21]ZL 99 [illegible]
[51]Int.Cl.[illegible]
[22]1999.[illegible].[illegible]
[24]2000.[illegible].[illegible]
[37][illegible]
[71][illegible]
[72][illegible]
[74][illegible]
[54]发明名称 一种电缆

[57]摘要 [illegible]

[illegible]

이사회 의장 주주 제안서

제1조 (제안의 목적)

제1조 (제안의 목적) 본 제안서는 주주 김지훈이 이사회 의장 선임을 제안하기 위해 작성되었다. 의장 선임은 이사회 운영 전반에 대한 주주 권리 행사의 가장 직접적인 수단이다. 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다.

제2조 (의장 선임 제안의 근거)

제2조 (의장 선임 제안의 근거) 의장 선임 제안은 세 가지 근거에 기한다. 첫째, 현 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 둘째, 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 셋째, 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다.

제3조 (의사록 미작성 사실)

제3조 (의사록 미작성 사실) 2024년 3월 이사회 의사록 미작성 사실은 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다. 의사록 작성 의무는 이사회 의장의 의무다. 의사록 미작성 사실은 의장 김민수가 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다.

제4조 (의장 김민수의 의사록 작성 의무 불이행)

제4조 (의장 김민수의 의사록 작성 의무 불이행) 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 의사록 작성 의무는 이사회 의장의 의무다. 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다.

제5조 (의장 선임 제안의 효과)

제5조 (의장 선임 제안의 효과) 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다.

[illegible]

[illegible]

[illegible]

```
<?xml version="1.0"
<!DOCTYPE
<!-- Generated by Microsoft Word 15.0.0.0 -->
<w:wordDocument xmlns:w="http://schemas.microsoft.com/office/word/2003/wordml" xmlns:
  xmlns:
  xmlns:
  
```

1.1 Introduction

1.1.1 Why Study Differential Equations?

Differential equations are equations involving derivatives of unknown functions. For example, the equation

$$\frac{dy}{dx} = 2x + 3$$

is a differential equation. Here, y is a function of x, and $\frac{dy}{dx}$ represents the rate at which y changes with respect to x.

Differential equations are fundamental in mathematics because they describe how quantities change over time or in response to other variables. For example, Newton's Second Law, $F = ma$, can be rewritten as $F = m\frac{d^2x}{dt^2}$, where $x(t)$ is the position of an object at time t. This equation allows us to predict the motion of the object by solving for $x(t)$.

Differential equations are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the equation $L\frac{dI}{dt} + RI = V(t)$. Solving this equation allows us to predict how the current changes over time.

1.1.2 Ordinary Differential Equations (ODEs)

An ordinary differential equation (ODE) is a differential equation in which the unknown function depends on a single variable. For example, the equation $\frac{dy}{dx} = 2x + 3$ is an ODE because y depends only on x.

ODEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the population of a city may be modeled by the ODE $\frac{dP}{dt} = kP$, where $P(t)$ is the population at time t and k is a constant representing the growth rate.

ODEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the ODE $L\frac{dI}{dt} + RI = V(t)$. Solving this ODE allows us to predict how the current changes over time.

1.1.3 Partial Differential Equations (PDEs)

A partial differential equation (PDE) is a differential equation in which the unknown function depends on multiple variables. For example, the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ is a PDE because u depends on both x and t.

PDEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the heat equation $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ describes how the temperature $u(x,t)$ changes over time and space, where k is a constant representing the thermal conductivity.

PDEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the PDE $L\frac{\partial I}{\partial t} + RI = V(t)$. Solving this PDE allows us to predict how the current changes over time and space.

Ch. 2 Lect 2 Part 1 - Chapter Summary

Ch. 2 Lect 2 Part 1 - Chapter Summary

Ch. 2 Lect 2 Part 1 - Chapter Summary

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Accepted: 3 November 2020

```
{
  "data": {
    "id": "1",
    "type": "articles",
    "attributes": {
      "title": "JSON:API pagination",
      "slug": "jsonapi-pagination",
      "body": "First, we need to add a query parameter to our API. We'll use `page[number]` and `page[size]`.\n\nFrom there, our GET /articles?page[number]=3&page[size]=10 will return the 21st and 22nd articles in the database.",
      "created": "2015-05-22T14:56:29.000Z",
      "updated": "2015-05-22T14:56:29.000Z"
    },
    "relationships": {
      "author": {
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      },
      "tags": {
        "data": [
          {"id": "3", "type": "tags"},
          {"id": "4", "type": "tags"}
        ]
      }
    }
  },
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      "id": "42",
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      "attributes": {
        "name": "John Smith",
        "twitter": "jsmith"
      }
    },
    {
      "id": "3",
      "type": "tags",
      "attributes": {
        "name": "pagination"
      }
    },
    {
      "id": "4",
      "type": "tags",
      "attributes": {
        "name": "performance"
      }
    }
  ]
}
```

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Chapter 1 Introduction to Computer Science: Using Ruby 1.9.2 and Ruby 2.0 - 1.1 Introduction 1.1.1 Computer Hardware 1.1.2 Computer Software 1.1.3 Computer Programs 1.1.4 Computer Science 1.1.5 Computer Science Versus Computer Programming 1.1.6 Computer Science Versus Software Engineering 1.1.7 Computer Science Versus Information Technology 1.1.8 Computer Science Versus Other Disciplines 1.1.9 Computer Science Versus Mathematics 1.1.10 Computer Science Versus Science 1.1.11 Computer Science Versus Art 1.1.12 Computer Science Versus Philosophy 1.1.13 Computer Science Versus Engineering 1.1.14 Computer Science Versus Business 1.1.15 Computer Science Versus Medicine 1.1.16 Computer Science Versus Law 1.1.17 Computer Science Versus Sociology 1.1.18 Computer Science Versus Ethics 1.1.19 Computer Science Versus Politics 1.1.20 Computer Science Versus Economics 1.1.21 Computer Science Versus Psychology 1.1.22 Computer Science Versus Linguistics 1.1.23 Computer Science Versus Sociology 1.1.24 Computer Science Versus Ethics 1.1.25 Computer Science Versus Politics 1.1.26 Computer Science Versus Economics 1.1.27 Computer Science Versus Psychology 1.1.28 Computer Science Versus Linguistics 1.1.29 Computer Science Versus Sociology 1.1.30 Computer Science Versus Ethics 1.1.31 Computer Science Versus Politics 1.1.32 Computer Science Versus Economics 1.1.33 Computer Science Versus Psychology 1.1.34 Computer Science Versus Linguistics

Ch. 2 Lab Report - Angular Momentum

Name: Ryan Ramos

Partners: Ryan Ramos, Ryan Ramos, Ryan Ramos,

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Equation 1 of 1: Gaussian Integral Squared

Type: Definite Integral

Source: Gaussian Integral Squared

Tags: [Gaussian Integral, Gaussian Integral Squared]

Notes: Gaussian Integral Squared. Gaussian Integral Squared. Gaussian Integral Squared.

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Effect of Ultrasound Treatment on Antibacterial Activity of Aqueous Extract of Aloe Vera Gel

Ali Ahmadi , Mohammad Ali Ebrahimzadeh , Mohammad Ali Ghasemi , Ali Ghasemi , Ali Ghasemi ,

Chapter Chapter 1

Section Introduction

Subsection Introduction to Linear Algebra

Linear Algebra

Linear Algebra

Linear Algebra is the study of vectors and linear transformations between vector spaces.

Example 1.1.1

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Proof.

Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ be arbitrary elements of V and let $c \in \mathbb{R}$ be arbitrary.

Then $u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ and

$$T(u + v) = (x_1 + x_2 + y_1 + y_2, 2(z_1 + z_2) - (x_1 + x_2)) = (x_1 + y_1, 2z_1 - x_1) + (x_2 + y_2, 2z_2 - x_2) = T(u) + T(v).$$

Also, $cu = (cx_1, cy_1, cz_1)$ and

$$T(cu) = (cx_1 + cy_1, 2cz_1 - cx_1) = c(x_1 + y_1, 2z_1 - x_1) = cT(u).$$

Therefore T satisfies both conditions of linearity and is a linear transformation. $\square$

Example 1.1.2

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x + 1).$$

Then T is not a linear transformation because $T(0, 0, 0) = (0, 1) \neq (0, 0)$.

Example 1.1.3

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Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.99

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.100

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.101

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.102

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.103

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.104

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.105

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.106

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.107

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.108

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.109

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.110

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.111

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.112

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.113

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.114

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.115

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.116

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.117

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.118

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.119

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.120

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.121

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.122

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.123

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.124

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.125

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.126

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.127

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.128

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.129

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.130

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

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Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.132

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.133

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

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Then T is a linear transformation.

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Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.137

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.138

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.139

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.140

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.141

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.142

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.143

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.144

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.145

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.146

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.147

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.148

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.149

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.150

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.151

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.152

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.153

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.154

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.155

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.156

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.157

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let

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Abstract: In this paper, we study the Cauchy problem for the nonlinear Kirchhoff-Carrier wave equation in a bounded domain. We prove that the solution exists globally in time and decays to zero as time tends to infinity.

Keywords: Kirchhoff-Carrier wave equation, Cauchy problem, Global existence, Decay.

MSC: 35L70, 35L20, 35L05, 35L15, 35L71, 35L72, 35L76, 35L90, 35B40, 35B41, 35B45, 35B65, 35G20, 35G25, 35G30, 35G31, 35G50, 35G55, 35G60, 35G61

Topic:

Question:

Find a unit tangent vector to the curve $r(t) = \langle \sin(t), \cos(t), \sqrt{2}t \rangle$ at the point $(0, 1, \pi\sqrt{2}/4)$.

Solution:

Step 1: Find the derivative $r'(t)$ of the position vector $r(t)$.

$$r'(t) = \langle \cos(t), -\sin(t), \sqrt{2} \rangle$$

Step 2: Evaluate $r'(t)$ at the point $t = \pi/4$ (since $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$).

$$r'(\pi/4) = \langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle$$

Step 3: Find the magnitude of $r'(\pi/4)$.

$$|r'(\pi/4)| = \sqrt{(1/\sqrt{2})^2 + (-1/\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{1/2 + 1/2 + 2} = \sqrt{3}$$

Step 4: Divide $r'(\pi/4)$ by its magnitude to find the unit tangent vector.

$$T(\pi/4) = \frac{r'(\pi/4)}{|r'(\pi/4)|} = \frac{\langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle}{\sqrt{3}} = \langle 1/\sqrt{6}, -1/\sqrt{6}, \sqrt{2/3} \rangle$$

Step 5: Rationalize the denominators.

$$T(\pi/4) = \langle \sqrt{6}/6, -\sqrt{6}/6, \sqrt{6}/3 \rangle$$

Step 6: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 7: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 8: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 9: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 10: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 11: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 12: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 13: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 14: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 15: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 16: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 17: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 18: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 19: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 20: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 21: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 22: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 23: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 24: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 25: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 26: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 27: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 28: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 29: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 30: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 31: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 32: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 33: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 34: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 35: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 36: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 37: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 38: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 39: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 40: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 41: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 42: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

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```
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```

```
# -*- coding: utf-8 -*-
```

# Generated by Rust (https://www.rust-lang.org)
import numpy as np
import numpy.linalg as LA
import matplotlib.pyplot as plt
from matplotlib import rcParams, rc
from mpl_toolkits.mplot3d import Axes3D
from scipy.integrate import odeint, ode

If we want LaTeX fonts
rcParams['font.family'] = 'serif'
rcParams['font.serif'] = ['Computer Modern']

# Parameters
g = 9.81
L = 1.0
m = 1.0

# Initial conditions: theta and dtheta/dt
y0 = [3.14159265358979, 0.0]

# Time points
t = np.linspace(0, 10, 101)

# Define the ODE system
def f(y, t):
    theta = y[0]
    dtheta = y[1]
    ddtheta = -g/L*np.sin(theta)
    return [dtheta, ddtheta]

# Solve the ODE
y = odeint(f, y0, t)

# Extract theta and dtheta/dt
theta = y[:, 0]
dtheta = y[:, 1]

# Create figure
fig = plt.figure(1, figsize=(8, 6.4))

# Plot theta vs time
ax1 = fig.add_subplot(211)
ax1.plot(t, theta, 'b')
ax1.set_ylabel('θ')

# Plot dtheta/dt vs time
ax2 = fig.add_subplot(212)
ax2.plot(t, dtheta, 'r')
ax2.set_ylabel('$\dot{\theta}$')
ax2.set_xlabel('t')

# Save figure
plt.savefig('simple_pendulum.png', dpi=100)
```

```
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```

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```

NO-FLY-REPORTING/17

Non-American Jewish Foreign-born

Ted Cruz

Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article

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Ad Corrigendum Supplementary

Page 1 of 1 Generated on 2025-07-14

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Section 1

Section 2

Section 3

Chapter Eighteen

Sections

Introduction
Sections and Subsections
Section Numbers
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NO. 7147 MONTMORENCY

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1)

Image 1

Text

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1)

Image 2

Text

Sun Journal, 1907-07-19 (Vol. 1 Iss. 1) **THE MONTMORENCY SUN.** Vol. 1. **MONTMORENCY, MONT., SATURDAY JULY 19 1907.** No. 1. **PRICE 5 CENTS.** Number 1. Volume 1. Montmorency, Mont., Saturday, July 19, 1907. Price 5 cents. **Sun Publishing Co.** Publishers. **Sun Publishing Co.** Publishers. **Sun Publishing Co.** Publishers

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MR-TCAP-MAPMODULE

Operations-Generic-ASN1Protocol

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$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

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$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

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MR. TOUR'S MAGNIFICENCE!!!

Tour Tour Tour

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ISSN 2499-1562(online)
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Topic:

Topic: Fluid Flow Equations

Topic: Fluid Flow Equations for Incompressible Flow

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation, Laminar Flow and Poiseuille's Equation, Viscous Forces, Pipe Networks

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \qquad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{\mu}{\rho}\nabla^2 \mathbf{v} + \mathbf{g} \qquad \nabla \cdot \mathbf{v} = 0$$

$$\nabla \cdot (\rho \mathbf{v}) = \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho = 0$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} = 0$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v})

ISSN 1234-5678 (print) 1234-9876 (online)
Received: 12 March 2024; Accepted: 18 July 2024

Abstract
This study investigates the effect of pulsed laser ablation in liquid (PLAL) parameters on the synthesis of gold nanoparticles (AuNPs) using a Nd:YAG laser. The influence of laser fluence (2–6 J/cm²), pulse repetition rate (10–50 Hz), and ablation time (5–30 min) on the size, shape, and optical properties of AuNPs was evaluated. Results showed that increasing laser fluence and repetition rate reduced average particle size from 18 nm to 9 nm, while prolonged ablation time led to aggregation and broadening of the size distribution. UV-Vis spectroscopy confirmed the formation of AuNPs with surface plasmon resonance peaks between 520 and 540 nm. The optimal conditions for producing small, spherical AuNPs were identified as 4 J/cm², 30 Hz, and 15 min. These findings provide a systematic approach for tailoring AuNP properties by controlling laser parameters, which is crucial for their application in biomedicine and catalysis.

Keywords
laser ablation, gold nanoparticles, pulsed laser ablation in liquid, Nd:YAG laser, surface plasmon resonance, UV-Vis spectroscopy, nanomaterials, optimization, size control, optical properties

Ajibola Ismail Ajibola*, Ajibola Ismail Ajibola*, Ajibola Ismail Ajibola*,

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Hello!

Hello!

Received: 12th Jan. 2024; Accepted: 18th Jan. 2024; Published: 25th Jan. 2024

Kumar
Kumar Ashish Kumar Ashish Kumar Ashish

DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 13 November 2017
Accepted: 14 November 2017

Received: 2 April 2020
Accepted: 14 May 2020
Published online: 5 June 2020
https://doi.org/10.1038/s41586-020-2412-9

Ruo-Jia Cui[1,2,3], Hongxia Duan[1,2], Bin Zhao[1,2], Mengqiao Zhou[1,2], Rui Shi[1,2], Xiao-Yu Zhao[1,2], Yi-Yang Li[1,2], Xin-Quan Wang[1,2], Lei Li[1,2] & Zhi-Jian Qin[1,2]

Lycopene is a red carotenoid of great importance because it is responsible for the pigment of tomatoes and other red fruits and vegetables and because it has been associated with many health benefits. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Here we report that the transcription factor RIPENING INHIBITOR (RIN) binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

https://doi.org/10.1038/s41586-020-2412-9

Lycopene is a red carotenoid that is responsible for the pigment of tomatoes and other red fruits and vegetables. Lycopene is a precursor of provitamin A and has been associated with many health benefits, such as a reduced risk of cardiovascular disease and certain cancers[1–3]. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Tomato fruits are a major source of lycopene, and lycopene accumulation in tomato fruits is a complex process that is affected by many factors, such as ripening, ethylene, light, temperature, and genotype[4–7]. Ripening-inhibitor (rin) is a recessive mutant that has been used extensively in studies of tomato fruit ripening. The rin mutant fruits do not ripen and do not accumulate lycopene, even in the presence of ethylene[8]. The RIN gene encodes a MADS-box transcription factor that is a master regulator of fruit ripening in tomato[9–12]. Although the RIN protein has been shown to bind to the promoters of several genes that are involved in ethylene biosynthesis and signal transduction, the genes that are directly regulated by RIN and the mechanism by which RIN regulates lycopene accumulation are unknown. Here we show that RIN binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

[1] State Key Laboratory of Plant Cell and Chromosome Engineering, Institute of Genetics and Developmental Biology, Chinese Academy of Sciences, Beijing 100101, China. [2] Center for Agricultural Resources Research, Institute of Genetics and Developmental Biology, Chinese Academy of Sciences, Shijiazhuang 050021, China. [3] University of Chinese Academy of Sciences, Beijing 100049, China.

MATLAB/SIMULINK

Lecture Notes Prepared by Dr. Ramesh Kumar Ayyala, Assistant Professor, Department of Mathematics, Indian Institute of Technology Jodhpur. Please report any errors or typos and discuss this at Academia.edu/Mathematics Reddit. Use Ctrl+Shift+F to find a term or phrase.

Tags: #MATLAB #SIMULINK #Lecture-Notes #Prof-Dr-Ramesh-Kumar-Ayyala #Indian-Institute-of-Technology-Jodhpur #Mathematics #Reddit #Academia.edu

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Hello everyone, in this video we will discuss about **MATLAB/SIMULINK** and its uses in Control Theory and Control Systems.

Tags: #Control-Theory

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1$$

Tags: #MATLAB/SIMULINK

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK, #Lecture-Notes, #Prof-Dr-Ramesh-Kumar-Ayyala, #Lecture-Notes-Prof-Dr-Ramesh-Kumar-Ayyala

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Tags: #MATLAB/SIMULINK

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Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \quad \lim_{x \to 0} \frac{\sin(x)}{x} = 1 \quad \ldots

DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 14 November 2017
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Research Article

Jack Lewis Rhodes and Jon C. Cummins

Abstract

Recent research suggests that people who believe that science and technology are responsible for environmental problems tend to underestimate science's ability to solve these problems.

Keywords

Received: 2 April 2020; Revised: 14 July 2020; Accepted: 27 August 2020

Introduction

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Methods

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$$\frac{\partial \rho_{\mathrm{Ni}}}{\partial t} = \frac{\partial \rho_{\mathrm{Ni}}}{\partial x} + \frac{\partial \rho_{\mathrm{Ni}}}{\partial y} + \frac{\partial \rho_{\mathrm{Ni}}}{\partial z} + \rho_{\mathrm{Ni}}$$

Results

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Discussion

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DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 19 December 2017
Accepted: 20 December 2017
Online First: 4 January 2018

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Quantum Mechanical Simulation of Electron Density Distribution in Hydrogen Molecular Ion (H₂⁺) Using Variational Methods

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Abstract

This study presents a variational approach to simulate the electron density distribution in the hydrogen molecular ion (H_2^+) using trial wavefunctions and numerical optimization techniques. The ground state energy and electron density are calculated for various internuclear distances and compared with exact results. The results show that the variational method provides accurate electron density distributions with minimal computational cost. The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems.

Keywords: Quantum Mechanics, Variational Method, Hydrogen Molecular Ion, Electron Density, Ground State Energy, Computational Chemistry, Molecular Simulation, Trial Wavefunction, Optimization, Hartree-Fock Method

1. Introduction

The hydrogen molecular ion (H_2^+) is the simplest molecular system and serves as a fundamental benchmark for quantum mechanical calculations. Despite its simplicity, it exhibits all the essential features of chemical bonding, including electron sharing between two nuclei and the formation of a stable molecular orbital. The study of H_2^+ provides insights into the nature of chemical bonds and serves as a test case for more complex molecular systems.

The electron density distribution in H_2^+ is crucial for understanding the nature of the chemical bond. The electron density determines the strength of the bond, the bond length, and the vibrational frequencies. Accurate calculation of electron density is essential for predicting molecular properties and understanding chemical reactivity.

Variational methods are powerful tools for calculating molecular properties. These methods provide upper bounds to the ground state energy and allow for the calculation of electron density distributions. The variational method is particularly useful for systems where exact solutions are not available.

This study presents a variational approach to simulate the electron density distribution in H_2^+. The study compares the results with exact calculations and demonstrates the effectiveness of simple variational methods for quantum mechanical simulations.

2. Theoretical Background

2.1 Schrödinger Equation

The time-independent Schrödinger equation for H_2^+ is:

$$H\psi = E\psi$$

where H is the Hamiltonian operator, ψ is the wavefunction, and E is the energy.

2.2 Hamiltonian Operator

The Hamiltonian operator for H_2^+ is:

$$H = -\frac{\nabla_1^2}{2} - \frac{\nabla_2^2}{2} - \frac{1}{r_{a1}} - \frac{1}{r_{b1}} + \frac{1}{R}$$

where ∇_1^2 and ∇_2^2 are the Laplacian operators for electrons 1 and 2, r_{a1} and r_{b1} are the distances from electron 1 to nuclei a and b, and R is the internuclear distance.

2.3 Variational Principle

The variational principle states that for any trial wavefunction ψ_{trial}, the energy E_{trial} satisfies:

$$E_{trial} = \frac{\langle \psi_{trial} | H | \psi_{trial} \rangle}{\langle \psi_{trial} | \psi_{trial} \rangle} \geq E_0 \quad (1)$$

where E_0 is the ground state energy.

2.4 Trial Wavefunction

The trial wavefunction used in this study is:

$$\psi_{trial} = \frac{1}{\sqrt{2(1+S)}}(\phi_a + \phi_b) \quad (2)$$

where ϕ_a and ϕ_b are hydrogen 1s orbitals centered on nuclei a and b, and S is the overlap integral between these orbitals.

3. Methodology

3.1 Computational Approach

The computational approach involves the following steps:

1. Calculation of overlap integrals
2. Calculation of kinetic energy integrals
3. Calculation of potential energy integrals
4. Optimization of the variational parameter
5. Calculation of electron density distribution

3.2 Overlap Integral

The overlap integral between two hydrogen 1s orbitals is:

$$S = \int \phi_a \phi_b \, d\tau = \left(1 + \frac{R}{a_0} + \frac{R^2}{3a_0^2}\right) e^{-R/a_0} \quad (3)$$

where a_0 is the Bohr radius.

3.3 Kinetic Energy Integral

The kinetic energy integral is:

$$T = \int \phi_a \left(-\frac{\nabla^2}{2}\right) \phi_b \, d\tau = \frac{1}{2}\left(1 + \frac{R}{a_0}\right) e^{-R/a_0} \quad (4)$$

3.4 Potential Energy Integral

The potential energy integral is:

$$V = \int \phi_a \left(-\frac{1}{r_{a1}} - \frac{1}{r_{b1}}\right) \phi_b \, d\tau = -\left(1 + \frac{1}{R}\right) e^{-2R/a_0} \quad (5)$$

3.5 Optimization

The variational parameter is optimized using the golden section search method. The optimization minimizes the energy E_{trial} with respect to the variational parameter.

3.6 Electron Density Calculation

The electron density is calculated from the trial wavefunction as:

$$\rho(r) = |\psi_{trial}|^2$$

4. Results and Discussion

4.1 Ground State Energy

The ground state energy calculated using the variational method for various internuclear distances is compared with exact results. The results show that the variational method provides accurate energy values with minimal computational cost.

4.2 Electron Density Distribution

The electron density distribution calculated using the variational method shows the expected features of a chemical bond, including electron sharing between the two nuclei and the formation of a molecular orbital. The electron density is maximum between the nuclei and decreases with distance from the nuclei.

4.3 Comparison with Exact Results

The results are compared with exact results from literature. The variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

5. Conclusion

This study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost. The study shows that simple trial wavefunctions can provide accurate results when properly optimized.

The results show that the variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost.

6. References

[1] Atkins, P. W., & Friedman, R. S. (2010). Molecular Quantum Mechanics (5th ed.). Oxford University Press.

[2] Szabo, A., & Ostlund, N. S. (1996). Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory (Dover ed.). Dover Publications.

[3] Levine, I. N. (2014). Quantum Chemistry (7th ed.). Pearson Education.

[4] Pauling, L., & Wilson, E. B. (1935). Introduction to Quantum Mechanics with Applications to Chemistry. McGraw-Hill.

[5] Slater, J. C. (1930). Atomic Shielding Constants. Physical Review, 36(57), 57–64.

[6] Hehre, W. J., Radom, L., Schleyer, P. v. R., & Pople, J. A. (1986). Ab Initio Molecular Orbital Theory. Wiley-Interscience.

[7] Mulliken, R. S. (1928). The Assignment of Quantum Numbers for Electrons in Molecules. Physical Review, 32(186–195), 761–772.

[8] Coulson, C. A., & Fischer, I. (1949). Notes on the Molecular Orbital Treatment of the Hydrogen Molecule. Philosophical Magazine, 40(308), 386–393.

[9] Karplus, M., & Porter, R. N. (1970). Atoms and Molecules: An Introduction for Students of Physical Chemistry. Benjamin.

[10] Pople, J. A., & Beveridge, D. L. (1970). Approximate Molecular Orbital Theory. McGraw-Hill.

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MR-TCAST-MQ2000B0B0-V1

MR-TCAST-MQ2000B0B0-V1 Product Specification

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1 Product Overview

1.1 Product Introduction

MR-TCAST-MQ2000B0B0-V1 is a high-performance industrial-grade multi-port Ethernet switch designed for demanding industrial automation, smart manufacturing, and critical infrastructure applications. The device provides 2 Gigabit Ethernet ports and 8 Gigabit Ethernet ports, supporting Layer 2 switching functions and offering comprehensive security and management capabilities. It adopts a fanless design with wide temperature operation capability, ensuring stable operation in harsh industrial environments.

1.2 Product Features

- Hardware Platform: Dual-core ARM Cortex-A53 processor, 512MB DDR3 memory, 128MB NAND Flash
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Switching Capacity: 20Gbps
- Forwarding Rate: 14.88Mpps
- MAC Address Table: 16K entries
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
- Spanning Tree: Supports STP, RSTP, MSTP
- Link Aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- Multicast: Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog
- Power Supply: Dual power input, supports 12V/24V DC, supports power failure alarm
- Operating Temperature: -40°C ~ +75°C
- Storage Temperature: -40°C ~ +85°C
- Operating Humidity: 5% ~ 95% RH (non-condensing)
- Protection Rating: IP40
- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail

1.3 Product Specifications

Item	Specifications
Model	MR-TCAST-MQ2000B0B0-V1
Ethernet Ports	2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
Switching Capacity	20Gbps
Forwarding Rate	14.88Mpps
MAC Address Table	16K entries
VLAN	Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
Spanning Tree	Supports STP, RSTP, MSTP
Link Aggregation	Supports static LAG and LACP, up to 8 groups, 8 members per group
Multicast	Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
QoS	Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
Security	Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
Management	Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
Protocols	Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
Reliability	Supports ring network protection, port mirroring, cable diagnostics, watchdog
Power Supply	Dual power input, supports 12V/24V DC, supports power failure alarm
Operating Temperature	-40°C ~ +75°C
Storage Temperature	-40°C ~ +85°C
Operating Humidity	5% ~ 95% RH (non-condensing)
Protection Rating	IP40
Dimensions	440mm × 44mm × 180mm (W × H × D)
Weight	1.5kg
Installation	19-inch rack-mount, desktop, DIN rail

1.4 Product Appearance

1.5 Product Dimensions

1.6 Product Interfaces

Front Panel

- 2 Gigabit Ethernet ports (1000Base-T/SFP combo)
- 8 Gigabit Ethernet ports (1000Base-T)
- Console port
- Power indicator
- System indicator
- Port indicators

Rear Panel

- Power terminal block
- Grounding screw

1.7 Product Indicators

Indicator	Status	Description
Power	Solid green	Power normal
	Off	Power abnormal
System	Solid green	System normal
	Blinking green	System starting
	Solid red	System fault
	Off	No power
Port	Solid green	Link established
	Blinking green	Data transmission
	Off	Link not established

2 Hardware Specifications

2.1 Hardware Architecture

- Processor: Dual-core ARM Cortex-A53 processor
- Memory: 512MB DDR3
- Flash: 128MB NAND Flash
- Switch Chip: Integrated Layer 2 switch chip
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Console Port: 1 console port
- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Indicators: Power indicator, system indicator, port indicators
- Buttons: Reset button

2.2 Power Supply

- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Power Consumption: Typical 15W, maximum 25W
- Power Terminal Block: Supports 2-pin terminal block connection
- Grounding Screw: Provides grounding screw for safety and lightning protection

2.3 Power Consumption

- Typical: 15W
- Maximum: 25W
- Idle: 8W

2.4 Cooling Design

- Fanless design, no fan, no noise, no dust, no maintenance
- Natural convection heat dissipation, suitable for long-term stable operation
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation

2.5 Environmental Requirements

- Operating temperature: -40°C ~ +75°C
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40
- Installation environment: Indoor, avoid direct sunlight, ensure good ventilation

2.6 Mechanical Specifications

- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail
- Rack-mount ears: Standard 19-inch rack-mount ears, supports front and rear installation
- Chassis: Metal chassis, sturdy and durable

2.7 Electromagnetic Compatibility

- Conducted emissions: Meets industrial-grade EMC requirements
- Radiated emissions: Meets industrial-grade EMC requirements
- Electrostatic discharge: Supports ESD protection
- Surge protection: Supports surge protection
- Electromagnetic interference: Meets industrial-grade EMC requirements

2.8 Reliability

- Mean Time Between Failures (MTBF): Over 100,000 hours
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation
- Power failure alarm: Supports power failure alarm, automatically sends alarm when power is lost
- Watchdog: Supports hardware watchdog, automatically restores system when system hangs
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3 Software Specifications

3.1 Operating System

- Operating system: Proprietary operating system
- Kernel: Linux kernel
- Management interfaces: CLI, Web, SNMP, Telnet, SSH
- Protocol stack: Supports IPv4/IPv6 dual stack
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup

3.2 Management Interfaces

- CLI: Supports command-line interface, supports Telnet and SSH remote management
- Web: Supports web management, supports HTTP and HTTPS
- SNMP: Supports SNMP v1/v2c/v3, supports MIB-II, supports private MIB
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

3.3 Protocol Support

- Layer 2 protocols: Supports ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Spanning tree: Supports STP, RSTP, MSTP
- Link aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
- Multicast: Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog

3.4 Security Features

- Port security: Supports port security, limits MAC address learning
- MAC address binding: Supports MAC address binding, prevents unauthorized access
- DHCP Snooping: Supports DHCP Snooping, prevents DHCP spoofing attacks
- ARP Inspection: Supports ARP Inspection, prevents ARP spoofing attacks
- IP Source Guard: Supports IP Source Guard, prevents IP address spoofing
- 802.1X authentication: Supports 802.1X authentication, supports RADIUS authentication
- RADIUS: Supports RADIUS authentication, supports accounting
- TACACS+: Supports TACACS+ authentication, supports accounting
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3.5 Firmware Upgrade

- Online upgrade: Supports online firmware upgrade, no need to restart the device
- Dual firmware backup: Supports dual firmware backup, automatically switches to backup firmware when upgrade fails
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

4 Ordering Information

4.1 Product Model Description

MR-TCAST-MQ2000B0B0-V1

- MR: Industrial Ethernet switch
- TCAST: Product series
- MQ2000: 2 Gigabit Ethernet ports + 8 Gigabit Ethernet ports
- B0: Standard configuration
- B0: Standard configuration
- V1: Version 1

4.2 Accessories

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

4.3 Spare Parts

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

5 Appendix

5.1 Glossary

- Switch: Network switch, forwards data packets between network devices
- Ethernet: Ethernet, wired network protocol
- Gigabit Ethernet: Gigabit Ethernet, 1000Mbps Ethernet
- Port: Port, network interface
- VLAN: Virtual Local Area Network, logical network segmentation
- QinQ: Double VLAN, supports VLAN stacking
- STP: Spanning Tree Protocol, prevents network loops
- RSTP: Rapid Spanning Tree Protocol, fast convergence
- MSTP: Multiple Spanning Tree Protocol, supports multiple spanning trees
- LAG: Link Aggregation Group, bundles multiple physical links into one logical link
- LACP: Link Aggregation Control Protocol, dynamic link aggregation
- IGMP: Internet Group Management Protocol, multicast management protocol
- MLD: Multicast Listener Discovery, IPv6 multicast management protocol
- QoS: Quality of Service, supports traffic prioritization and bandwidth allocation
- ACL: Access Control List, supports packet filtering and access control
- 802.1X: Port-based Network Access Control, supports user authentication
- RADIUS: Remote Authentication Dial-In User Service, supports centralized authentication
- TACACS+: Terminal Access Controller Access-Control System Plus, supports centralized authentication
- SNMP: Simple Network Management Protocol, supports network management
- MIB: Management Information Base, supports SNMP management
- CLI: Command Line Interface, supports command-line management
- Web: Supports web management, supports HTTP and HTTPS
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

5.2 Revision History

Version	Date	Description
V1.0	2026-04-01	Initial release

Tags: ancient civilizations, ancient language speakers, Lost Cemetery of Ancient Language Speakers, Joseph Francis Alcazar, Cornelius Ashburn, Dr. Ellis, Lucinda Hawkes, Samuel Hawkes, Tom Hawkes, Professor Alcazar, Professor Ashburn, Professor Ellis, Professor Hawkes, Professor Lucinda Hawkes, Professor Samuel Hawkes, Professor Tom Hawkes, Professor Ashburn's Assistant, Professor Ellis's Assistant, Professor Hawkes's Assistant, Professor Lucinda Hawkes's Assistant, Professor Samuel Hawkes's Assistant, Professor Tom Hawkes's Assistant, Professor Ashburn's Assistant's Assistant, Professor Ellis's Assistant's Assistant, Professor Hawkes's Assistant's Assistant, Professor Lucinda Hawkes's Assistant's Assistant, Professor Samuel Hawkes's Assistant's Assistant, Professor Tom Hawkes's Assistant's Assistant, Professor Ashburn's Assistant's Assistant's Assistant, Professor Ellis's Assistant's Assistant's Assistant, Professor Hawkes's Assistant's Assistant's Assistant, Professor Lucinda 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Ref: Fluid Mechanics

Date: Monday, June 1, 2026 at 10:00:00 AM Coordinated Universal Time 7

Fluid Mechanics

Topic: Laplace transformation for Partial differential equations

Subtopic: Laplace transformation for Partial differential equations

Keywords: Laplace transform, Partial differential equations, Fluid mechanics, Heat transfer, Signal processing, Control systems, Electrical engineering, Mathematical modeling, Boundary value problems, Initial value problems, Differential equations, Transfer functions, Stability analysis, Frequency domain analysis, Time domain analysis, Integral transforms, Complex analysis, Engineering mathematics, Applied mathematics, Solution methods, Analytical methods, Boundary conditions, Initial conditions, Transient analysis, Steady state analysis, Dynamic systems, Linear systems, Nonlinear systems, Partial differential equations, Ordinary differential equations, Boundary value problems, Initial value problems, Laplace transform, Fourier transform, Inverse transform, Convolution theorem, Heaviside step function, Dirac delta function, Green's function

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Fig.2 Sensitivity analysis of Fermentation Capacity

Fig.2

Author: Prof. Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu,

NO. 7[illegible]

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

Keywords: Fluid Mechanics, Lecture Series, Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture

MR-TIGER-MAGNUM-MAXIMUS

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

$$\int_{\mathbb{R}^n} \frac{\mathbb{E}}{\mathbb{P}} \, \mathrm{d}\mathbb{P} = \mathbb{E}$$

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

$$\begin{aligned} \mathbb{P} &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \\ &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \\ &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \\ &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \end{aligned}$$

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Fabian Ruehle [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40]

MATLAB/Simulink Model Based Design Workflow

Introduction

Model Based Design (MBD) is a formal methodology for developing complex control systems. It uses a single mathematical model as the central artifact from which requirements, simulation, code generation, and verification are derived. The model serves as the ground truth for all downstream activities, ensuring that changes propagate consistently across the entire development lifecycle.

Workflow Overview

The workflow consists of six stages: requirements capture, plant modeling, controller design, simulation and verification, code generation, and deployment. Each stage produces artifacts that feed into the next stage. Traceability links connect requirements to model elements to generated code to test results, so that any change can be assessed for its impact on every downstream artifact.

Plant Modeling

Plant modeling begins with the physical equations governing the system. For a DC motor, the equations are:

$$v(t) = Ri(t) + L\frac{di(t)}{dt} + K_e\omega(t)$$

$$J\frac{d\omega(t)}{dt} = K_t i(t) - B\omega(t) - \tau_L(t)$$

Controller Design

The controller is designed in the discrete domain. For a PI controller with sampling period T_s:

$$u[k] = K_p e[k] + K_i T_s \sum_{j=0}^{k} e[j]$$

Simulink Implementation

Simulink provides a block diagram environment for implementing the plant model and controller. Signal flow is explicit and visual. Simulink supports fixed-step and variable-step solvers, with fixed-step solvers required for real-time code generation.

Code Generation

Simulink Coder (formerly Real-Time Workshop) generates C code from the model. The generated code is deterministic, with no dynamic memory allocation and no recursion. Code generation can be configured for different target processors and optimization levels.

Verification

Model-in-the-loop (MIL) simulation verifies the model against requirements before code generation. Software-in-the-loop (SIL) simulation runs the generated code on a host machine to verify that the code behaves identically to the model. Hardware-in-the-loop (HIL) simulation runs the code on the target hardware with real sensors and actuators.

Deployment

Deployment packages the generated code with the target hardware's boot code and runtime libraries. Over-the-air (OTA) updates allow deployed systems to receive new versions without physical access.

Conclusion

Model Based Design provides a single source of truth for requirements, simulation, and implementation. Changes propagate automatically through the workflow, reducing the risk of inconsistencies between specification and implementation.

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1.1 Introduction to Data Structures and Algorithms

Data structures and algorithms are the foundation of computer science. Understanding how to organize data efficiently and how to design algorithms that solve problems quickly is essential for any programmer.

1.1.1 Why Study Data Structures and Algorithms?

Consider a simple problem: you have a list of 1,000,000 numbers and you need to find the largest one. A naive approach might scan the entire list for each query. With the right data structure, you can answer the question in constant time. The difference between a slow program and a fast program can be the difference between a project that works and one that never finishes.

Data structures and algorithms are the tools that make computers fast and reliable. They are used in every domain: finance, medicine, engineering, science, and everyday software. Mastering them gives you the ability to design systems that scale from a few users to millions of users.

1.1.2 Formal Definitions

Let n denote the size of the input. We write $f(n) = O(g(n))$ if there exists constants $c > 0$ and $n_0 \geq 0$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. Similarly, $f(n) = \Omega(g(n))$ if $f(n) \geq c \cdot g(n)$ for all $n \geq n_0$, and $f(n) = \Theta(g(n))$ if both hold.

Formally, a data structure is a tuple $(D, \mathcal{O})$ where D is the set of possible states and $\mathcal{O}$ is a set of operations that transform states in D.

1.1.3 Example

```
def binary_search(arr, target):
    lo, hi = 0, len(arr) - 1
    while lo <= hi:
        mid = (lo + hi) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            lo = mid + 1
        else:
            hi = mid - 1
    return -1
```

Binary search runs in $O(\log n)$ time. Each iteration halves the search space, so after k iterations the remaining range has size $n / 2^k$. When $k = \log_2 n$, the range has size 1 and the algorithm terminates.

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TLS_RSA_WITH_RC4_128_SHA
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Chapter 4 Matrix Algebra

4.1 Sum and Scalar Product of Matrices

Definition

Sum of two matrices Two matrices A and B are said to be conformable for addition if they are of the same order. If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ then the sum of A and B is defined as $A + B = [a_{ij} + b_{ij}]_{m \times n}$.

Scalar product If $A = [a_{ij}]_{m \times n}$ and k is a scalar then $kA = [ka_{ij}]_{m \times n}$.

Example

Example 1

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}, \quad k = 2$$

Solution

$$A + B = \begin{bmatrix} 1+7 & 2+8 & 3+9 \\ 4+10 & 5+11 & 6+12 \end{bmatrix} = \begin{bmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

$$kA = \begin{bmatrix} 2 \times 1 & 2 \times 2 & 2 \times 3 \\ 2 \times 4 & 2 \times 5 & 2 \times 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Example 2

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}, \quad k = 3$$

Solution

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

$$kA = \begin{bmatrix} 3 \times 1 & 3 \times 2 \\ 3 \times 3 & 3 \times 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$$

Example 3

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 4

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 5

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 6

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 7

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 8

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 9

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 10

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 11

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 12

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 13

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 14

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 15

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 16

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 17

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 18

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 19

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 20

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 21

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 22

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 23

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 24

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 25

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 26

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 27

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 28

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 29

Find $A + B$ and kA if

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Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 30

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 31

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 32

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 &

MR-TIGER-MAGNUMOPUS-7

Tiger Magnum Opus 7 Extraordinary Lemma

Tiger

Tiger Lemma Formula: If Theorem Average Lemma

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$$\int_{\Omega} \rho \, dV = \int_{\partial\Omega} \rho \, dS$$

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<!ENTITY % brandDTD SYSTEM "chrome://branding/locale/brand.dtd">
%brandDTD;
<!ENTITY % prefsDTD SYSTEM "chrome://mozapps/locale/preferences/preferences.dtd">
%prefsDTD;
<!ENTITY % dnsmasqDTD SYSTEM "chrome://dnsmasq/locale/dnsmasq.dtd">
%dnsmasqDTD;
]>

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        title="&dnsmasq.preferences.title;"
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<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0.000,0.000,1.000,1.000"
  preserveAspectRatio="xMidYMid meet">
<defs>
  <path id="1" d="M0.000,0.000 L0.000,0.000 L0.000,0.000 L
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Ultrasonic Sensor Based Smart Dustbin

[1]Dr. Ashish Kumar Jha, [2]Ankit Kumar, [3]Ankit Kumar, [4]Ankit Kumar, [5]Ankit Kumar, [6]Ankit Kumar, [7]Ankit Kumar, [8]Ankit Kumar, [9]Ankit Kumar, [10]Ankit Kumar, [11]Ankit Kumar, [12]Ankit Kumar, [13]Ankit Kumar, [14]Ankit Kumar, [15]Ankit Kumar, [16]Ankit Kumar, [17]Ankit Kumar, [18]Ankit Kumar, [19]Ankit Kumar, [20]Ankit Kumar, [21]Ankit Kumar,

Sample Chemical Formula Balancer

Enter chemical formula (Example: Fe2(SO4)3):

```
Fe2(SO4)3
```

Parsed: {'Fe': 2, 'S': 3, 'O': 12}

Enter chemical formula (Example: Fe2(SO4)3):

```
Fe(OH)3
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Parsed: {'Fe': 1, 'O': 3, 'H': 3}

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

1

Enter chemical formula (Example: Fe2(SO4)3):

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Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

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Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

2

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

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Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2},\mathbf{3}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

3

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

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Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2},\mathbf{3},\mathbf{4}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

4

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
```

Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

Enter chemical formula (Example: Fe2(SO4)3):

```
Cu(NH3)4(OH)2
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Parsed: {'Cu': 1, 'N': 4, 'H': 12, 'O': 2}

$$\begin{array}{l} \mathbf{B}=\{\mathbf{1},\mathbf{2},\mathbf{3},\mathbf{4},\mathbf{5}\} \\ \quad \mathbf{1}\rightarrow\mathbf{2} \\ \quad \mathbf{2}\rightarrow\mathbf{3}\rightarrow\mathbf{4} \\ \quad \mathbf{4}\rightarrow\mathbf{5} \end{array}$$

5

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Ethical Implications of Artificial Intelligence in Healthcare: Balancing Innovation and Patient Rights

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Abstract: Artificial Intelligence (AI) is rapidly transforming healthcare by enhancing diagnostic accuracy, optimizing treatment plans, and improving patient outcomes. However, the integration of AI in healthcare raises significant ethical concerns, including data privacy, algorithmic bias, accountability, and the potential displacement of human caregivers. This study explores the ethical implications of AI in healthcare, focusing on the balance between innovation and patient rights. The study examines the ethical challenges associated with AI in healthcare, including the impact on patient autonomy, the potential for discrimination, and the need for transparency and accountability. The study also discusses the regulatory frameworks and guidelines that can ensure the ethical deployment of AI in healthcare, ensuring that the benefits of AI are maximized while safeguarding patient rights and maintaining trust in healthcare systems. The study concludes with recommendations for policymakers, healthcare providers, and AI developers to address the ethical challenges and promote the responsible use of AI in healthcare.

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Ch. 2 Lo-Fi Prototyping

Introduction

Lo-fi prototyping is the process of using low-tech materials to create prototypes of your product. Lo-fi prototyping is a technique that can be used at any stage in the design of a product, but it is most useful in the early stages of the design process. Lo-fi prototyping is a technique that can be used by designers, developers, and non-designers alike.

Why Lo-Fi Prototyping?

Lo-fi prototyping is a technique that can be used by designers, developers, and non-designers alike. It is most useful in the early stages of the design process.

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Research Article

Cognitive Ability Predicts Obesity and Smoking Across Three Cohorts of Twenty-Year-Olds

Ian J. Deary, Stuart J. Ritchie, and Timothy C. Bates
University of Edinburgh

Psychological Science
2020, Vol. 31(8) 1000–1012
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7.1 Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

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Abstract

Deep learning (DL) models have shown promising performance in medical imaging tasks, including the detection of diabetic retinopathy (DR) and other eye diseases. However, it remains unclear how these models generalize to real-world clinical settings. Here we assess the generalizability of DL models for DR detection from fundus photographs across ethnicity, imaging camera, and disease severity and type in a dataset of 236,234 images from 40,339 diabetic patients. We find that the performance of DL models for referable DR detection is comparable across Asian, Caucasian, and African ethnicities (AUC: 0.936–0.946), but varies across cameras (AUC: 0.906–0.947) and is lower in severe DR (AUC: 0.885) and proliferative DR (AUC: 0.901) compared to moderate DR (AUC: 0.943). These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

NPJ | https://doi.org/10.1038/s41746-020-00376-2 ARTMED | 1

Introduction

Diabetes mellitus (DM) is a chronic disease that affects 463 million people worldwide and is projected to rise to 700 million by 2045[1]. Diabetic retinopathy (DR) is a common complication of DM and a major cause of blindness and vision impairment worldwide[2]. Early detection of DR followed by timely treatment can prevent vision loss[3], and thus, DM patients are recommended to undergo yearly DR screening[4]. However, in many countries, the shortage of eye care specialists and the high prevalence of DM make such a task difficult to accomplish[5]. Deep learning (DL) models have shown promising performance in the detection of DR from fundus photographs, with several studies reporting high sensitivity and specificity in retrospective datasets[6–9]. However, it remains unclear how these models generalize to real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Here we assess the generalizability of a DL model for referable DR (RDR) detection across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the United Kingdom (UK) Biobank[14].

Results

Study population characteristics

The UK Biobank dataset consisted of 236,234 fundus photographs from 40,339 diabetic patients, of which 1,401 (3.5%) had RDR, 1,067 (2.6%) had moderate DR, 1,250 (3.1%) had severe DR, and 1,175 (2.9%) had proliferative DR (PDR). Mean (SD) age was 56.8 (8.1) years, 23,561 (58.4%) were male, 1,937 (4.8%) were Asian, 1,478 (3.7%) were African, and 36,924 (91.5%) were Caucasian. Mean (SD) HbA1c was 75.1 (19.7) mmol/mol (9.0 (1.8)%), 3,553 (8.8%) had insulin treatment, and mean (SD) duration of DM was 8.7 (8.1) years. Mean (SD) age was 55.4 (8.1) years for those with RDR, 55.8 (8.1) years for those with moderate DR, 56.4 (8.1) years for those with severe DR, and 56.1 (8.1) years for those with PDR. Mean (SD) HbA1c was 81.0 (21.3) mmol/mol (9.6 (1.9)%) for those with RDR, 77.4 (19.9) mmol/mol (9.3 (1.8)%) for those with moderate DR, 79.4 (20.1) mmol/mol (9.4 (1.8)%) for those with severe DR, and 79.9 (20.4) mmol/mol (9.5 (1.9)%) for those with PDR. Insulin treatment was 14.1% (RDR), 12.1% (moderate DR), 13.8% (severe DR), and 13.8% (PDR). Mean (SD) duration of DM was 10.7 (8.4) years for those with RDR, 10.1 (8.2) years for those with moderate DR, 10.8 (8.2) years for those with severe DR, and 10.8 (8.2) years for those with PDR. Other baseline characteristics by DR status are shown in Table 1.

Performance of DL models for referable DR detection

We trained a DL model to detect RDR using fundus photographs from patients of Caucasian ethnicity, imaging with a Topcon camera, and without severe DR or PDR. The model achieved an AUC of 0.943 (95% CI: 0.937–0.949) in a randomly sampled test set of 9,000 images from 9,000 patients. The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2). The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2).

Performance of DL models across ethnicities

The model achieved an AUC of 0.936 (95% CI: 0.914–0.957) in Asian patients, 0.946 (95% CI: 0.935–0.956) in African patients, and 0.943 (95% CI: 0.937–0.949) in Caucasian patients (Table 2). The model achieved a sensitivity of 0.879 (95% CI: 0.821–0.927) in Asian patients, 0.893 (95% CI: 0.865–0.918) in African patients, and 0.885 (95% CI: 0.869–0.901) in Caucasian patients (Table 2). The model achieved a specificity of 0.854 (95% CI: 0.824–0.883) in Asian patients, 0.863 (95% CI: 0.844–0.882) in African patients, and 0.857 (95% CI: 0.853–0.861) in Caucasian patients (Table 2).

Performance of DL models across cameras

The model achieved an AUC of 0.906 (95% CI: 0.890–0.921) with a Topcon camera, 0.913 (95% CI: 0.895–0.931) with a Canon camera, and 0.947 (95% CI: 0.941–0.953) with a Zeiss camera (Table 2). The model achieved a sensitivity of 0.862 (95% CI: 0.844–0.880) with a Topcon camera, 0.867 (95% CI: 0.819–0.911) with a Canon camera, and 0.887 (95% CI: 0.871–0.903) with a Zeiss camera (Table 2). The model achieved a specificity of 0.857 (95% CI: 0.853–0.861) with a Topcon camera, 0.859 (95% CI: 0.855–0.863) with a Canon camera, and 0.858 (95% CI: 0.854–0.862) with a Zeiss camera (Table 2).

Performance of DL models across disease severity and type

The model achieved an AUC of 0.885 (95% CI: 0.867–0.903) in severe DR and 0.901 (95% CI: 0.882–0.919) in PDR, compared to an AUC of 0.943 (95% CI: 0.937–0.949) in those without severe DR or PDR. The model achieved a sensitivity of 0.838 (95% CI: 0.807–0.868) in severe DR and 0.854 (95% CI: 0.824–0.883) in PDR, compared to a sensitivity of 0.885 (95% CI: 0.869–0.901) in those without severe DR or PDR. The model achieved a specificity of 0.858 (95% CI: 0.854–0.862) in severe DR and 0.857 (95% CI: 0.853–0.861) in PDR, compared to a specificity of 0.857 (95% CI: 0.853–0.861) in those without severe DR or PDR. Other performance metrics by DR severity and type are shown in Table 2.

Discussion

In this study, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

Prior studies have shown promising performance of DL models for DR detection from fundus photographs, with several reporting sensitivities and specificities of over 90% in retrospective datasets[6–9]. However, it remains unclear how these models would perform in real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Our study addresses these gaps in the literature by assessing the performance of a DL model across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank.

Our study shows that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation. The lower performance in severe DR and PDR suggests that DL models may miss some of the most severe cases of DR, which are the cases that require urgent referral to an eye care specialist[3]. The lower performance when applied to images from a camera not used in the development dataset suggests that DL models may be sensitive to the specific characteristics of the cameras used in the development dataset, and that they may not generalize well to other cameras. The lower performance in some ethnicities suggests that DL models may be biased toward the ethnicities in the development dataset, and that they may not generalize well to other ethnicities.

Our study has several strengths. We used a large dataset of diabetic patients from the UK Biobank, which is a prospective cohort study of over 500,000 participants from the United Kingdom (UK) aged between 40 and 69 at recruitment[14]. The UK Biobank dataset is representative of the UK population in terms of ethnicity, socioeconomic status, and health status[15], and thus, our study results may be generalizable to other populations with similar characteristics. We used a state-of-the-art DL model for DR detection[16], and we assessed its performance across ethnicities, cameras, and disease severities and types, which has not been done in prior studies. We also used a large sample size to ensure that our results were precise and that we had enough power to detect differences in performance across subgroups.

Our study has several limitations. First, we used only one DL model for DR detection, and our results may not generalize to other DL models. Second, we used only one camera for the development dataset, and our results may not generalize to other cameras. Third, we used only one ethnicity for the development dataset, and our results may not generalize to other ethnicities. Fourth, we used only one disease severity and type for the development dataset, and our results may not generalize to other disease severities and types. Fifth, we used only one dataset for our study, and our results may not generalize to other datasets. Sixth, we did not assess the impact of other factors on the performance of the DL model, such as age, sex, and comorbidities, and these factors may affect the performance of the DL model.

In conclusion, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

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1.1 Introduction

Computer science is the study of algorithms, including their efficiency, their limitations, and their mathematical properties. Related fields examine computer hardware, programming languages, and computer applications in various disciplines. This chapter introduces some of the mathematical notation and techniques used in the study of algorithms, and it reviews some of the data structures that are used in the algorithms presented in later chapters.

1.2 Mathematical Notation and Functions

Let $\mathbb{R}$ denote the set of real numbers, $\mathbb{Z}$ denote the set of integers, and $\mathbb{N}$ denote the set of natural numbers (integers greater than or equal to zero). If x is a real number, then $\lfloor x \rfloor$ denotes the largest integer less than or equal to x, and $\lceil x \rceil$ denotes the smallest integer greater than or equal to x. For example, $\lfloor 3.14 \rfloor = 3$ and $\lceil 3.14 \rceil = 4$. Given two integers a and b, $a \bmod b$ denotes the remainder when a is divided by b, and $a \equiv b \pmod{c}$ means that $a - b$ is a multiple of c. For example, $7 \bmod 3 = 1$ and $7 \equiv 1 \pmod{3}$.

1.2.1 Logarithms and Exponents

Unless otherwise specified, all logarithms are base 2. For $x > 0$, $\log(x)$ denotes the base-2 logarithm of x. For $x > 0$ and $y > 0$, $\log(xy) = \log(x) + \log(y)$. For $x > 0$ and $y > 0$, $\log(x/y) = \log(x) - \log(y)$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$.

1.2.2 Factorial and Binomial Coefficients

For $n \in \mathbb{N}$, $n!$ denotes the factorial of n, defined as $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$. For $n \in \mathbb{N}$, $n! = n \cdot (n-1)!$. For $n \in \mathbb{N}$, $n! = \prod_{i=1}^{n} i

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UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

Dr. Sarah Mitchell-Rodriguez, Ms. Jessica Tran, Mr. Daniel Okafor, and Ms. Priyanka Desai,
Plaintiffs,
v.
Dr. Robert Hayes, in his official capacity as Commissioner of the Massachusetts Department of Public Health,
Defendant.

Civil Action No. 1:20-cv-12345-RGS

COMPLAINT FOR DECLARATORY AND INJUNCTIVE RELIEF

Jury Demand Not Applicable

Plaintiffs Dr. Sarah Mitchell-Rodriguez, Ms. Jessica Tran, Mr. Daniel Okafor, and Ms. Priyanka Desai (collectively, "Plaintiffs") bring this complaint against Defendant Dr. Robert Hayes, in his official capacity as Commissioner of the Massachusetts Department of Public Health (the "Commissioner"), and allege as follows:

I. INTRODUCTION

1. Plaintiffs seek to compel the Commissioner to comply with the Medicaid Act's requirement that state agencies "promptly" process Medicaid applications. Despite this unambiguous mandate, the Commissioner has adopted a practice of holding applications for weeks or months while requesting documents that are not required by law, leaving low-income residents without coverage for medically necessary care. This practice violates the Medicaid Act, 42 U.S.C. § 1396a(a)(8), and the Fourteenth Amendment to the United States Constitution. Plaintiffs seek a declaration that the challenged practice is unlawful and an injunction requiring the Commissioner to process applications promptly and to stop requesting documents not required by the Medicaid Act.

II. JURISDICTION AND VENUE

2. This Court has jurisdiction under 28 U.S.C. § 1331 (federal question) and 28 U.S.C. § 1343(a)(3) (civil rights). Plaintiffs' claims for declaratory and injunctive relief are authorized by 28 U.S.C. § 2201 and Rule 65 of the Federal Rules of Civil Procedure.

3. Venue is proper in this District under 28 U.S.C. § 1391(b)(2) because a substantial part of the events or omissions giving rise to the claims occurred in this District, including the denial of Plaintiff Jessica Tran's application while she was a resident of Worcester County, Massachusetts.

III. PARTIES

4. Plaintiff Dr. Sarah Mitchell-Rodriguez is a resident of Worcester County, Massachusetts. She is a practicing family medicine physician and the mother of a 9-year-old daughter who is enrolled in MassHealth. She brings this action on her own behalf and on behalf of her minor daughter.

5. Plaintiff Ms. Jessica Tran is a resident of Worcester County, Massachusetts. She is a single parent of two children and a part-time community college student. She applied for MassHealth in March 2024 and was denied coverage for her children after the Department requested documents she had already provided.

6. Plaintiff Mr. Daniel Okafor is a resident of Hampden County, Massachusetts. He is a self-employed carpenter and the father of a 14-year-old son who has asthma and requires regular medication. His family's application has been pending since February 2024.

7. Plaintiff Ms. Priyanka Desai is a resident of Middlesex County, Massachusetts. She is a graduate student and a mother of one. Her application was denied because she could not produce a document that the Department requested but that is not required by the Medicaid Act.

8. Defendant Dr. Robert Hayes is the Commissioner of the Massachusetts Department of Public Health. He is sued in his official capacity. The Commissioner is responsible for the administration of MassHealth and the processing of applications for benefits.

IV. LEGAL BACKGROUND

9. The Medicaid Act requires that a state plan provide that "all individuals wishing to make application for medical assistance under the plan shall have opportunity to file such applications, and that such applications shall be promptly processed." 42 U.S.C. § 1396a(a)(8). The Act does not authorize a state agency to deny or delay a complete application because it requests documents not required by the statute.

10. The Fourteenth Amendment to the United States Constitution guarantees that no state shall deprive any person of life, liberty, or property without due process of law. A state agency's practice of withholding benefits for reasons not authorized by the governing statute violates the right to due process.

V. FACTS

11. Dr. Mitchell-Rodriguez's daughter qualified for MassHealth when her family's income fell below the statutory threshold in January 2023. The family renewed coverage in December 2023 and provided all documents required by the Department. The renewal was approved in 14 days.

12. Ms. Tran applied for MassHealth for herself and her two children in March 2024. She provided her pay stubs, her children's birth certificates, and her lease. The Department requested a sixth document, a letter from her employer, that is not required by the Medicaid Act. The application has been pending for 5 months. Ms. Tran's children have been uninsured since their previous coverage ended in February 2024.

13. Mr. Okafor's son has been prescribed a daily inhaler and a rescue inhaler. The family's income is approximately $31,200 per year, well below the MassHealth threshold. The application has been pending since February 2024. The family has been paying out of pocket for the inhaler, which costs $[illegible] per month.

14. Ms. Desai's application was denied because she could not produce a document the Department requested. She then filed a new application and provided the document. The second application was approved within 10 days. The delay caused her to miss two months of coverage.

15. The Department's practice of requesting documents not required by the Medicaid Act has caused the denial, delay, and loss of coverage for hundreds of applicants in the past two years, according to Department of Public Health data produced in discovery in a separate matter.

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Ethnobotanical study of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan

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Abstract: This study aimed to investigate the ethnobotanical uses of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan. Data were collected through questionnaires and interviews with local people. A total of 10 medicinal plants belonging to 10 families were recorded. The most frequently used plants were Calotropis procera and Tribulus terrestris. The results indicate that the local people rely on these plants for various ailments, including digestive disorders, skin diseases, and respiratory problems. The study highlights the importance of traditional knowledge in the region and suggests the need for further pharmacological and phytochemical investigations to validate the traditional uses and ensure sustainable utilization of these resources.

Keywords: ethnobotany, medicinal plants, Tharparkar, Sindh, Pakistan, traditional knowledge, Calotropis procera, Tribulus terrestris

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7.2 Cluster Representations

7.2.1 Centroids and Prototypes (Medoids)

Definition. Let $C_k = \{x_1, x_2, \ldots, x_{n_k}\}$ be a cluster with n_k points. The **centroid** (or **mean**) of C_k is the point

$$\mu_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_i$$

Example 7.2.1. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1)\}$. Then

$$\mu_1 = \frac{1}{3}\left((1,2) + (2,3) + (3,1)\right) = \frac{1}{3}(6,6) = (2,2)$$

Proposition 7.2.1. The centroid μ_k minimizes the sum of squared distances to the points in C_k:

$$\mu_k = \arg\min_{\mu \in \mathbb{R}^d} \sum_{i=1}^{n_k} \|x_i - \mu\|^2$$

Proof. Expand the objective:

$$\sum_{i=1}^{n_k} \|x_i - \mu\|^2 = \sum_{i=1}^{n_k} \|x_i\|^2 - 2\mu^T \sum_{i=1}^{n_k} x_i + n_k \|\mu\|^2$$

This is a quadratic in μ. Setting the gradient to zero:

$$\nabla_\mu \left[\sum_{i=1}^{n_k} \|x_i - \mu\|^2\right] = -2\sum_{i=1}^{n_k} x_i + 2n_k\mu = 0$$

$$\Rightarrow \mu = \frac{1}{n_k}\sum_{i=1}^{n_k} x_i$$

Remark. The centroid is not necessarily a data point. For example, the centroid of $\{(0,0), (1,0)\}$ is $(0.5, 0)$, which is not a data point.

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.2 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.3 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.4 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.5 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

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Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

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7.2.8 Medoids

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Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.9 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

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7.2.10 Medoids

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$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

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Lesson Twenty-one | Matrix Multiplication and Inverses

Matrix multiplication is the heart of linear algebra. It is how we combine two transformations into one, how we solve systems of equations, how we compute eigenvalues, how we train neural networks. Today we derive it from first principles and understand why it works the way it does.

Author: Prof. Elena V. Kozlova | Computational Mathematics Group

---

1. Motivation: Why Do We Need Matrix Multiplication?

Suppose we have two linear transformations $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^p$. Their composite $T_2 \circ T_1 : \mathbb{R}^n \to \mathbb{R}^p$ is also linear. If A represents T_1 and B represents T_2, we want a matrix C such that

$$(T_2 \circ T_1)(\mathbf{x}) = C\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

We want C to be the product BA.

---

2. Definition

Definition 2.1 (Matrix Multiplication). Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. The product $C = AB \in \mathbb{R}^{m \times p}$ is defined by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad 1 \le i \le m, \; 1 \le j \le p.$$

---

3. Properties

Theorem 3.1 (Properties of Matrix Multiplication). Assuming the sizes match, the following hold:

1. **Associativity:** $(AB)C = A(BC)$.
2. **Distributivity:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
3. $(AB)^T = B^T A^T$.
4. $\det(AB) = \det(A)\det(B)$.
5. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof. (1) Associativity follows from the fact that $(AB)C$ and $A(BC)$ both compute the (i,j) entry of the triple product $\sum_{k,l} a_{ik} b_{kl} c_{lj}$. (2) Distributivity follows from the distributivity of scalar multiplication. (3) $(AB)^T_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k b_{ki} a_{jk} = (B^T A^T)_{ij}$. (4) Taken for granted for now; we will prove it in Lesson Twenty-two. (5) $\operatorname{tr}(AB) = \sum_i \sum_k a_{ik} b_{ki} = \sum_k \sum_i b_{ki} a_{ik} = \operatorname{tr}(BA)$. □

Warning. Matrix multiplication is not commutative. In general $AB \neq BA$. Even when both products exist and have the same size, $AB = BA$ only if A and B commute.

---

4. Inverses

Definition 4.1 (Inverse). A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if there exists A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

Theorem 4.2 (Uniqueness). If A is invertible, A^{-1} is unique.

Proof. If B and C both satisfy $AB = BA = I$ and $AC = CA = I$, then $B = BI = B(AC) = (BA)C = IC = C$. □

Theorem 4.3 (Properties of Inverses).

1. $(AB)^{-1} = B^{-1}A^{-1}$.
2. $(A^T)^{-1} = (A^{-1})^T$.
3. $\det(A^{-1}) = 1/\det(A)$.

---

5. Exercises

Exercise 5.1. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ by hand.

Exercise 5.2. Show that $(AB)^{-1} = B^{-1}A^{-1}$ when A and B are invertible.

Exercise 5.3. Prove that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ without using the formula $c_{ij} = \sum_k a_{ik} b_{kj}$.

---

6. Closing Remarks

Matrix multiplication is the engine of linear algebra. Every theorem we will prove in the next few lessons depends on it. Lesson Twenty-two: determinants. Lesson Twenty-three: eigenvalues. Lesson Twenty-four: the spectral theorem. Lesson Twenty-five: applications to machine learning.

Lost Museum Direct Acquisition Log

Agent: Claude (Anthropic) via Claude Code, Opus 4.6
Date: 2026-03-14
Mission: Acquire 3D models for the Lost Museum collection
Source: Smithsonian 3D Collections (https://3d.si.edu/)
Method: Direct HTTP download via Smithsonian Open Access API (https://api.si.edu/openaccess/api/v1.0/search?q=type:3d_model&rows=10&api_key=DEMO_KEY)
Rights: CC0 only

Object	Rights	Size
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB

Total: 10 models, 0.0 MB

第一章 Flutter 基础概念和设计

1.1 Widget 是一切 UI 的基本单位

Flutter 的核心哲学是一切皆 Widget。

```
Widget ::= StatelessWidget | StatefulWidget | InheritedWidget | RenderObjectWidget
StatelessWidget ::= 不可变，build() 只依赖 Widget 自身和 BuildContext
StatefulWidget ::= 可变，通过 State 对象持有状态，State.build() 依赖 Widget 自身、BuildContext 和 State 内部状态
BuildContext ::= 指向 Element 树中对应节点的只读句柄
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.2 三棵树的关系

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{inflate}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject\ Tree}$$

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{diff}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{layout,\ paint}} \mathrm{Screen}$$

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.3 Widget 不可变性

Widget 一旦创建，其属性全部 `final`。`build()` 每次调用返回全新 Widget 实例，旧实例丢弃。Element 通过 `canUpdate(oldWidget, newWidget)` 判断能否复用：`runtimeType` 相同且 `key` 相同则复用，否则销毁重建。

1.4 Element 生命周期

```
Element ::= 挂载 (mount) → 活跃 (active) → 卸载 (unmount)
mount ::= 调用 Widget.createElement() → 子节点递归 mount → 插入 RenderObject 树
unmount ::= 递归 unmount 子节点 → 从 RenderObject 树移除 → 释放 State.dispose()
```

1.5 RenderObject 层

```
RenderObject ::= 布局 (layout) → 绘制 (paint) → 合成 (composite)
layout ::= 父节点传入 BoxConstraints → 子节点计算自己的 Size → 父节点决定子节点的 offset
paint ::= 生成 Layer 树 → 提交给 Skia/Impeller 渲染引擎
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.6 Key 的作用

```
Key ::= 唯一标识符，用于 Element 复用判断
LocalKey ::= ValueKey | ObjectKey | UniqueKey
GlobalKey ::= 全局唯一，跨 Widget 树移动时保持 State 不丢失
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.7 BuildContext 的本质

```
BuildContext ::= 指向 Element 树中对应节点的只读句柄
context.findAncestorWidgetOfExactType<T>() ::= 沿 Element 树向上遍历，返回类型匹配的祖先 Widget
context.findRenderObject() ::= 返回 Element 对应的 RenderObject
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

MR. TIGER MAGNIFICENT

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

In what distant deeps or skies

Burnt the fire of thine eyes?

On what wings dare he aspire?

What the hand dare seize the fire?

And what shoulder, and what art,

Could twist the sinews of thy heart?

And when thy heart began to beat,

What dread hand? And what dread feet?

What the hammer? What the chain,

In what furnace was thy brain?

What the anvil? What dare

Grasp thy deadly terrors clare?

When the stars threw down their spears

And water'd heaven with their tears,

Did he smile his work to see?

Did he who made the lamb make thee?

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

$$f \circ \mathcal{F}$$

$$\mathcal{F}$$

$$\mathcal{F}^{-1}$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x)$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)

Ans:

Ans:

Ans:

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1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur

1.1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur, Adipiscing

1.1.1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur, Adipiscing, Elit

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Q1.

Subject:Mathematics Chapter:Trigonometry Topic:Trigonometric Ratios and Identities

Question:If $\sin\theta + \cos\theta = \sqrt{2}$, then $\sin\theta - \cos\theta =$ Options:(1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 0 (4) 1

Answer:(3) 0

Solution:$(\sin\theta + \cos\theta)^2 = 2 \Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 2$

$\Rightarrow 1 + 2\sin\theta\cos\theta = 2 \Rightarrow 2\sin\theta\cos\theta = 1$

$(\sin\theta - \cos\theta)^2 = \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 1 - 1 = 0$

$\Rightarrow \sin\theta - \cos\theta = 0$

Q2.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, then $\tan\theta =$

Options:(1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{5}{3}$ (4) $\frac{5}{4}$

Answer:(1) $\frac{3}{4}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$

Q3.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cos\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(2) $\frac{\sqrt{3}}{2}$

Solution:$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{1}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

Q4.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q5.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q6.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Q7.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\operatorname{cosec}\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 2

Answer:(4) 2

Solution:$\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{1}{1/2} = 2$

Q8.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q9.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q10.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Disclaimer:This PDF is generated from the Question Bank. Please verify critical details before use.

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Abstract

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MR. TIGGY-WINKLE'S

Mr. Tiggy-Winkle's Wash-Tub (Frontispiece)

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub (Frontispiece) 1

Mrs. Tiggy-Winkle's Cottage 2

Mrs. Tiggy-Winkle Ironing 3

Mrs. Tiggy-Winkle Hanging Up Her Lace 4

Mrs. Tiggy-Winkle's Kitchen 5

Mrs. Tiggy-Winkle's Cottage Door 6

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub

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MR-TCOT-MATHEMATICS

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Example 1.1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Example 1.130. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.131. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.132. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.133. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.134. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.135. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.136. Let $A = \begin{

Gaggenau Service Manual

Gaggenau Service Manual

Gaggenau Service Manual Gaggenau Service Manual Gaggenau Service Manual

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Author: Dr. Lucas Thompson
Department: Chemical Engineering
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Abstract: This document provides a comprehensive analysis of the Antoine equation, a fundamental tool in chemical engineering for calculating vapor pressure. We begin with the derivation of the equation from the Clausius-Clapeyron relation, discuss its parameters and their physical significance, and provide a step-by-step guide for fitting the equation to experimental data using nonlinear regression. The document includes a critical discussion of the equation's limitations, including its accuracy over different temperature ranges and its extrapolation behavior, and concludes with a review of alternative models such as the Wagner equation and the Clarke-Glew equation. This work is intended for students and practitioners who need a robust understanding of vapor pressure calculations.

Keywords: Antoine equation, vapor pressure, thermodynamics, parameter fitting, nonlinear regression, chemical engineering, Clausius-Clapeyron, Wagner equation, Clarke-Glew equation

1.1 Introduction

$$e = V \cdot I$$

1.1.1

Subsection 1.1.1
Subsubsection 1.1.1.1
Paragraph 1.1.1.1.1

1.1.2

Subsection 1.1.2
Subsubsection 1.1.2.1
Paragraph 1.1.2.1.1

1.1.3

Subsection 1.1.3
Subsubsection 1.1.3.1
Paragraph 1.1.3.1.1

1.1.4

Subsection 1.1.4
Subsubsection 1.1.4.1
Paragraph 1.1.4.1.1

1.2 Section 1.2

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

1.2.1

Subsection 1.2.1
Subsubsection 1.2.1.1
Paragraph 1.2.1.1.1

1.2.2

Subsection 1.2.2
Subsubsection 1.2.2.1
Paragraph 1.2.2.1.1

1.2.3

Subsection 1.2.3
Subsubsection 1.2.3.1
Paragraph 1.2.3.1.1

1.2.4

Subsection 1.2.4
Subsubsection 1.2.4.1
Paragraph 1.2.4.1.1

1.3 Section 1.3

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \\ \frac{\partial \rho u_x}{\partial x} &= \\ \frac{\partial \rho u_y}{\partial y} &= \\ \frac{\partial \rho u_z}{\partial z} &= \end{aligned}$$

1.3.1

Subsection 1.3.1
Subsubsection 1.3.1.1
Paragraph 1.3.1.1.1

1.3.2

Subsection 1.3.2
Subsubsection 1.3.2.1
Paragraph 1.3.2.1.1

1.3.3

Subsection 1.3.3
Subsubsection 1.3.3.1
Paragraph 1.3.3.1.1

1.3.4

Subsection 1.3.4
Subsubsection 1.3.4.1
Paragraph 1.3.4.1.1

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1.1 Sets and Subsets

Let A and B be sets. If every element of A is also an element of B, we say that A is a subset of B, and we write $A \subseteq B$.

1.2 Operations on Sets

Let A and B be sets.

The union of A and B, denoted $A \cup B$, is the set of all elements that are in A or in B (or in both).

The intersection of A and B, denoted $A \cap B$, is the set of all elements that are in both A and B.

The difference of A and B, denoted $A \setminus B$, is the set of all elements that are in A but not in B.

The complement of A, denoted A^c, is the set of all elements that are not in A.

1.3 Cartesian Product

Let A and B be sets.

The Cartesian product of A and B, denoted $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

1.4 Relations

Let A and B be sets.

A relation from A to B is a subset $R \subseteq A \times B$.

If $(a, b) \in R$, we write aRb.

1.5 Equivalence Relations

Let A be a set.

A relation R on A is reflexive if aRa for all $a \in A$.

R is symmetric if $aRb \Rightarrow bRa$.

R is transitive if aRb and $bRc \Rightarrow aRc$.

R is an equivalence relation if it is reflexive, symmetric, and transitive.

1.6 Equivalence Classes

Let R be an equivalence relation on A.

The equivalence class of a, denoted $[a]$, is the set $\{x \in A : xRa\}$.

1.7 Partitions

Let A be a set.

A partition of A is a family $\{A_i\}_{i \in I}$ of subsets of A such that:

- $A_i \neq \emptyset$ for all $i \in I$,
- $A_i \cap A_j = \emptyset$ for $i \neq j$,
- $\bigcup_{i \in I} A_i = A$.

1.8 Equivalence Relations and Partitions

Let A be a set.

Every equivalence relation on A induces a partition of A into equivalence classes.

Conversely, every partition of A defines an equivalence relation on A by $aRb \iff a, b$ belong to the same part.

1.9 Functions

Let A and B be sets.

A function $f : A \to B$ is a relation $f \subseteq A \times B$ such that for every $a \in A$, there exists exactly one $b \in B$ such that $(a, b) \in f$.

We write $f(a) = b$.

1.10 Injectivity, Surjectivity, Bijectivity

Let $f : A \to B$.

f is injective if $f(a) = f(b) \Rightarrow a = b$.

f is surjective if $\operatorname{Im}(f) = B$.

f is bijective if it is injective and surjective.

1.11 Composition of Functions

Let $f : A \to B$ and $g : B \to C$.

The composition $g \circ f : A \to C$ is defined by $(g \circ f)(a) = g(f(a))$.

1.12 Inverse Functions

Let $f : A \to B$.

f has an inverse $f^{-1} : B \to A$ if and only if f is bijective.

1.13 Cardinality

Let A and B be sets.

$|A| = |B|$ if there exists a bijection $f : A \to B$.

$|A| \leq |B|$ if there exists an injective function $f : A \to B$.

$|A| \geq |B|$ if there exists a surjective function $f : A \to B$.

1.14 Finite Sets

Let A be a set.

A is finite if $|A| = |\{1, 2, \ldots, n\}|$ for some $n \in \mathbb{N}$.

1.15 Countable Sets

Let A be a set.

A is countable if $|A| \leq |\mathbb{N}|$.

A is countable if and only if there exists a surjective function $f : \mathbb{N} \to A$.

1.16 Countable Unions

Let $\{A_n\}_{n \in \mathbb{N}}$ be a family of countable sets.

$\bigcup_{n \in \mathbb{N}} A_n$ is countable.

1.17 Cartesian Products of Countable Sets

Let A and B be countable sets.

$A \times B$ is countable.

1.18 Finite Cartesian Products

Let $A_1, A_2, \ldots, A_n$ be countable sets.

$A_1 \times A_2 \times \cdots \times A_n$ is countable.

1.19 Infinite Cartesian Products

Let A be a countable set.

$A^{\mathbb{N}}$ is uncountable.

1.20 Power Sets

Let A be a set.

$\mathcal{P}(A)$ is uncountable.

1.21 Cantor's Theorem

Let A be a set.

$|A| < |\mathcal{P}(A)|$.

1.22 Cantor's Theorem Applied to $\mathbb{N}$

$|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$.

1.23 Cardinality of $\mathcal{P}(\mathbb{N})$

$|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$.

1.24 Cardinality of $\mathbb{R}$

$|\mathbb{R}| > |\mathbb{N}|$.

1.25 Cardinality of $\mathbb{Q}$

$|\mathbb{Q}| = |\mathbb{N}|$.

1.26 Cardinality of $\mathbb{Z}$

$|\mathbb{Z}| = |\mathbb{N}|$.

1.27 Cardinality of $\mathbb{C}$

$|\mathbb{C}| = |\mathbb{R}|$.

1.28 Cardinality of $\mathbb{R}^n$

$|\mathbb{R}^n| = |\mathbb{R}|$.

1.29 Cardinality of $\mathbb{R}^{\mathbb{N}}$

$|\mathbb{R}^{\mathbb{N}}| > |\mathbb{R}|$.

1.30 Cardinality of $\mathbb{R}^{\mathbb{R}}$

$|\mathbb{R}^{\mathbb{R}}| > |\mathbb{R}^{\mathbb{N}}|$.

1.31 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}| > |\mathbb{R}^{\mathbb{R}}|$.

1.32 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}|$.

1.33 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}|$.

1.34 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}|$.

1.35 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}|$.

1.36 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}|$.

1.37 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}|$.

1.38 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}|$.

1.39 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}}|$.

1.40 Cardinality of $\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}}}$

$|\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}}}| > |\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}^{\mathbb{R}}}}}}}}}}}}|$.

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Online First: 27 November 2017

Chapter 1 Introduction

1.1 Overview

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

1.2 Scope

At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident, similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

```
SELECT COUNT(DISTINCT c.customer_id) FROM customers c
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.1}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.2}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.3}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.4}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.5}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.6}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.7}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.8}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.9}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.10}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

Sample Distribution

Sample Distribution of Sample Means, $\bar{x}$ - Central Limit Theorem

Sample Distribution of Sample Proportions, $\hat{p}$ - Central Limit Theorem

```
NO: 7147-MAURITANIA-17
```

Ms Samira Zaid Al-Khawaja (gender expert)

```
Dear
```

Dear Lucie Tunzelmann (Ms Alexandra Gilmour) and
Dear Ms Samira Zaid Al-Khawaja (gender expert) Dear Ms Alexandra Gilmour, Dear Lucie Tunzelmann (Ms Alexandra Gilmour) Dear Ms Samira Zaid Al-Khawaja (gender expert) Dear Ms

MR-TIGER-MAGNETOSPHERIC-7

Magnetosphere and Ionosphere coupling model.

Type

Model

Model type:Magnetosphere and Ionosphere coupling model

Model subtype:Magnetosphere and Ionosphere coupling model

Model name:Magnetosphere and Ionosphere coupling model

Model description:Magnetosphere and Ionosphere coupling model,Magnetosphere and Ionosphere coupling model. Single fluids MHD equations (Ideal MHD) is used for the magnetosphere and a 2-D Ionosphere model.

Model Time Domain:Transient,Steady-state,Frequency

Model Dimension:3D

Equations

$$\nabla \times \vec{B} = \vec{J}$$

Variables

Independent

Name:Magnetic field

Description:

Dependent

Name:Current density

Description:

Constants

Name:

Description:

Name:

Description:

Name:

Description:

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Online First Publication: July 17, 2019

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Publication Language: English

Publication Country: United States of America

Error: Unrecognized function or variable 'gamma'.

>>

Error: Unrecognized function or variable 'gamma'.

>>

Error using ode45 (line 73)
Integration tolerance 1.0e-06 cannot be met at t = 1.00000e-05. Try increasing the tolerances.

Error in ode45_ivp_solver (line 5)
[tout, yout] = ode45(ode, tspan, y0, options);

>>

$$-\int \frac{1}{x^2}\,dx$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$\int \frac{1}{x^2}\,dx + C_1$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

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Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

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Warning: Unable to find explicit form.
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> In sym/int (line 46)

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Warning: Unable to find explicit form.
> In sym/int (line 46)

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-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

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-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$-\int \frac{1}{x^2}\,dx - \int \frac{1}{x^2}\,dx + C_1$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

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ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$\int \frac{1}{x^2}\,dx - \int \frac{1}{x^2}\,dx + C_1 \qquad 1.0.0.$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

Topic:

Question:

Prompt: Create a flashcard for a Physics exam question.
Output: Output ONE question in LaTeX format. Use ONLY the following LaTeX commands: \begin{enumerate}, \item, \textbf{...}, \textit{...}, \begin{equation}, \begin{aligned}, \begin{table}, \begin{tabular}, \begin{tikzcd}, \begin{tikzpicture}, \begin{axis}, \addplot, \begin{figure}, \begin{minipage}, \begin{center}, \begin{enumerate}, \begin{itemize}, \begin{description}, \begin{proof}, \begin{lemma}, \begin{theorem}, \begin{definition}, \begin{example}, \begin{remark}, \begin{corollary}, \begin{proposition}, \begin{claim}, \begin{case}, \begin{subproof}, \begin{subcase}, \begin{subproof}, \begin{subcase}, \begin{subproof}, \begin{subcase

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$$f(x)$$

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$$f(x)$$

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Mr-Tiger-Magnetoscopy-Magnetoscopy-Magnetoscopy

Mr-Tiger-Magnetoscopy-Magnetoscopy-Magnetoscopy

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MR-TIGER-MAGNUM-MATCH
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Mr-Tiger-Magnum-Match-Strategy-agent
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Mr-Tiger-Magnum-Match-Strategy-agent-Mr-Tiger-Magnum-Match-Strategy-agent-Mr-Tiger-Magnum-Match-Strategy-agent-
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MR-TIGER-MAGNUM-OPUS-7

Mr. Tiger Magnum Opus 7 (Tiger Magnum Opus)

Type: Tiger

Rank: Grand Grandmaster of Universal Knowledge, Master of Mathematics, Logic, Linguistics, Philosophy, Computer Science, Cryptography, Sigint, Crypto, Ai, Langsec

Tagline: Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

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$$\neg\Box\Box\bot$$

Tiger

Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

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1.1 Cartesian coordinates in the plane

Let O be a fixed point and $\mathbf{i}, \mathbf{j}$ two unit vectors perpendicular to one another. Every point P in the plane is identified with the ordered pair (x, y) of real numbers for which

$$\overrightarrow{OP} = x\,\mathbf{i} + y\,\mathbf{j}.$$

The number x is the abscissa and y the ordinate of P. Conversely, every ordered pair $(x, y) \in \mathbb{R}^2$ determines a point P by the same formula.

Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$. The distance between them is

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \tag{1.1.1}$$

Equation (1.1.1) is the Euclidean metric on $\mathbb{R}^2$. It satisfies the triangle inequality $d(P_1, P_3) \le d(P_1, P_2) + d(P_2, P_3)$, with equality if and only if P_2 lies on the segment $[P_1, P_3]$.

1.2 Lines and their equations

A line in the plane is the set of points satisfying a linear equation $ax + by = c$, $(a, b) \neq (0, 0)$. Two lines are parallel if their normal vectors are proportional, perpendicular if the dot product of their normal vectors vanishes.

1.3 Circles

A circle of radius $r > 0$ centred at (h, k) is the locus $(x - h)^2 + (y - k)^2 = r^2$. Completing the square in $x^2 + y^2 + Ax + By + C = 0$ decides whether it is a circle, a point, or the empty set.

1.4 Conic sections

The ellipse, parabola and hyperbola are the curves of second degree in two variables. Their canonical forms are $x^2/a^2 + y^2/b^2 = 1$, $y^2 = 4px$, and $x^2/a^2 - y^2/b^2 = 1$. Each admits a focal and a directrix characterisation, which we prove in Chapter 4.

MC-TCAP-MAPMODULE

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<!-- Generated by SVGO v3.0.2 (https://svgo.dev) -->
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  xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0 0 1000 1000"
  preserveAspectRatio="xMidYMid meet"
  data-name="Layer 1" />
```

Ultimate Bacon Bits Nutrition Facts - Calories, Carbs, Fat, Protein, Fiber, Cholesterol, Iron, Sodium & More | Fatsecret

Ultimate Bacon Bits

Serving Size: 1 tbsp

Calories 35 kcal | **Fat** 2.5 g | **Carbs** 0.5 g | **Protein** 3 g

Vitamin A 0 IU, Vitamin C 0 mg, Calcium 0 mg, Iron 0.3 mg, Potassium 0 mg, Fiber 0 g, Sugar 0 g, Sodium 150 mg, Cholesterol 5 mg, Saturated Fat 0.5 g, Trans Fat 0 g, Monounsaturated Fat 1 g, Polyunsaturated Fat 0.5 g, Omega-3 0 g, Omega-6 0 g, Vitamin D 0 IU, Vitamin E 0 mg, Vitamin K 0 mcg, Vitamin B1 0 mg, Vitamin B2 0 mg, Vitamin B3 0 mg, Vitamin B5 0 mg, Vitamin B6 0 mg, Vitamin B12 0 mcg, Folate 0 mcg, Phosphorus 0 mg, Magnesium 0 mg, Zinc 0 mg, Selenium 0 mcg, Copper 0 mg, Manganese 0 mg, Iodine 0 mcg, Caffeine 0 mg, Water 0 g, Alcohol 0 g, Choline 0 mg, Betaine 0 mg, Sugar Alcohols 0 g, Net Carbs 0.5 g, Starch 0 g, Monosodium Glutamate 0 mg, Phenylalanine 0 mg, Tryptophan 0 mg, Tyrosine 0 mg, Leucine 0 mg, Isoleucine 0 mg, Lysine 0 mg, Methionine 0 mg, Threonine 0 mg, Valine 0 mg, Histidine 0 mg, Arginine 0 mg, Cystine 0 mg, Glycine 0 mg, Proline 0 mg, Serine 0 mg, Alanine 0 mg, Aspartic Acid 0 mg, Glutamic Acid

제1장 일반사항

1.1 일반사항

1.1.1 일반사항

1.1.1.1 일반사항

1.1.1.1.1 일반사항

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1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

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Abstract

Mr. Scott Cunningham, in Statistical Language Modeling, Mr. Scott Cunningham's Statistical Language Modeling, Chapter 3

MR. FLOOR MAGNIFICENCE?

MR. FLOOR MAGNIFICENCE?

Mr. Floor Magnificence Floorington Magnificence

Mr. Floor Floorington Floorington Floorington Floorington

Mr. Floor Floorington Floorington Floorington Floorington

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet href="/assets/rss.xsl" type="text/xsl"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sitemap="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xmlns:image="http://www.google.com/schemas/sitemap-image/1.1" xmlns:media="http://search.yahoo.com/mrss/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:wp="http://wordpress.org/export/1.2/" xmlns:content="http://purl.org/rss/1.0/modules/content
```

```
# -*- coding: utf-8 -*-
# ---------------------------------------------------------------------
# Gmsh SDK
# ---------------------------------------------------------------------
# This file is generated automatically.
# ---------------------------------------------------------------------
# Version: 4.13.1
# ---------------------------------------------------------------------
# Generated on: Wed Jan 15 17:11:11 2025
# ---------------------------------------------------------------------
```

Gmsh

Gmsh is a free 3D finite element grid generator with a built-in CAD engine and post-processing facilities. Gmsh aims at being the fastest and most light mesh generator, easy to install and easy to use. Gmsh accepts parameters from command line, text files and interactively from its graphical user interface. Gmsh uses its own ASCII file format by default, but can also read and write many other mesh formats.

Gmsh SDK

The Gmsh SDK provides a stable API to Gmsh's mesh generation and post-processing facilities. Gmsh's SDK can be used from the C++, C, Python, Julia and Lua APIs. The Python API is built on top of the C API, which is itself built on top of the C++ API. The Python API uses ctypes to call the C API functions, and does not require any compilation. The Julia API is built on top of the Lua API, which is itself built on top of the C API. The Lua API is built on top of the C++ API.

Python API

The Python API is distributed as a set of Python modules. The main module is `gmsh`, which defines the functions and classes of the Gmsh SDK. The Python API is built on top of the C API, which is itself built on top of the C++ API. The Python API uses ctypes to call the C API functions, and does not require any compilation.

```
import os, sys, platform, ctypes, ctypes.pointer, ctypes.ctypes_lib, ctypes.util, ctypes.wintypes, ctypes.c_void_p, ctypes.c_char_p, ctypes.c_wchar_p, ctypes.c_int, ctypes.c_uint, ctypes.c_long, ctypes.c_ulong, ctypes.c_longlong, ctypes.c_ulonglong, ctypes.c_double, ctypes.c_float, ctypes.c_bool, ctypes.c_char, ctypes.c_wchar, ctypes.c_byte, ctypes.c_ubyte, ctypes.c_short, ctypes.c_ushort, ctypes.c_size_t, ctypes.c_ssize_t, ctypes.c_ptrdiff_t, ctypes.c_intptr_t, ctypes.c_uintptr_t, ctypes.c_int8, ctypes.c_uint8, ctypes.c_int16, ctypes.c_uint16, ctypes.c_int32, ctypes.c_uint32, ctypes.c_int64, ctypes.c_uint64, ctypes.c_complex, ctypes.c_dcomplex, ctypes.c_fcomplex, ctypes.c_ldouble, ctypes.c_lcomplex
```

```
MAT-73207 MACHINE LEARNING II

Jan Snellman, Jesper Carlsson, Joakim Lundborg, Andreas Lundell

Imports
from sympy import *, Matrix, Rational, sqrt, ceiling, floor
from sympy.plotting import plot, plot_parametric, plot_implicit, plot3d, plot3d_parametric_line, plot_parametric_region, Image, Poly, Integer, Rational, Ex, exp, factorial, ln, sin, cos, tan, pi, E, I, oo, gamma, zeta
from sympy.stats import *, Normal, Binomial, Poisson, Exponential, Uniform, Beta, Gamma, Chi, Chi2, Cauchy, Laplace, Logistic, LogNormal, Pareto, Weibull, Gumbel, Maxwell, Rayleigh, Dagum, Erlang, Kumaraswamy, Triangular, Arcsin, Cosine, Exponential, Logarithmic, Sine, Uniform, Density, E, P, variance, density, cdf, sample, pdf, cdf, sample, E, P, variance, density
from sympy.stats.rv import *, RandomMatrix, RandomSymbol, random_symbols, pspace, given, sample_iter, expectation, where, quantile
from sympy.stats.crv_types import *, ContinuousDistributionHandmade

In [1]:
```

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

```
In [2]:
Sum(k**2, (k, 1, n)).doit().factor()
```

$$\frac{n\left(n + 1\right)\left(2 n + 1\right)}{6}$$

```
In [3]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10)
```

$$385$$

```
In [4]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100)
```

$$\frac{338350}{3}$$

```
In [5]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000)
```

$$\frac{334167500}{3}$$

```
In [6]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000)
```

$$\frac{333383350000}{3}$$

```
In [7]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000)
```

$$\frac{333033350000000}{3}$$

```
In [8]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000)
```

$$\frac{333333500000000000}{3}$$

```
In [9]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000)
```

$$\frac{333333350000000000000}{3}$$

```
In [10]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000)
```

$$\frac{333333333500000000000000}{3}$$

```
In [11]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000)
```

$$\frac{333333333333500000000000000}{3}$$

```
In [12]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000)
```

$$\frac{333333333333350000000000000000}{3}$$

```
In [13]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000)
```

$$\frac{333333333333335000000000000000000}{3}$$

```
In [14]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000)
```

$$\frac{333333333333333500000000000000000000}{3}$$

```
In [15]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000)
```

$$\frac{333333333333333350000000000000000000000}{3}$$

```
In [16]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000)
```

$$\frac{333333333333333335000000000000000000000000}{3}$$

```
In [17]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000)
```

$$\frac{333333333333333333500000000000000000000000000}{3}$$

```
In [18]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000)
```

$$\frac{333333333333333333350000000000000000000000000000}{3}$$

```
In [19]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000)
```

$$\frac{333333333333333333335000000000000000000000000000000}{3}$$

```
In [20]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000)
```

$$\frac{333333333333333333333500000000000000000000000000000000}{3}$$

```
In [21]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000)
```

$$\frac{333333333333333333333350000000000000000000000000000000000}{3}$$

```
In [22]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000)
```

$$\frac{333333333333333333333335000000000000000000000000000000000000}{3}$$

```
In [23]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000)
```

$$\frac{333333333333333333333333500000000000000000000000000000000000000}{3}$$

```
In [24]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000)
```

$$\frac{333333333333333333333333350000000000000000000000000000000000000000}{3}$$

```
In [25]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000)
```

$$\frac{333333333333333333333333335000000000000000000000000000000000000000000}{3}$$

```
In [26]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000)
```

$$\frac{333333333333333333333333333500000000000000000000000000000000000000000000}{3}$$

```
In [27]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000)
```

$$\frac{333333333333333333333333333350000000000000000000000000000000000000000000000}{3}$$

```
In [28]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000)
```

$$\frac{333333333333333333333333333335000000000000000000000000000000000000000000000000}{3}$$

```
In [29]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000)
```

$$\frac{333333333333333333333333333333500000000000000000000000000000000000000000000000000}{3}$$

```
In [30]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000)
```

$$\frac{333333333333333333333333333333350000000000000000000000000000000000000000000000000000}{3}$$

```
In [31]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000)
```

$$\frac{333333333333333333333333333333335000000000000000000000000000000000000000000000000000000}{3}$$

```
In [32]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000}{3}$$

```
In [33]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [34]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [35]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [36]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [37]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [38]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [39]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [40]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [41]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [42]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [43]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [44]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [45]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [46]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [47]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [48]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [49]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [50]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [51]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [52]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [53]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [54]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [55]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [56]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [57]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [58]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [59]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [60]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [61]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [62]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [63]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [64]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [65]:
Sum(k**2, (k, 1, n)).doit().factor
```

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Online First: 27 November 2017

Date: 2025-12-31 00:00:00 +0000
Source: https://www.ecfr.gov/
Title: 10 Energy
Part: PART 1021—NATIONAL ENVIRONMENTAL POLICY ACT IMPLEMENTATIONS
Section: 1021.410 Categorical exclusions.
Subsection: (b)(1)(i)(A)(1)(i)(a)(1)(i)(A)(1)(i)(a)(1)(i)(A)(1)(i)(a

```
<?xml version="1.0" encoding="UTF-8"?>

<?xml-stylesheet href="/sitemap.xsl"?>

<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9"
xmlns:news="http://www.google.com/schemas/sitemap-news/0.9"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.sitemaps.org/schemas/sitemap/0.9 http://www.sitemaps.org/schemas/sitemap/0.9/sitemap.xsd">
<sitemap>
<loc>
https://
</loc>
<lastmod>
2025-08-22
</lastmod>
</sitemap>
<sitemap>
<loc>
https://
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</lastmod>
</sitemap>
<sitemap>
<loc>
https://
</loc>
<lastmod>
2025-08-22
</lastmod>
</sitemap>
</urlset>
```

```
{"id":"0000-00-00T00:00:00.000Z","_id":"
{"_id":"
{"_id":"
{"
```

```
<?xml version="1.0" encoding="UTF-8"?>
<rss version="2.0">
<channel>
<title>Latest Updates</title>
<item>
<guid isaccepted="false">http://dl.dbpedia.org/data/Turnin_Turnin_Dance.ogg</guid>
<enclosure_
length="
type="
</item>
<item>
<guid isaccepted="false">http://dl.dbpedia.org/data/Turnin_Turnin_Dance.ogg</guid>
<enclosure_
length="
type="
</item>
<item_
<guid_isaccepted="false">http://dl.dbpedia.org/data/Turnin_Turnin_Dance.ogg</guid_
<enclosure_
length="
type="
```

```
{"id":"0000-00-00Bqxq1-Z0Ldqq-Bqxq-1",
"ts":"0000-00-00T00:00:00.000Z",
"kind":"Bqxq",
"name":"Bqxq",
"pubkey":"0000000000000000000000000000000000000000000000000000000000000000",
"tags":[["d","Bqxq"],["v","1"],["k","0"],["c","Bqxq"],["p","0000000000000000000000000000000000000000000000000000000000000000"],["r","wss://relay.damus.io"],["r","wss://nos.lol"],["r","wss://relay.nostr.band"],["r","wss://relay.primal.net"],["r","wss://relay.snort.social"],["r","wss://nostr.wine"],["r","wss://relay.nostrplebs.com"],["r","wss://nostr.mom"],["r","wss://nostr.bitcoiner.social"],["r","wss://nostr.oxtr.dev"],["r","wss://nostr.bitcoiner.social"],["r","wss://nostr.oxtr.dev"],["r","wss://nostr.mom"],["r","wss://nostr.wine"],["r","wss://relay.nostrplebs.com"],["r","wss://relay.snort.social"],["r","wss://relay.primal.net"],["r","wss://relay.nostr.band"],["r","wss://nos.lol"],["r","wss://relay.damus.io"],["r","wss://relay.nostr.bg
```

```
BEGIN:VCARD
VERSION:3.0
N:;Ivan Ivanovich;;;
FN:Ivan Ivanovich
ORG:;Ivan Ivanovich Corporation Service Center Storage
TITLE:
TEL;TYPE=CELL:
TEL;TYPE=WORK:
TEL;TYPE=HOME:
EMAIL;TYPE=INTERNET:
ADR;TYPE=WORK;LABEL="Ivan Ivanovich
Ivan Ivanovich
Ivan Ivanovich Corporation Service Center Storage
Ivan Ivanovich
Ivan Ivanovich
Ivan Ivanovich
```

Generated on 2026-03-15 14:00:00 UTC
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§ 141.62 — Maximum contaminant levels (MCLs) for inorganic chemicals

Subpart G — National Primary Drinking Water Regulations

Part 141 — National Primary Drinking Water Regulations

Subchapter D — Public Water Systems

Chapter I — Environmental Protection Agency

Title 40 — Protection of Environment

(a) Maximum contaminant levels (MCLs) for inorganic chemicals. The following MCLs for inorganic chemicals, specified in milligrams per liter (mg/L), are the maximum permissible levels in water delivered to any user of a public water system. Compliance with these MCLs is determined by the analytical results obtained at the point of entry to the distribution system, as specified in subsequent sections of this part.

(1) Antimony — 0.006 mg/L.

(2) Arsenic — 0.010 mg/L. Effective date: January 23, 2006. Compliance with this MCL is required no later than January 23, 2006 for all public water systems.

(3) Asbestos — 7 million fibers per liter (MFL) (fibers longer than 10 micrometers).

(4) Barium — 2.0 mg/L.

(5) Beryllium — 0.004 mg/L.

(6) Cadmium — 0.005 mg/L.

(7) Chromium (total) — 0.100 mg/L.

(8) Cyanide (as free cyanide) — 0.200 mg/L.

(9) Fluoride — 4.0 mg/L. Secondary maximum contaminant level (SMCL): 2.0 mg/L.

(10) Mercury (inorganic) — 0.002 mg/L.

(11) Nitrate (as nitrogen) — 10.0 mg/L.

(12) Nitrite (as nitrogen) — 1.0 mg/L.

(13) Total nitrate and nitrite (as nitrogen) — 10.0 mg/L.

(14) Selenium — 0.050 mg/L.

(15) Thallium — 0.002 mg/L.

(b) Compliance. Compliance with the MCLs for inorganic chemicals is determined based on a running annual average at each sampling point. If the running annual average at any sampling point exceeds the MCL, the system is in violation. If any one sample would cause the running annual average to exceed the MCL, the system is in violation immediately.

(c) Reporting. Systems must report the results of inorganic chemical monitoring to the State within 10 days after the end of the month in which the results are received.

1.1 Motivation

Ever wondered about the mechanics behind

Ever wondered about the mechanics behind

Ever wondered about the mechanics behind

ISSN 1337-7027
Year IX
Number 4
http://www.kvasy.upol.sk
pp. 1 – 4

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 System information as of Sun Jun 15 07:17:01 UTC 2025

  System load:  0.0               Processes:             104
  Usage of /:   18.4% of 7.57GB   Users logged in:       0
  Memory usage: 24%               IPv4 address for eth0: 172.31.17.0
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabled ESM Apps service at https://ubuntu.com/esm

Last login: Sun Jun 15 07:17:01 2025 from 13.126.210.16
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/
total 28
drwxr-x--- 3 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-xr-x 3 root   root   4096 Jun 15 07:17 ..
-rw-r--r-- 1 ubuntu ubuntu  220 Jun 15 07:17 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Jun 15 07:17 .bashrc
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .cache
-rw-r--r-- 1 ubuntu ubuntu  807 Jun 15 07:17 .profile
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/.ssh/
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-x--- 3 ubuntu ubuntu 
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Subject: Cosmic Ray Shower Simulation

Author: Dr. Ashish Kumar Jha, Assistant Professor, Department of Physics, Government Degree College Kathua, Jammu and Kashmir, India

Abstract: This study presents a Cosmic Ray Shower Simulation, which models the interaction of high-energy cosmic rays with Earth's atmosphere. The simulation uses a Monte Carlo approach to model the random nature of particle interactions and decays, providing a probabilistic representation of the shower development. The simulation tracks the number of particles in the shower as it propagates through the atmosphere, and it calculates the shower's maximum size and the depth at which it occurs. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Keywords:
Cosmic Ray Shower Simulation, Monte Carlo Simulation, Particle Interactions, Atmospheric Physics, Cosmic Ray Properties, Shower Maximum, Shower Depth, Probabilistic Modeling, Theoretical Predictions, Experimental Data, Accuracy, Reliability, Implications, Applications, Medical Physics, Radiation Protection

Introduction:
Cosmic rays are high-energy particles that originate from outer space and interact with the Earth's atmosphere, producing a cascade of secondary particles known as a cosmic ray shower. The study of cosmic ray showers is important for understanding the properties of cosmic rays and the Earth's atmosphere, and it has applications in fields such as medical physics and radiation protection. The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Methodology:
The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The simulation tracks the number of particles in the shower as it propagates through the atmosphere, and it calculates the shower's maximum size and the depth at which it occurs. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Results:
The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Conclusion:
The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

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Is the Chinese Communist Party an ideology or interest-led? Evidence from Chinese Spatial Data

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Abstract

Is the Chinese Communist Party (CCP) an ideology or interest-led? We exploit the Chinese case to test the "ideology versus interest" hypothesis. We find that the CCP's ideology is more important than its interest in determining the allocation of resources. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the non-state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector.

JEL Codes: D72; D73; D74; H11; H12; H13; H14; H15; H16; H17; H18; H19; H20; H21; H22; H23; H24; H25; H26; H27; H28; H29; H30; H31; H32; H33; H34; H35; H36; H37; H38; H39; H40; H41; H42; H43; H44; H45; H46; H47; H48; H49; H50;

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Received: 12-04-2024, Revised: 25-04-2024, Accepted: 10-05-2024
Abstract: This study examines the impact of Ayushman Bharat on healthcare access and financial protection among rural households in Uttarakhand, India. Using a sample of 1,200 households across three districts, the study finds that the scheme has increased hospitalization rates by 18% and reduced out-of-pocket expenditure by an average of ₹4,200 per year. However, awareness remains low among the most vulnerable households, and implementation gaps in empanelled hospitals limit the scheme's reach in remote areas. The study concludes that while the scheme has made a significant contribution to rural healthcare, strengthening awareness and improving hospital empanelment are critical for its effectiveness.

```
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ffmpeg version 7.1.1-1ubuntu1 Copyright (c) 2000-2025 the FFmpeg developers
  built with gcc 14.2.0 (Ubuntu 14.2.0-4ubuntu2)
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Date: 2/22/07 10:00:00
Source: Kyodo News International
Copyright: Copyright 2007 Kyodo News International, Inc.
Subject: Japan, Asia, Pacific, Government, Crime, Law enforcement, Health care, Health care industry, Drug abuse, Drug trafficking, Medical, Medical facilities, Medical industry, Medical research, Medical services, Medical technology, Pharmaceuticals, Research, Research and development, Technology, Industry, Business, Drug abuse, Drug trafficking, Medical, Medical facilities, Medical industry, Medical research, Medical services, Medical technology, Pharmaceuticals, Research, Research and development, Technology, Industry, Business
Length: 1 words
Type: Brief
Language: English

No terms apply.

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Sample Invoice

Sample Customer

Reference

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Payments

Date	Amount

Prepayments

Date	Amount

Overpayments

Date	Amount

Credit Notes

Date	Amount

Subtotal 0.00

Total Tax 0.00

Total 0.00

Total Discount

Amount Due 0.00

Amount Paid 0.00

Amount Credited 0.00

Fully Paid on or before 01 Jan 2017

Currency Rounding 0.00

Xero (NZ) Limited, 19 Aitken Street, Wellington, New Zealand

Xero (UK) Limited, 2 Cathedral Piazza, London, EC4V 4BE, United Kingdom

Xero (AU) Pty Ltd, Level 5, 151 Clarence Street, Sydney, NSW 2000, Australia

Xero (US) LLC, 1099 Wall Street West, Lyndhurst, NJ 07071, United States

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Chapter 1

Introduction

Introduction to Cooking

Introduction to Cooking - Chapter 1 Page - Page 1

Introduction to Cooking

Cooking is a fundamental skill that every person should possess.

Cooking is a fundamental skill that every person should possess. Whether you're a beginner or a seasoned cook, mastering the art of cooking can open up a world of delicious possibilities and bring joy to your life. This chapter will be a comprehensive guide to the world of cooking, covering the basics, techniques, and some of the most popular dishes from various cuisines.

Cooking is not just about following a recipe; it's about understanding the ingredients, the techniques, and the science behind the process. It's about being able to adapt and improvise when things don't go as planned. This chapter will be a guide to help you develop your cooking skills and confidence in the kitchen.

Cooking - Basic Ingredients

Salt

Pepper

Oil

Butter

Flour

Sugar

Eggs

Milk

Cooking - Basic Techniques

Boiling

Steaming

Frying

Baking

Roasting

Grilling

Cooking - Popular Dishes

Pasta

Rice

Soup

Salad

Stir-fry

Curry

Cooking - Popular Cuisines

Italian

Mexican

Chinese

Indian

Thai

Cooking - Tips and Tricks

Mise en place

Taste as you cook

Don't be afraid to experiment

Keep a clean workspace

Learn from your mistakes

Cooking - Conclusion

In conclusion, cooking is a fundamental skill that every person should possess. It's a skill that can bring joy to your life, save you money, and allow you to enjoy a wide range of delicious dishes from various cuisines. By mastering the basics, techniques, and some of the most popular dishes, you can become a confident and skilled cook in the kitchen.

Cooking - Resources

For more information and inspiration, you can visit websites like Allrecipes, Food Network, and BBC Good Food. These websites have a wide range of recipes, cooking tips, and videos to help you learn and improve your cooking skills.

Cooking - Conclusion

In conclusion, cooking is a fundamental skill that every person should possess. It's a skill that can bring joy to your life, save you money, and allow you to enjoy a wide range of delicious dishes from various cuisines. By mastering the basics, techniques, and some of the most popular dishes, you can become a confident and skilled cook in the kitchen.

NO. 7147 MONTMORENCY

Ann Arbor, Mich. 48104 Kensington Court

Apt. 3

Dear Lauren Kensington: Our Electric Storage Water Heat

Dear Lauren Kensington: Our Electric Storage Water Heat

Dear Lauren Kensington: Our Electric Storage Water Heat

ISSN: 2349-7688

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Dr. Ashish Kumar Sinha, Associate Professor, Department of Commerce, Patna University, Patna, India. E-mail: ashishkumarsinha@gmail.com

Vol. 8 No. 2 July - Sep 2021

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1.1 Exercises

1.1.1 Find the slope of the line through the points $(-2, 3)$ and $(4, -1)$.

Solution: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-3}{4-(-2)} = \frac{-4}{6} = -\frac{2}{3}$

1.1.2 Find the slope of the line through the points $(-2, 3)$ and $(4, 3)$.

Solution: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-3}{4-(-2)} = \frac{0}{6} = 0$

1.1.3 Find the slope of the line through the points $(-2, 3)$ and $(-2, -1)$.

Solution: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-3}{-2-(-2)} = \frac{-4}{0}$ undefined

1.1.4 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$.

Solution: $y = -\frac{2}{3}x + \frac{5}{3}$

1.1.5 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$.

Solution: $y = 3$

1.1.6 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$.

Solution: $x = -2$

1.1.7 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in point-slope form.

Solution: $y - 3 = -\frac{2}{3}(x + 2)$

1.1.8 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in point-slope form.

Solution: $y - 3 = 0(x + 2)$

1.1.9 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in point-slope form.

Solution: $x + 2 = 0(y - 3)$

1.1.10 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in slope-intercept form.

Solution: $y = -\frac{2}{3}x + \frac{5}{3}$

1.1.11 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in slope-intercept form.

Solution: $y = 3$

1.1.12 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in slope-intercept form.

Solution: $x = -2$

1.1.13 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in standard form.

Solution: $2x + 3y = 5$

1.1.14 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in standard form.

Solution: $y = 3$

1.1.15 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in standard form.

Solution: $x = -2$

1.1.16 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in intercept form.

Solution: $\frac{x}{5/2} + \frac{y}{5/3} = 1$

1.1.17 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in intercept form.

Solution: $y = 3$

1.1.18 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in intercept form.

Solution: $x = -2$

1.1.19 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in vector form.

Solution: $\langle x, y \rangle = \langle -2, 3 \rangle + t\langle 6, -4 \rangle$

1.1.20 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in vector form.

Solution: $\langle x, y \rangle = \langle -2, 3 \rangle + t\langle 6, 0 \rangle$

1.1.21 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in vector form.

Solution: $\langle x, y \rangle = \langle -2, 3 \rangle + t\langle 0, -4 \rangle$

1.1.22 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in parametric form.

Solution: $x = -2 + 6t$, $y = 3 - 4t$

1.1.23 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in parametric form.

Solution: $x = -2 + 6t$, $y = 3$

1.1.24 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in parametric form.

Solution: $x = -2$, $y = 3 - 4t$

1.1.25 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in symmetric form.

Solution: $\frac{x+2}{6} = \frac{y-3}{-4}$

1.1.26 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in symmetric form.

Solution: $y = 3$

1.1.27 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in symmetric form.

Solution: $x = -2$

1.1.28 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in normal form.

Solution: $\frac{2}{\sqrt{13}}x + \frac{3}{\sqrt{13}}y = \frac{5}{\sqrt{13}}$

1.1.29 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in normal form.

Solution: $y = 3$

1.1.30 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in normal form.

Solution: $x = -2$

1.1.31 Find the equation of the line through the point $(-2, 3)$ and $(4, -1)$ in general form.

Solution: $2x + 3y = 5$

1.1.32 Find the equation of the line through the point $(-2, 3)$ and $(4, 3)$ in general form.

Solution: $y = 3$

1.1.33 Find the equation of the line through the point $(-2, 3)$ and $(-2, -1)$ in general form.

Solution: $x = -2$

```
BEG 71437 MEDIUMBOXOFT
```


Item Number: 71437 Medium Shipping Box

Weight: 0.0

Dimensions: 12.0 inches long x 10.0 inches wide x 8.0 inches deep

Description: Medium Shipping Box Made of Kraft Corrugated Cardboard. Lock-Bottom Construction. Dimensions: 12 x 10 x 8 inches

Image: https://www.uline.com/BL_11320/S-71437.png?mh=1000

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[illegible]
[illegible]
[illegible]
[illegible]
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Item Number: 71437 Medium Shipping Box Made of Kraft Corrugated Cardboard

Weight: 0.0

Dimensions: 12.0 inches long x 10.0 inches wide x 8.0 inches deep

Image: https://www.uline.com/BL_11320/S-71437.png?mh=1000

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[illegible]
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[illegible] [illegible] [illegible] [

Q1. Prove $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$, $x \in [0,1]$.

Sol. Let $\sin^{-1}x = \theta$, $\theta \in [0, \frac{\pi}{2}]$

$\Rightarrow \sin\theta = x$

$\Rightarrow \cos\theta = \sqrt{1-x^2}$

$\Rightarrow \theta = \cos^{-1}\sqrt{1-x^2}$

$\Rightarrow \sin^{-1}x = \cos^{-1}\sqrt{1-x^2}$

Q2. Prove $\sin^{-1}x = \tan^{-1}\frac{x}{\sqrt{1-x^2}}$, $x \in [-1,1]$.

Sol. Let $\sin^{-1}x = \theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\Rightarrow \sin\theta = x$

$\Rightarrow \cos\theta = \sqrt{1-x^2}$

$\Rightarrow \tan\theta = \frac{x}{\sqrt{1-x^2}}$

$\Rightarrow \theta = \tan^{-1}\frac{x}{\sqrt{1-x^2}}$

$\Rightarrow \sin^{-1}x = \tan^{-1}\frac{x}{\sqrt{1-x^2}}$

Q3. Prove $\sin^{-1}x = \sec^{-1}\frac{1}{\sqrt{1-x^2}}$, $x \in [-1,1]$.

Sol. Let $\sin^{-1}x = \theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\Rightarrow \sin\theta = x$

$\Rightarrow \cos\theta = \sqrt{1-x^2}$

$\Rightarrow \sec\theta = \frac{1}{\sqrt{1-x^2}}$

$\Rightarrow \theta = \sec^{-1}\frac{1}{\sqrt{1-x^2}}$

$\Rightarrow \sin^{-1}x = \sec^{-1}\frac{1}{\sqrt{1-x^2}}$

1.1 Overview

1.1.1 Introduction to Linux

1.1.1.1 Introduction to Linux

Linux is a free operating system for computers. It is a UNIX-like operating system that was developed by a group of people led by Linus Torvalds.

Linux

Linux is a free operating system for computers. It is a UNIX-like operating system that was developed by a group of people led by Linus Torvalds.

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Received: 2 August 2021
Revised: 14 December 2021
Accepted: 20 December 2021
DOI: 10.1177/0000000000000000

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua: Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Author name, author name, author name,

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Objective

Use angular momentum and the conservation of angular momentum to determine the velocity of a projectile. The projectile is fired at a fixed rotational speed, then it is struck by a mass and the projectile and the mass stick together and continue to rotate.

Data

Mass of projectile (kg)	0.0
Mass of clay (kg)	0.0
Radius of projectile (m)	0.0
Angular velocity before impact (rad/s)	0.0
Angular velocity after impact (rad/s)	0.0

Calculations

Show all work, including formulas and substitutions.

$$I_{projectile} = \frac{1}{2} m_{projectile} r_{projectile}^2 = \frac{1}{2}(0.0\ kg)(0.0\ m)^2 = 0.0\ kg \cdot m^2$$

$$L_{before} = I_{projectile}\omega_{before} = (0.0\ kg \cdot m^2)(0.0\ rad/s) = 0.0\ kg \cdot m^2/s$$

$$L_{after} = (I_{projectile} + m_{clay} r_{projectile}^2)\omega_{after} = (0.0\ kg \cdot m^2 + (0.0\ kg)(0.0\ m)^2)(0.0\ rad/s) = 0.0\ kg \cdot m^2/s$$

$$L_{before} = L_{after}$$

$$0.0\ kg \cdot m^2/s = 0.0\ kg \cdot m^2/s$$

Conclusion

Angular momentum is conserved because the angular momentum before and after the impact are equal. The angular velocity of the projectile decreased from 0.0 rad/s to 0.0 rad/s, or by 0.0 rad/s. The angular velocity of the projectile decreased because the mass of the clay added to the projectile increased the moment of inertia of the projectile, which increased the angular momentum of the projectile. The angular momentum of the projectile remained constant because angular momentum is conserved.

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Hello!

Hi!

I'm Gilbert Bigabe and I live in

Wow!

So interesting!

I'm learning Spanish since two years. Do you think, that I speak Spanish well?

Yes, you speak Spanish very well

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Affiliation: Professor, Department of Computer Science, Jain Vishva Bharati Institute, Ladnun, Rajasthan, India.

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URL: http://dx.doi.org/10.5430/jnep.v2n2p1

Hello!

Hi Kiran!Thanks again and cheers!

Hi Kiran!Hello!

Hi Kiran!Thanks! Looking at the Kiran's Kitchen - Mon 12/11/2023, Mondays - Mon 12/11/2023 (1:00-2:00)
Hi Kiran!Hi Kiran!Hello!
Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!
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Hi Kiran!Hello!
Hi Kiran!Hi Kiran!Looking at the Kiran's Kitchen - Mon 12/11/2023, Mondays - Mon 12/11/2023 (1:00-2:00)
Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran

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Author name, Author name, Author name,

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Received: 12 Jan 2022 Revised: 22 Feb 2022 Accepted: 10 Mar 2022
https://doi.org/10.54216/FMR.010101

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1.2.1 Equivalence Relations

Definition 1.2.1. Let S be a set. A relation $\sim$ on S is an equivalence relation if it satisfies the following three conditions:

1. $a \sim a$ for all $a \in S$ (reflexivity),
2. $a \sim b$ implies $b \sim a$ (symmetry),
3. $a \sim b$ and $b \sim c$ implies $a \sim c$ (transitivity).

Example 1.2.1. Let $S = \{1, 2, 3, 4\}$. The relation $\sim$ defined by $a \sim b$ if $a - b$ is even is an equivalence relation. For example, $1 \sim 3$ and $2 \sim 4$ but $1 \not\sim 2$. Here, $\{1, 3\}$ and $\{2, 4\}$ are the two equivalence classes.

$$a \sim b \text{ if } \frac{a-b}{2} \in \mathbb{Z}$$

Proof.

Reflexivity: $a - a = 0$ is even, so $a \sim a$.

Symmetry: If $a - b$ is even, then $b - a = -(a - b)$ is also even, so $b \sim a$.

Transitivity: If $a - b$ and $b - c$ are even, then $a - c = (a - b) + (b - c)$ is even, so $a \sim c$.

Example 1.2.2. Let $S = \mathbb{Z}$. The relation defined by $a \sim b$ if $a - b$ is divisible by 3 is an equivalence relation. For example, $7 \sim 1$ because $7 - 1 = 6$ is divisible by 3, but $7 \not\sim 2$ because $7 - 2 = 5$ is not divisible by 3.

$$a \sim b \text{ if } \frac{a-b}{3} \in \mathbb{Z}$$

Proof.

Reflexivity: $a - a = 0$ is divisible by 3, so $a \sim a$.

Symmetry: If $a - b$ is divisible by 3, then $b - a = -(a - b)$ is also divisible by 3, so $b \sim a$.

Transitivity: If $a - b$ and $b - c$ are divisible by 3, then $a - c = (a - b) + (b - c)$ is divisible by 3, so $a \sim c$.

Example 1.2.3. Let $S = \mathbb{R}$. The relation defined by $a \sim b$ if $a - b \in \mathbb{Q}$ is an equivalence relation. For example, $\sqrt{2} \sim 1$ because $\sqrt{2} - 1 \in \mathbb{Q}$, but $\sqrt{2} \not\sim \pi$ because $\sqrt{2} - \pi \notin \mathbb{Q}$.

$$a \sim b \text{ if } a - b \in \mathbb{Q}$$

Proof.

Reflexivity: $a - a = 0 \in \mathbb{Q}$, so $a \sim a$.

Symmetry: If $a - b \in \mathbb{Q}$, then $b - a = -(a - b) \in \mathbb{Q}$, so $b \sim a$.

Transitivity: If $a - b \in \mathbb{Q}$ and $b - c \in \mathbb{Q}$, then $a - c = (a - b) + (b - c) \in \mathbb{Q}$, so $a \sim c$.

Example 1.2.4. Let $S = \mathbb{C}$. The relation defined by $a \sim b$ if $a - b \in \mathbb{Q}$ is an equivalence relation. For example, $1 + i \sim 1$ because $(1 + i) - 1 = i \in \mathbb{Q}$, but $1 + i \not\sim 2$ because $(1 + i) - 2 = -1 + i \notin \mathbb{Q}$.

$$a \sim b \text{ if } a - b \in \mathbb{Q}$$

Proof.

Reflexivity: $a - a = 0 \in \mathbb{Q}$, so $a \sim a$.

Symmetry: If $a - b \in \mathbb{Q}$, then $b - a = -(a - b) \in \mathbb{Q}$, so $b \sim a$.

Transitivity: If $a - b \in \mathbb{Q}$ and $b - c \in \mathbb{Q}$, then $a - c = (a - b) + (b - c) \in \mathbb{Q}$, so $a \sim c$.

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Gilbert[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144,145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168,169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192,193,194,195,196,197,198,199,200,201,202,203,204,205,206,207,208,209,210,211,212,213,214,215,216,217,218,219,220,221,222,223,224,225,226,227,228,229,230,231,232,233,234,235,236,237,238,239,240,241,242,243,244,245,246,247,248,249,250,251,252,253,254,255,256,257,258,259,260,261,262,263,264,265,266,267,268,269,270,271,272,273,274,275,276,277,278,279,280,281,282,283,284,285,286,287,288,289,290,291,292,293,294,295,296,297,298,299,300,301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,322,323,324,325,326,327,328,329,330,331,332,333,334,335,336,337,338,339,340,341,342,343,344,345,346,347,348,349,350,351,352,353,354,355,356,357,358,359,360,361,362,363,364,365,366,367,368,369,370,371,372,373,374,375,376,377,378,379,380,381,382,383,384,385,386,387,388,389,390,391,392,393,394,395,396,397,398,399,400,401,402,403,404,405,406,407,408,409,410,411,412,413,414,415,416,417,418,419,420,421,422,423,424,425,426,427,428,429,430,431,432,433,434,435,436,437,438,439,440,441,442,443,444,445,446,447,448,449,450,451,452,453,454,455,456,457,458,459,460,461,462,463,464,465,466,467,468,469,470,471,472,473,474,475,476,477,478,479,480,481,482,483,484,485,486,487,488,489,490,491,492,493,494,495,496,497,498,499,500,501,502,503,504,505,506,507,508,509,510,511,512,513,514,515,516,517,518,519,520,521,522,523,524,525,526,527,528,529,530,531,532,533,534,535,536,537,538,539,540,541,542,543,544,545,546,547,548,549,550,551,552,553,554,555,556,557,558,559,560,561,562,563,564,565,566,567,568,569,570,571,572,573,574,575,576,577,578,579,580,581,582,583,584,585,586,587,588,589,590,591,592,593,594,595,596,597,598,599,600,601,602,603,604,605,606,607,608,609,610,611,612,613,614,615,616,617,618,619,620,621,622,623,624,625,626,627,628,629,630,631,632,633,634,635,636,637,638,639,640,641,642,643,644,645,646,647,648,649,650,651,652,653,654,655,656,657,658,659,660,661,662,663,664,665,666,667,668,669,670,671,672,673,674,675,676,677,678,679,680,681,682,683,684,685,686,687,688,689,690,691,692,693,694,695,696,697,698,699,700,701,702,703,704,705,706,707,708,709,710,711,712,713,714,715,716,717,718,719,720,721,722,723,724,725,726,727,728,729,730,731,732,733,734,735,736,737,738,739,740,741,742,743,744,745,746,747,748,749,750,751,752,753,754,755,756,757,758,759,760,761,762,763,764,765,766,767,768,769,770,771,772,773,774,775,776,777,778,779,780,781,782,783,784,785,786,787,788,789,790,791,792,793,794,795,796,797,798,799,800,801,802,803,804,805,806,807,808,809,810,811,812,813,814,815,816,817,818,819,820,821,822,823,824,825,826,827,828,829,830,831,832,833,834,835,836,837,838,839,840,841,842,843,844,845,846,847,848,849,850,851,852,853,854,855,856,857,858,859,860,861,862,863,864,865,866,867,868,869,870,871,872,873,874,875,876,877,878,879,880,881,882,883,884,885,886,887,888,889,890,891,892,893,894,895,896,897,898,899,900,901,902,903,904,905,906,907,908,909,910,911,912,913,914,915,916,917,918,919,920,921,922,923,924,925,926,927,928,929,930,931,932,933,934,935,936,937,938,939,940,941,942,943,944,945,946,947,948,949,950,951,952,953,954,955,956,957,958,959,960,961,962,963,964,965,966,967,968,969,970,971,972,973,974,975,976,977,978,979,980,981,982,983,984,985,986,987,988,989,990,991,992,993,994,995,996,997,998,999,1000,1001,1002,1003,1004,1005,1006,1007,1008,1009,1010,1011,1012,1013,1014,1015,1016,1017,1018,1019,1020,1021,1022,1023,1024,1025,1026,1027,1028,1029,1030,1031,1032,1033,1034,1035,1036,1037,1038,1039,1040,1041,1042,1043,1044,1045,1046,1047,1048,1049,1050,1051,1052,1053,1054,1055,1056,1057,1058,1059,1060,1061,1062,1063,1064,1065,1066,1067,1068,1069,1070,1071,1072,1073,1074,1075,1076,1077,1078,1079,1080,1081,1082,1083,1084,1085,1086,1087,1088,1089,1090,1091,1092,1093,1094,1095,1096,1097,1098,1099,1100,1101,1102,1103,1104,1105,1106,1107,1108,1109,1110,1111,1112,1113,1114,1115,1116,1117,1118,1119,1120,1121,1122,1123,1124,1125,1126,1127,1128,1129,1130,1131,1132,1133,1134,1135,1136,1137,1138,1139,1140,1141,1142,1143,1144,1145,1146,1147,1148,1149,1150,1151,1152,1153,1154,1155,1156,1157,1158,1159,1160,1161,1162,1163,1164,1165,1166,1167,1168,1169,1170,1171,1172,1173,1174,1175,1176,1177,1178,1179,1180,1181,1182,1183,1184,1185,1186,1187,1188,1189,1190,1191,1192,1193,1194,1195,1196,1197,1198,1199,1200,1201,1202,1203,1204,1205,1206,1207,1208,1209,1210,1211,1212,1213,1214,1215,1216,1217,1218,1219,1220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Cost-Sensitive Learning for Imbalanced Medical Data

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Abstract

In medical diagnosis, misclassifying a sick patient is more costly than sending a healthy patient for further tests. We introduce a simple, yet effective, class-weighted term for deep neural networks that penalizes mistakes in proportion to clinical cost. We pair this with three ablations—uniform weights, inverse-frequency weights, and no weighting—to isolate the effect of cost sensitivity. Across two benchmarks, the proposed method consistently shifts predictions toward lower clinical cost while maintaining competitive accuracy. Ablation results confirm that gains arise from cost-aligned weighting rather than from class rebalancing alone. Qualitative analysis shows fewer high-cost errors at the expected margin. We release code and configurations to reproduce all results.

Keywords

Cost-sensitive learning, class imbalance, medical diagnosis, deep neural networks, ablation study, reproducible research

Bibliography

[1] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 321–357.
[2] He, H., & Garcia, E. A. (2009). Learning from Imbalanced Data. IEEE Transactions on Knowledge and Data Engineering, 21(9), 1263–1284.
[3] Elkan, C. (2001). The Foundations of Cost-Sensitive Learning. In Proceedings of the Seventeenth International Joint Conference on Artificial Intelligence (IJCAI), 973–978.
[4] Zhou, Z. H., & Liu, X. Y. (2006). Training Cost-sensitive Neural Networks with Methods Addressing the Class Imbalance Problem. IEEE Transactions on Knowledge and Data Engineering, 18(1), 63–77.
[5] Ling, C. X., & Sheng, V. S. (2008). Cost-sensitive Learning and the Class Imbalance Problem. In Encyclopedia of Machine Learning, 231–235. Springer.
[6] Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal Loss for Dense Object Detection. IEEE International Conference on Computer Vision (ICCV), 2980–2988.
[7] Cui, Y., Jia, M., Lin, T. Y., Song, Y., & Belongie, S. (2019). Class-balanced Loss Based on Effective Number of Samples. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 9268–9277.
[8] Buda, M., Maki, A., & Mazurowski, M. A. (2018). A Systematic Study of the Class Imbalance Problem in Convolutional Neural Networks. Neural Networks, 106, 249–259.
[9] Johnson, J. M., & Khoshgoftaar, T. M. (2019). Survey on Deep Learning with Class Imbalance. Journal of Big Data, 6, 27.
[10] Ganin, Y., & Lempitsky, V. (2015). Unsupervised Domain Adaptation by Backpropagation. In Proceedings of the 32nd International Conference on Machine Learning (ICML), 1180–1189.
[11] Tzeng, E., Hoffman, J., Saenko, K., & Darrell, T. (2017). Adversarial Discriminative Domain Adaptation. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 7167–7176.
[12] Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., & Lempitsky, V. (2016). Domain-adversarial Training of Neural Networks. Journal of Machine Learning Research, 17(59), 1–35.
[13] Long, M., Cao, Y., Wang, J., & Jordan, M. I. (2015). Learning Transferable Features with Deep Adaptation Networks. In Proceedings of the 32nd International Conference on Machine Learning (ICML), 97–105.
[14] Sun, B., & Saenko, K. (2016). Deep CORAL: Correlation Alignment for Deep Domain Adaptation. In Proceedings of the European Conference on Computer Vision (ECCV) Workshops, 443–450.
[15] Tzeng, E., Hoffman, J., Zhang, N., Saenko, K., & Darrell, T. (2014). Deep Domain Confusion: Maximizing for Domain Invariance. arXiv preprint arXiv:1412.3474.
[16] Pan, S. J., & Yang, Q. (2010). A Survey on Transfer Learning. IEEE Transactions on Knowledge and Data Engineering, 22(10), 1345–1359.
[17] Weiss, K., Khoshgoftaar, T. M., & Wang, D. D. (2016). A Survey of Transfer Learning. Journal of Big Data, 3, 9.
[18] Bengio, Y., Ducharme, R., Vincent, P., & Jauvin, C. (2003). A Neural Probabilistic Language Model. Journal of Machine Learning Research, 3, 1137–1155.
[19] Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed Representations of Words and Phrases and their Compositionality. Advances in Neural Information Processing Systems (NIPS), 26, 3111–3119.
[20] Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global Vectors for Word Representation. Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 1532–1543.
[21] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 4171–4186.
[22] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention Is All You Need. Advances in Neural Information Processing Systems (NIPS), 30, 5998–6008.
[23] Kingma, D. P., & Ba, J. (2015). Adam: A Method for Stochastic Optimization. International Conference on Learning Representations (ICLR).
[24] Loshchilov, I., & Hutter, F. (2017). Decoupled Weight Decay Regularization. arXiv preprint arXiv:1711.05101.
[25] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., & Chintala, S. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. Advances in Neural Information Processing Systems (NeurIPS), 32, 8026–8037.
[26] Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., & Oliphant, T. E. (2020). Array Programming with NumPy. Nature, 585(7825), 357–362.
[27] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
[28] Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering, 9(3), 90–95.
[29] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., Davison, J., Shleifer, S., von Platen, P., Ma, C., Jernite, Y., Plu, J., Xu, C., Le Scao, T., Gugger, S., Drame, M., Lhoest, Q., & Rush, A. M. (2020). Transformers: State-of-the-art Natural Language Processing. Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP), 38–45.
[30] Bengio, Y., Simard, P., & Frasconi, P. (1994). Learning Long-term Dependencies with Gradient Descent is Difficult. IEEE Transactions on Neural Networks, 5(2), 157–166.
[31] Glorot, X., & Bengio, Y. (2010). Understanding the Difficulty of Training Deep Feedforward Neural Networks. Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics (AISTATS), 249–256.
[32] He, K., Zhang, X., Ren, S., & Sun, J. (2015). Delving Deep into Rectifiers: Surpassing Human-level Performance on ImageNet Classification. IEEE International Conference on Computer Vision (ICCV), 1026–1034.
[33] Ioffe, S., & Szegedy, C. (2015). Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. International Conference on Machine Learning (ICML), 448–456.
[34] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Journal of Machine Learning Research, 15(1), 1929–1958.
[35] Zeiler, M. D., & Fergus, R. (2014). Visualizing and Understanding Convolutional Networks. European Conference on Computer Vision (ECCV), 818–833.
[36] Simonyan, K., & Zisserman, A. (2014). Very Deep Convolutional Networks for Large-scale Image Recognition. arXiv preprint arXiv:1409.1556.
[37] Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2015). Going Deeper with Convolutions. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 1–9.
[38] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778.
[39] Huang, G., Liu, Z., Van der Maaten, L., & Weinberger, K. Q. (2017). Densely Connected Convolutional Networks. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 4700–4708.
[40] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. International Conference on Machine Learning (ICML), 6105–6114.
[41] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. International Conference on Learning Representations (ICLR).
[42] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. IEEE/CVF International Conference on Computer Vision (ICCV), 9992–10009.
[43] Touvron, H., Cord, M., Sablayrolles, A., Synnaeve, G., & Jégou, H. (2021). Training data-efficient image transformers & distillation through attention. International Conference on Machine Learning (ICML), 10347–10357.
[44] Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-end Object Detection with Transformers. European Conference on Computer Vision (ECCV), 213–229.
[45] Zhu, X., Su, W., Lu, L., Li, B., Wang, X., & Dai, J. (2021). Deformable DETR: Deformable Transformers for End-to-end Object Detection. International Conference on Learning Representations (ICLR).
[46] Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A Simple Framework for Contrastive Learning of Visual Representations. International Conference on Machine Learning (ICML), 1597–1607.
[47] He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020). Momentum Contrast for Unsupervised Visual Representation Learning. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 9729–9738.
[48] Grill, J. B., Strub, F., Altché, F., Tallec, C., Richemond, P. H., Buchatskaya, E., Doersch, C., Pires, B. A., Guo, Z., Gheshlaghi Azar, M., Piot, B., Kavukcuoglu, K., Munos, R., & Valko, M. (2020). Bootstrap Your Own Latent: A New Approach to Self-supervised Learning. Advances in Neural Information Processing Systems (NeurIPS), 33, 21271–21284.
[49] Chen, X., Fan, H., Girshick, R., & He, K. (2020). Improved Baselines with Momentum Contrastive Learning. arXiv preprint arXiv:2003.04297.
[50] Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., & Joulin, A. (2020). Unsupervised Learning of Visual Features by Contrasting Cluster Assignments. Advances in Neural Information Processing Systems (NeurIPS), 33, 9912–9924.
[51] Oord, A. van den, Li, Y., & Vinyals, O. (2018). Representation Learning with Contrastive Predictive Coding. arXiv preprint arXiv:1807.03748.
[52] Hjelm, R. D., Fedorov, A., Lavoie-Marchildon, S., Grewal, K., Bachman, P., Trischler, A., & Bengio, Y. (2019). Learning Deep Representations by Mutual Information Estimation and Maximization. International Conference on Learning Representations (ICLR).
[53] Tian, Y., Krishnan, D., & Isola, P. (2020). Contrastive Multivariate Coding. European Conference on Computer Vision (ECCV), 776–794.
[54] Chen, T., Kornblith, S., Swersky, K., Norouzi, M., & Hinton, G. (2020). Big Self-supervised Models are Strong Semi-supervised Learners. Advances in Neural Information Processing Systems (NeurIPS), 33, 22243–22255.
[55] Bachman, P., Hjelm, R. D., & Buchwalter, W. (2019). Learning Representations by Maximizing Mutual Information across Views. Advances in Neural Information Processing Systems (NeurIPS), 32, 15509–15519.
[56] Henaff, O. J., Srinivas, A., De Fauw, J., Razavi, A., Doersch, C., Eslami, S. M. A., & Oord, A. van den. (2019). Data-efficient Image Recognition with Contrastive Predictive Coding. International Conference on Machine Learning (ICML), 4182–4192.
[57] Tian, Y., Krishnan, D., & Isola, P. (2019). Contrastive Representation Distillation. arXiv preprint arXiv:1910.10699.
[58] Chen, T., Kornblith, S., Swersky, K., Norouzi, M., & Hinton, G. (2020). On Mutual Information in Contrastive Learning for Visual Representations. arXiv preprint arXiv:2005.00820.
[59] Xie, Q., Luong, M. T., Hovy, E., & Le, Q. V. (2020). Self-training with Noisy Student improves ImageNet classification. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 10687–10698.
[60] Sohn, K., Berthelot, D., Carlini, N., Zhang, Z., Zhang, H., Raffel, C., Cubuk, E. D., Kurakin, A., & Li, C. L. (2020). FixMatch: Simplifying Semi-supervised Learning with Consistency and Confidence. Advances in Neural Information Processing Systems (NeurIPS), 33, 596–608.
[61] Berthelot, D., Carlini, N., Goodfellow, I., Papernot, N., Oliver, A., & Raffel, C. A. (2019). MixMatch: A Holistic Approach to Semi-supervised Learning. Advances in Neural Information Processing Systems (NeurIPS), 32, 5049–5059.
[62] Laine, S., & Aila, T. (2017). Temporal Ensembling for Semi-supervised Learning. International Conference on Learning Representations (ICLR).
[63] Tarvainen, A., & Valpola, H. (2017). Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. Advances in Neural Information Processing Systems (NeurIPS), 30, 1195–1204.
[64] Miyato, T., Maeda, S. I., Koyama, M., & Ishii, S. (2018). Virtual adversarial training: a regularization method for supervised and semi-supervised learning. IEEE Transactions on Pattern Analysis and Machine Intelligence, 41(8), 1979–1993.
[65] Berthelot, D., Carlini, N., Cubuk, E. D., Kurakin, A., Sohn, K., Zhang, H., & Raffel, C. (2019). ReMixMatch: Semi-supervised Learning with Distribution Matching and Augmentation Anchoring. International Conference on Learning Representations (ICLR).
[66] Xie, Q., Dai, Z., Hovy, E., Luong, M. T., & Le, Q. V. (2020). Unsupervised Data Augmentation for Consistency Training. Advances in Neural Information Processing Systems (NeurIPS), 33, 6256–6268.
[67] Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2018). mixup: Beyond Empirical Risk Minimization. International Conference on Learning Representations (ICLR).
[68] Yun, S., Han, D., Oh, S. J., Chun, S., Choe, J., & Yoo, Y. (2019). CutMix: Regularization strategy to train strong classifiers with localizable features. IEEE/CVF International Conference on Computer Vision (ICCV), 6023–6032.
[69] DeVries, T., & Taylor, G. W. (2017). Improved Regularization of Convolutional Neural Networks with Cutout. arXiv preprint arXiv:1708.04552.
[70] Cubuk, E. D., Zoph, B., Mane, D., Vasudevan, V., & Le, Q. V. (2019). AutoAugment: Learning Augmentation Strategies from Data. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 113–123.
[71] Cubuk, E. D., Zoph, B., Shlens, J., & Le, Q. V. (2020). RandAugment: Practical automated data augmentation with a reduced search space. Advances in Neural Information Processing Systems (NeurIPS), 33, 18613–18624.
[72] Lim, S., Kim, I., Lee, T., Kim, H., Jang, S., & Lee, K. (2019). Fast AutoAugment. Advances in Neural Information Processing Systems (NeurIPS), 32, 6665–6675.
[73] Hataya, R., Zdenek, J., Yoshikawa, K., Imakura, A., & Sakurai, T. (2020). Faster AutoAugment: Learning Augmentation Strategies using Backpropagation. European Conference on Computer Vision (ECCV), 1–16.
[74] Ho, D., Liang, E., Chen, X., Stoica, I., & Abbeel, P. (2019). Population Based Augmentation: Efficient Learning of Augmentation Policy Schedules. International Conference on Machine Learning (ICML), 2731–2741.
[75] Lim, S. H., Elrod, I. N., & Kanan, C. (2021). Adversarial Auto-Augment with Label Preservation: A Surprisingly Simple Baseline. Advances in Neural Information Processing Systems (NeurIPS), 34, 30962–30975.
[76] Li, Y., Hu, G., Wang, Y., Hospedales, T., Robertson, N. M., & Yang, Y. (2020). DADA: Differentiable Automatic Data Augmentation. European Conference on Computer Vision (ECCV), 1–18.
[77] Zhang, X., Wang, Q., Zhang, J., & Zhong, Z. (2019). Adversarial AutoAugment. arXiv preprint arXiv:1912.11188.
[78] Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2019). The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 586–595.
[79] Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image Quality Assessment: From Error Visibility to Structural Similarity. IEEE Transactions on Image Processing, 13(4), 600–612.
[80] Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. Advances in Neural Information Processing Systems (NeurIPS), 30, 6626–6637.
[81] Salimans, T., Goodfellow, I., Zaremba, W., Cheung, V., Radford, A., & Chen, X. (2016). Improved Techniques for Training GANs. Advances in Neural Information Processing Systems (NeurIPS), 29, 2234–2242.
[82] Bińkowski, M., Sutherland, D. J., Arbel, M., & Gretton, A. (2018). Demystifying MMD GANs. International Conference on Learning Representations (ICLR).
[83] Lucic, M., Kurach, K., Michalski, M., Gelly, S., & Bousquet, O. (2018). Are GANs Created Equal? A Large-scale Study. Advances in Neural Information Processing Systems (NeurIPS), 31, 700–709.
[84] Brock, A., Donahue, J., & Simonyan, K. (2019). Large Scale GAN Training for High Fidelity Natural Image Synthesis. International Conference on Learning Representations (ICLR).
[85] Karras, T., Laine, S., Aila, T., Hellsten, J., Lehtinen, J., & Aila, T. (2020). Training Generative Adversarial Networks with Limited Data. Advances in Neural Information Processing Systems (NeurIPS), 33, 12104–12114.
[86] Karras, T., Aila, T., Laine, S., & Lehtinen, J. (2018). Progressive Growing of GANs for Improved Quality, Stability, and Variation. International Conference on Learning Representations (ICLR).
[87] Karras, T., Laine, S., & Aila, T. (2019). A Style-based Generator Architecture for Generative Adversarial Networks. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 4401–4410.
[88] Karras, T., Laine, S., Aila, T., Hellsten, J., Lehtinen, J., & Aila, T. (2020). Analyzing and Improving the Image Quality of StyleGAN. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 8110–8119.
[89] Karras, T., Laine, S., Aila, T., Lehtinen, J., & Aila, T. (2021). Alias-free Generative Adversarial Networks. Advances in Neural Information Processing Systems (NeurIPS), 34, 852–863.
[90] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2022). Training Generative Adversarial Networks with Limited Data. Advances in Neural Information Processing Systems (NeurIPS), 35, 26480–26493.
[91] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2023). Alias-free Generative Adversarial Networks. Advances in Neural Information Processing Systems (NeurIPS), 36, 12345–12357.
[92] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2024). Training Generative Adversarial Networks with Limited Data. Advances in Neural Information Processing Systems (NeurIPS), 37, 12345–12357.
[93] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2025). Alias-free Generative Adversarial Networks. Advances in Neural Information Processing Systems (NeurIPS), 38, 12345–12357.
[94] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2026). Training Generative Adversarial Networks with Limited Data. Advances in Neural Information Processing Systems (NeurIPS), 39, 12345–12357.
[95] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2027). Alias-free Generative Adversarial Networks. Advances in Neural Information Processing Systems (NeurIPS), 40, 12345–12357.
[96] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2028). Training Generative Adversarial Networks with Limited Data. Advances in Neural Information Processing Systems (NeurIPS), 41, 12345–12357.
[97] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2029). Alias-free Generative Adversarial Networks. Advances in Neural Information Processing Systems (NeurIPS), 42, 12345–12357.
[98] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2030). Training Generative Adversarial Networks with Limited Data. Advances in Neural Information Processing Systems (NeurIPS), 43, 12345–12357.
[99] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2031). Alias-free Generative Adversarial Networks. Advances in Neural Information Processing Systems (NeurIPS), 44, 12345–12357.
[100] Karras, T., Aila, T., Laine, S., Lehtinen, J., & Aila, T. (2032). Training Generative Adversarial Networks with Limited Data. Advances in Neural Information Processing Systems (NeurIPS), 45, 12345–12357.

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Abstract Abstract Abstract

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Abstract
Cryptanalysis of a chaotic image encryption scheme based on permutation-diffusion structure, Recently a chaotic image encryption scheme based on permutation-diffusion structure was proposed by Chong Fu et al. In this paper, we show that the scheme is not secure against chosen-plaintext attack and chosen-ciphertext attack.

Keywords
Chaotic cryptanalysis, chosen-plaintext attack, chosen-ciphertext attack, Chong Fu et al.'s scheme, permutation-diffusion structure

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Section 1

Section 1 Section 1 Section 1

Exploring the Impact of Digital Transformation on Consumer Behavior in Modern Retail Industries

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Section 1.2 Sets

Introduction

Sets are fundamental to mathematics. We will use sets throughout this course to describe and study mathematical ideas. Sets are used in the definition of the integers, the rational numbers, and the real numbers. Sets are used in the definition of functions, which are used to describe the relationship between two quantities. Sets are used in probability theory to describe the outcomes of an experiment.

Definition 1.2.1 Set

A set is a collection of objects, called the elements of the set. We denote the set by capital letters such as A, B, C, S, T, X, Y, Z.

Notation 1.2.2 Set Notation

We denote the set A by $A = \{a, b, c, d, e\}$ and we write $a \in A$ to denote that a is an element of A. We write $a \notin A$ to denote that a is not an element of A.

Example 1.2.3 Set Notation

$A = \{1, 2, 3, 4, 5\}$, $B = \{a, b, c, d, e\}$, $C = \{\text{red}, \text{blue}, \text{green}\}$

$1 \in A$, $2 \in A$, $3 \in A$, $4 \in A$, $5 \in A$

$a \in B$, $b \in B$, $c \in B$, $d \in B$, $e \in B$

$\text{red} \in C$, $\text{blue} \in C$, $\text{green} \in C$

$6 \notin A$, $f \notin B$, $\text{yellow} \notin C$

Definition 1.2.4 Equal Sets

Two sets A and B are equal if they have the same elements. We denote this by $A = B$.

Example 1.2.5 Equal Sets

$A = \{1, 2, 3\}$, $B = \{3, 2, 1\}$, $A = B$

$A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$, $A \neq B$

Definition 1.2.6 Subset

A set A is a subset of a set B if every element of A is also an element of B. We denote this by $A \subseteq B$.

Example 1.2.7 Subset

$A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$, $A \subseteq B$

$A = \{1, 2, 3\}$, $B = \{1, 2, 3\}$, $A \subseteq B$

$A = \{1, 2, 3\}$, $B = \{1, 2, 4\}$, $A \not\subseteq B$

Definition 1.2.8 Proper Subset

A set A is a proper subset of a set B if $A \subseteq B$ and $A \neq B$. We denote this by $A \subset B$.

Example 1.2.9 Proper Subset

$A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$, $A \subset B$

$A = \{1, 2, 3\}$, $B = \{1, 2, 3\}$, $A \not\subset B$

$A = \{1, 2, 3\}$, $B = \{1, 2, 4\}$, $A \not\subset B$

Definition 1.2.10 Empty Set

The empty set is the set that has no elements. We denote the empty set by $\emptyset$ or $\{\}$.

Example 1.2.11 Empty Set

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $A = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $A \neq \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $A = \{\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $A \neq \{\}$

Definition 1.2.12 Cardinality

The cardinality of a set A is the number of elements in A. We denote the cardinality of A by $|A|$.

Example 1.2.13 Cardinality

$A = \{1, 2, 3\}$, $|A| = 3$

$A = \{a, b, c, d, e\}$, $|A| = 5$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $|A| = 3$

$A = \emptyset$, $|A| = 0$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $|A| = 0$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $|A| = 2$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $|A| = 0$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $|A| = 2$

Definition 1.2.14 Power Set

The power set of a set A is the set of all subsets of A. We denote the power set of A by $\mathcal{P}(A)$.

Example 1.2.15 Power Set

$A = \{1, 2, 3\}$, $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

$A = \{a, b, c, d, e\}$, $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \{a, b\}, \{a, c\}, \{a, d\}, \{a, e\}, \{b, c\}, \{b, d\}, \{b, e\}, \{c, d\}, \{c, e\}, \{d, e\}, \{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\}, \{a, d, e\}, \{b, c, d\}, \{b, c, e\}, \{b, d, e\}, \{c, d, e\}, \{a, b, c, d\}, \{a, b, c, e\}, \{a, b, d, e\}, \{a, c, d, e\}, \{b, c, d, e\}, \{a, b, c, d, e\}\}$

$A = \emptyset$, $\mathcal{P}(A) = \{\emptyset\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset, \{1\}, \{-1\}, \{1, -1\}\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset, \{1\}, \{-1\}, \{1, -1\}\}$

Definition 1.2.16 Cartesian Product

The Cartesian product of two sets A and B is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$. We denote the Cartesian product of A and B by $A \times B$.

Example 1.2.17 Cartesian Product

$A = \{1, 2, 3\}$, $B = \{a, b, c, d, e\}$, $A \times B = \{(1, a), (1, b), (1, c), (1, d), (1, e), (2, a), (2, b), (2, c), (2, d), (2, e), (3, a), (3, b), (3, c), (3, d), (3, e)\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{1, 2, 3, 4, 5\}$, $A \times B = \{(\text{red}, 1), (\text{red}, 2), (\text{red}, 3), (\text{red}, 4), (\text{red}, 5), (\text{blue}, 1), (\text{blue}, 2), (\text{blue}, 3), (\text{blue}, 4), (\text{blue}, 5), (\text{green}, 1), (\text{green}, 2), (\text{green}, 3), (\text{green}, 4), (\text{green}, 5)\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \times B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \times B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \times B = \{(1, 1), (1, 2), (1, 3), (-1, 1), (-1, 2), (-1, 3)\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \emptyset$, $A \times B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \emptyset$, $A \times B = \emptyset$

Definition 1.2.18 Union

The union of two sets A and B is the set of all elements that are in A or in B. We denote the union of A and B by $A \cup B$.

Example 1.2.19 Union

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $A \cup B = \{1, 2, 3, 4, 5\}$

$A = \{a, b, c, d, e\}$, $B = \{c, d, e, f, g\}$, $A \cup B = \{a, b, c, d, e, f, g\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{green}, \text{yellow}, \text{orange}\}$, $A \cup B = \{\text{red}, \text{blue}, \text{green}, \text{yellow}, \text{orange}\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \cup B = \{1, 2, 3\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cup B = \{1, 2, 3\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cup B = \{1, 2, 3, -1\}$

Definition 1.2.20 Intersection

The intersection of two sets A and B is the set of all elements that are in A and in B. We denote the intersection of A and B by $A \cap B$.

Example 1.2.21 Intersection

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $A \cap B = \{3\}$

$A = \{a, b, c, d, e\}$, $B = \{c, d, e, f, g\}$, $A \cap B = \{c, d, e\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{green}, \text{yellow}, \text{orange}\}$, $A \cap B = \{\text{green}\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \{1\}$

Definition 1.2.22 Difference

The difference of two sets A and B is the set of all elements that are in A but not in B. We denote the difference of A and B by $A \setminus B$.

Example 1.2.23 Difference

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $A \setminus B = \{1, 2\}$

$A = \{a, b, c, d, e\}$, $B = \{c, d, e, f, g\}$, $A \setminus B = \{a, b\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{green}, \text{yellow}, \text{orange}\}$, $A \setminus B = \{\text{red}, \text{blue}\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \setminus B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \setminus B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \setminus B = \{-1\}$

Definition 1.2.24 Complement

The complement of a set A is the set of all elements that are not in A. We denote the complement of A by A^c.

Example 1.2.25 Complement

$A = \{1, 2, 3\}$, $A^c = \{x \in \mathbb{R} : x \neq 1, 2, 3\}$

$A = \{a, b, c, d, e\}$, $A^c = \{x \in \mathbb{R} : x \neq a, b, c, d, e\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $A^c = \{x \in \mathbb{R} : x \neq \text{red}, \text{blue}, \text{green}\}$

$A = \emptyset$, $A^c = \{x \in \mathbb{R}\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $A^c = \{x \in \mathbb{R}\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $A^c = \{x \in \mathbb{R} : x \neq 1, -1\}$

Definition 1.2.26 Disjoint

Two sets A and B are disjoint if they have no elements in common. We denote this by $A \cap B = \emptyset$.

Example 1.2.27 Disjoint

$A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$, $A \cap B = \emptyset$

$A = \{a, b, c, d, e\}$, $B = \{f, g, h, i, j\}$, $A \cap B = \emptyset$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{yellow}, \text{orange}\}$, $A \cap B = \emptyset$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

Definition 1.2.28 Partition

A partition of a set A is a collection of disjoint subsets of A whose union is A. We denote a partition of A by $\{A_1, A_2, \ldots, A_n\}$.

Example 1.2.29 Partition

$A = \{1, 2, 3, 4, 5, 6\}$, $\{A_1, A_2, A_3\} = \{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$

$A = \{a, b, c, d, e, f, g, h, i, j\}$, $\{A_1, A_2, A_3, A_4, A_5\} = \{\{a, b\}, \{c, d\}, \{e, f\}, \{g, h\}, \{i, j\}\}$

$A = \{\text{red}, \text{blue}, \text{green}, \text{yellow}, \text{orange}\}$, $\{A_1, A_2, A_3\} = \{\{\text{red}, \text{blue}\}, \{\text{green}, \text{yellow}\}, \{\text{orange}\}\}$

$A = \emptyset$, $\{A_1, A_2, A_3\} = \{\emptyset, \emptyset, \emptyset\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $\{A_1, A_2, A_3\} = \{\emptyset, \emptyset, \emptyset\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $\{A_1, A_2, A_3\} = \{\{1\}, \{-1\}, \emptyset\}$

Definition 1.2.30 Cardinality of a Set

The cardinality of a set A is the number of elements in A. We denote the cardinality of A by $|A|$.

Example 1.2.31 Cardinality of a Set

$A = \{1, 2, 3\}$, $|A| = 3$

$A = \{a, b, c, d, e\}$, $|A| = 5$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $|A| = 3$

$A = \emptyset$, $|A| = 0$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $|A| = 0$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $|A| = 2$

Definition 1.2.32 Power Set

The power set of a set A is the set of all subsets of A. We denote the power set of A by $\mathcal{P}(A)$.

Example 1.2.33 Power Set

$A = \{1, 2, 3\}$, $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

$A = \{a, b, c, d, e\}$, $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \{a, b\}, \{a, c\}, \{a, d\}, \{a, e\}, \{b, c\}, \{b, d\}, \{b, e\}, \{c, d\}, \{c, e\}, \{d, e\}, \{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\}, \{a, d, e\}, \{b, c, d\}, \{b, c, e\}, \{b, d, e\}, \{c, d, e\}, \{a, b, c, d\}, \{a, b, c, e\}, \{a, b, d, e\}, \{a, c, d, e\}, \{b, c, d, e\}, \{a, b, c, d, e\}\}$

$A = \emptyset$, $\mathcal{P}(A) = \{\emptyset\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset, \{1\}, \{-1\}, \{1, -1\}\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $\mathcal{P}(A) = \{\emptyset, \{1\}, \{-1\}, \{1, -1\}\}$

Definition 1.2.34 Cartesian Product

The Cartesian product of two sets A and B is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$. We denote the Cartesian product of A and B by $A \times B$.

Example 1.2.35 Cartesian Product

$A = \{1, 2, 3\}$, $B = \{a, b, c, d, e\}$, $A \times B = \{(1, a), (1, b), (1, c), (1, d), (1, e), (2, a), (2, b), (2, c), (2, d), (2, e), (3, a), (3, b), (3, c), (3, d), (3, e)\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{1, 2, 3, 4, 5\}$, $A \times B = \{(\text{red}, 1), (\text{red}, 2), (\text{red}, 3), (\text{red}, 4), (\text{red}, 5), (\text{blue}, 1), (\text{blue}, 2), (\text{blue}, 3), (\text{blue}, 4), (\text{blue}, 5), (\text{green}, 1), (\text{green}, 2), (\text{green}, 3), (\text{green}, 4), (\text{green}, 5)\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \times B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \times B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \times B = \{(1, 1), (1, 2), (1, 3), (-1, 1), (-1, 2), (-1, 3)\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \emptyset$, $A \times B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \emptyset$, $A \times B = \emptyset$

Definition 1.2.36 Union

The union of two sets A and B is the set of all elements that are in A or in B. We denote the union of A and B by $A \cup B$.

Example 1.2.37 Union

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $A \cup B = \{1, 2, 3, 4, 5\}$

$A = \{a, b, c, d, e\}$, $B = \{c, d, e, f, g\}$, $A \cup B = \{a, b, c, d, e, f, g\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{green}, \text{yellow}, \text{orange}\}$, $A \cup B = \{\text{red}, \text{blue}, \text{green}, \text{yellow}, \text{orange}\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \cup B = \{1, 2, 3\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cup B = \{1, 2, 3\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cup B = \{1, 2, 3, -1\}$

Definition 1.2.38 Intersection

The intersection of two sets A and B is the set of all elements that are in A and in B. We denote the intersection of A and B by $A \cap B$.

Example 1.2.39 Intersection

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $A \cap B = \{3\}$

$A = \{a, b, c, d, e\}$, $B = \{c, d, e, f, g\}$, $A \cap B = \{c, d, e\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{green}, \text{yellow}, \text{orange}\}$, $A \cap B = \{\text{green}\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \{1\}$

Definition 1.2.40 Difference

The difference of two sets A and B is the set of all elements that are in A but not in B. We denote the difference of A and B by $A \setminus B$.

Example 1.2.41 Difference

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $A \setminus B = \{1, 2\}$

$A = \{a, b, c, d, e\}$, $B = \{c, d, e, f, g\}$, $A \setminus B = \{a, b\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{green}, \text{yellow}, \text{orange}\}$, $A \setminus B = \{\text{red}, \text{blue}\}$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \setminus B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \setminus B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \setminus B = \{-1\}$

Definition 1.2.42 Complement

The complement of a set A is the set of all elements that are not in A. We denote the complement of A by A^c.

Example 1.2.43 Complement

$A = \{1, 2, 3\}$, $A^c = \{x \in \mathbb{R} : x \neq 1, 2, 3\}$

$A = \{a, b, c, d, e\}$, $A^c = \{x \in \mathbb{R} : x \neq a, b, c, d, e\}$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $A^c = \{x \in \mathbb{R} : x \neq \text{red}, \text{blue}, \text{green}\}$

$A = \emptyset$, $A^c = \{x \in \mathbb{R}\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $A^c = \{x \in \mathbb{R}\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $A^c = \{x \in \mathbb{R} : x \neq 1, -1\}$

Definition 1.2.44 Disjoint

Two sets A and B are disjoint if they have no elements in common. We denote this by $A \cap B = \emptyset$.

Example 1.2.45 Disjoint

$A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$, $A \cap B = \emptyset$

$A = \{a, b, c, d, e\}$, $B = \{f, g, h, i, j\}$, $A \cap B = \emptyset$

$A = \{\text{red}, \text{blue}, \text{green}\}$, $B = \{\text{yellow}, \text{orange}\}$, $A \cap B = \emptyset$

$A = \emptyset$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $B = \{1, 2, 3\}$, $A \cap B = \emptyset$

Definition 1.2.46 Partition

A partition of a set A is a collection of disjoint subsets of A whose union is A. We denote a partition of A by $\{A_1, A_2, \ldots, A_n\}$.

Example 1.2.47 Partition

$A = \{1, 2, 3, 4, 5, 6\}$, $\{A_1, A_2, A_3\} = \{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$

$A = \{a, b, c, d, e, f, g, h, i, j\}$, $\{A_1, A_2, A_3, A_4, A_5\} = \{\{a, b\}, \{c, d\}, \{e, f\}, \{g, h\}, \{i, j\}\}$

$A = \{\text{red}, \text{blue}, \text{green}, \text{yellow}, \text{orange}\}$, $\{A_1, A_2, A_3\} = \{\{\text{red}, \text{blue}\}, \{\text{green}, \text{yellow}\}, \{\text{orange}\}\}$

$A = \emptyset$, $\{A_1, A_2, A_3\} = \{\emptyset, \emptyset, \emptyset\}$

$A = \{x \in \mathbb{R} : x^2 + 1 = 0\}$, $\{A_1, A_2, A_3\} = \{\emptyset, \emptyset, \emptyset\}$

$A = \{x \in \mathbb{R} : x^2 - 1 = 0\}$, $\{A_1, A_2, A_3\} = \{\{1\}, \{-1\}, \emptyset\}$

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Abstract—This paper presents a novel approach to quantum error correction using topological codes. We demonstrate that our method achieves a threshold error rate of 1.2% under realistic noise models, outperforming existing surface code implementations by a factor of 3.4. The proposed architecture requires only nearest-neighbor interactions and is compatible with current superconducting qubit platforms. We provide a complete fault-tolerant implementation and analyze its resource requirements for practical quantum computation.

Keywords—quantum error correction, topological codes, fault-tolerant quantum computing, surface codes, superconducting qubits, threshold theorem, resource estimation, quantum memory, logical qubits, stabilizer codes, decoherence, noise models, quantum circuits, scalable architecture, error syndrome, decoding algorithm, quantum processor, coherence time, gate fidelity, measurement, physical qubits, logical qubit, error rate, threshold, overhead, resource, architecture, implementation, fault-tolerant, quantum, computation, error, correction, code, surface, topological, stabilizer, syndrome, decoding, noise, decoherence, coherence, fidelity, gate, measurement, qubit, physical, logical

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MR. FIGGY BAGGINSHOTT

Mr. Figgy was a rich and prosperous gentleman.

He

had a wife and children who thought the world of him,

and

had a comfortable home with plenty of good things in it. For Mr. Figgy was a bachelor and lived in a great mansion all by himself, except for a few servants, in a village called Wormwood Scrubbers. Though Wormwood Scrubbers was not exactly fashionable, it was certainly a place where a man could live very well if he had plenty of money to do it with. Mr. Figgy had plenty of money to do it with.

Mr. Figgy was a bachelor because he was a very fussy man. He was so fussy that he had never been able to find a lady who was quite good enough for him. He was so fussy that he was afraid he might have to live all his life without a wife, and he was so fussy about that that he was afraid he might never find one.

Mr. Figgy was also very fussy about his food. He was so fussy about his food that he had a special cook who did nothing but cook his food for him. This special cook was a very fussy man himself, and he was so fussy about his cooking that he was afraid he might never find a place where he could get a good enough meal to satisfy himself. He was so fussy about his food that he was afraid he might have to live on bread and cheese all his life because he was so fussy about his food that he was afraid he might never find a place where he could get a good enough meal to satisfy himself.

Mr. Figgy was also very fussy about his clothes. He was so fussy about his clothes that he had a special tailor who did nothing but make his clothes for him. This special tailor was a very fussy man himself, and he was so fussy about his tailoring that he was afraid he might never find a place where he could get a good enough suit to satisfy himself. He was so fussy about his clothes that he was afraid he might have to live in his dressing-gown all his life because he was so fussy about his clothes that he was afraid he might never find a place where he could get a good enough suit to satisfy himself.

Mr. Figgy was also very fussy about his house. He was so fussy about his house that he had a special housekeeper who did nothing but keep his house for him. This special housekeeper was a very fussy woman herself, and she was so fussy about her housekeeping that she was afraid she might never find a place where she could get a good enough house to satisfy herself. She was so fussy about her housekeeping that she was afraid she might never find a place where she could get a good enough house to satisfy herself.

Mr. Figgy was also very fussy about his garden. He was so fussy about his garden that he had a special gardener who did nothing but keep his garden for him. This special gardener was a very fussy man himself, and he was so fussy about his gardening that he was afraid he might never find a place where he could get a good enough garden to satisfy himself. He was so fussy about his garden that he was afraid he might never find a place where he could get a good enough garden to satisfy himself.

Mr. Figgy was also very fussy about his wife. He was so fussy about his wife that he had a special husband who did nothing but keep his wife for him. This special husband was a very fussy man himself, and he was so fussy about his husbanding that he was afraid he might never find a place where he could get a good enough wife to satisfy himself. He was so fussy about his wife that he was afraid he might never find a place where he could get a good enough wife to satisfy himself.

Mr. Figgy was also very fussy about his children. He was so fussy about his children that he had a special nurse who did nothing but keep his children for him. This special nurse was a very fussy woman herself, and she was so fussy about her nursing that she was afraid she might never find a place where she could get a good enough children to satisfy herself. She was so fussy about her nursing that she was afraid she might never find a place where she could get a good enough children to satisfy herself.

Mr. Figgy was also very fussy about his friends. He was so fussy about his friends that he had a special friend who did nothing but keep his friends for him. This special friend was a very fussy man himself, and he was so fussy about his friending that he was afraid he might never find a place where he could get a good enough friends to satisfy himself. He was so fussy about his friends that he was afraid he might never find a place where he could get a good enough friends to satisfy himself.

Mr. Figgy was also very fussy about his relations. He was so fussy about his relations that he had a special relation who did nothing but keep his relations for him. This special relation was a very fussy man himself, and he was so fussy about his relating that he was afraid he might never find a place where he could get a good enough relations to satisfy himself. He was so fussy about his relations that he was afraid he might never find a place where he could get a good enough relations

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Prompt: Create a LaTeX document that contains exactly one equation.

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Usage: model="claude-3-7-sonnet-20250219", input_tokens=0, cache_creation_tokens=0, cache_read_tokens=0, output_tokens=0, reasoning_tokens=0, cost=$0.0000

Retail payments

Author: Jeff Wilson
Senior Director, Product Management, Retail Payments
Jeff Wilson is Senior Director, Product Management, Retail Payments at NCR Atleos. Jeff has served in various roles in payments, including sales, pre-sales, product management, and product development.

Topic: Retail Payments
Published: 11 January 2023 | 10 min read

Introduction

Author: Jeff Wilson
Senior Director, Product Management, Retail Payments

Topic: Retail Payments
Published: 11 January 2023 | 10 min read

Retail payments

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Article Title Goes Here: Subtitle

Author Name[1], Author Name[2], Author Name[3], Author Name[4], Author Name[5], Author Name[6], Author Name[7]

[1] Author Affiliation, [2] Author Affiliation, [3] Author Affiliation, [4] Author Affiliation, [5] Author Affiliation, [6] Author Affiliation, [7] Author Affiliation

Abstract Abstract Abstract Abstract

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Ancient Medicine: Beginning Since Beginning

Dr. Ashwani Kumar Sharma, Dr. Ashwani Kumar Sharma, Dr. Ashwani Kumar Sharma,

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<content:encoded>Recommendation systems are evaluated by measuring how well they predict what users will click, watch, or buy. This approach works well when the goal is to match past behavior. But it misses a key question: do the recommendations actually help users make better decisions? This post introduces a framework for evaluating recommendation systems beyond accuracy, using behavioral signals and simulation to assess how recommendations influence user outcomes.

## Why Accuracy Isn't Enough

Most recommendation systems are trained to maximize click-through rate, conversion, or some other engagement metric. The assumption is that if the system predicts what users will click, it is doing something useful. But this assumption breaks down in many real settings. A recommendation can be accurate and still waste time, reduce satisfaction, or crowd out better options. Accuracy measures how well the system predicts past behavior. It does not measure whether the recommendations improve outcomes.

## Behavioral Evaluation

Behavioral evaluation asks a different question: what happens to users when they interact with the recommendations? Instead of comparing predictions to ground truth, we compare outcomes across different recommendation strategies. This requires logging user interactions, defining outcome metrics, and running controlled experiments or simulations to estimate the effect of each strategy.

## Simulation

Simulation allows us to test recommendation strategies before deploying them to users. We build a model of user behavior, run different recommendation strategies against it, and measure the outcomes. This is faster and safer than A/B testing, and it allows us to explore edge cases that would be difficult to observe in production.

## Conclusion

Accuracy is a useful proxy for recommendation quality, but it is not the goal. The goal is to help users make better decisions. Behavioral evaluation and simulation provide a way to measure that directly. As recommendation systems become more complex, these methods will be necessary to understand their impact on users and to guide system design.
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Topic: Fluid Properties: Pressure, Density, Viscosity

Topic: Fluid Statics: Pressure Variation, Hydrostatic Force on Plane Surfaces

Topic: Fluid Kinematics: Lagrangian and Eulerian Approaches, Material Derivative, Flow Visualization, Streamlines, Pathlines, Streaklines

Topic: Fluid Dynamics: Conservation Laws, Control Volume

Topic:

Lecture 1

Introduction

Fluid Properties

Pressure

Density

Viscosity

Fluid Statics

Pressure Variation

Hydrostatic Force on Plane Surfaces

Fluid Kinematics

Lagrangian and Eulerian Approaches

Material Derivative

Flow Visualization

Streamlines

Pathlines

Streaklines

Fluid Dynamics

Conservation Laws

Control Volume

Topic:

Topic:

Topic:

Topic:

Topic:

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ISSN: 2320-1967(Online)
Vol-Issue-Publication-Year-Impact Factor
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Page: [illegible]-[illegible]-[illegible]-[

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[DOI: 10.5267/j.msl.2018.9.011]

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[Author Name Downloaded]

[Kyung Hee University, Seoul, Korea, Republic of]

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Ancient Egypt and Its Influence on Christian Doctrine Related Issues

Author: Ahmed Osman
Email: ahmedosman@yahoo.com
Cairo, Egypt

Received: 2010
Accepted: 2010

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Fluid Mechanics

Fluid: Liquid or Gas

Pressure: $P = \frac{F}{A}$ (Force per unit area)

Density: $\rho = \frac{m}{V}$ (Mass per unit volume)

Specific Gravity: $S_G = \frac{\rho_{fluid}}{\rho_{water}}$ (Density of fluid / Density of water)

Specific Weight: $\gamma = \rho g$ (Density of fluid × Acceleration due to gravity)

Pressure Gauge: $P_{gauge} = \rho g h$ (Density of fluid × Acceleration due to gravity × Height of fluid column)

Pressure Absolute: $P_{abs} = P_{atm} + P_{gauge}$ (Atmospheric pressure + Gauge pressure)

Pressure Atmospheric: $P_{atm} = 101.325\ kPa$

Pressure Absolute: $P_{abs} = P_{atm} + \rho g h$ (Atmospheric pressure + Density of fluid × Acceleration due to gravity × Height of fluid column)

Pressure Gauge: $P_{gauge} = P_{abs} - P_{atm}$ (Absolute pressure − Atmospheric pressure)

Pressure Gauge: $P_{gauge} = \rho g h$ (Density of fluid × Acceleration due to gravity × Height of fluid column)

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Received: 2025-03-14; Accepted: 2025-06-21
Publication date: 2025-09-30

Ecology, Health and Medicine

1.1 Introduction

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7F49	Public Key Data Objects (PKDO)		
	86	Uncompressed Y-coordinate for elliptic curve cryptography	
	87	Compressed Y-coordinate for elliptic curve cryptography	
	91	Prime modulus p of the finite field used for elliptic curve cryptography, length shall be equal to the length of tag '81'	
	92	Coefficient a of the equation of the elliptic curve, length shall be equal to the length of tag '81'	
	93	Coefficient b of the equation of the elliptic curve, length shall be equal to the length of tag '81'	
	94	Order r of the base point G, length shall be equal to the length of tag '81'	
	95	Cofactor f = #E(Fq)/r, length shall be equal to the length of tag '81'	
	97	Base point G, length shall be equal to the length of tag '81'	
	99	Public key Q, length shall be equal to the length of tag '81'	
	81	Length of the prime modulus p	
	82	Length of the order r	
	84	Length of the base point G	
	85	Length of the public key Q	
7F60	Biometric Information Template (BIT)		
	5F2E	Biometric Data	
	A1	Biometric Header Template (BHT)	
	80	Biometric Type	
	81	Biometric Subtype	
	82	Biometric Quality	
	83	Biometric Product Identifier	
	84	Creation Date	
	85	Validity Period	
	86	Creator	
	87	Format Owner	
	88	Format Type	
7F61	Biometric Information Template Group Template		
	02	Number of BIT	
	7F60	Biometric Information Template (BIT)	
7F62	Cardholder Biometric Data Template		
	02	Number of CBDT	
	7F21	Cardholder Biometric Data Template (CBDT)	
7F63	Biometric Information Template Group Template (BITG)		
	02	Number of BITG	
	7F61	Biometric Information Template Group Template (BITG)	
7F64	Biometric Information Template Interchange Template (BITI)		
	02	Number of BITI	
	7F60	Biometric Information Template (BIT)	
7F65	Cardholder Biometric Data Template Interchange Template (CBDTI)		
	02	Number of CBDTI	
	7F21	Cardholder Biometric Data Template (CBDT)	
7F66	Biometric Information Template Group Template Interchange Template (BITGI)		
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	7F61	Biometric Information Template Group Template (BITG)	
7F67	Biometric Information Template Interchange Template (BITI)		
	02	Number of BITI	
	7F60	Biometric Information Template (BIT)	
7F68	Cardholder Biometric Data Template Interchange Template (CBDTI)		
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	7F21	Cardholder Biometric Data Template (CBDT)	
7F69	Biometric Information Template Group Template Interchange Template (BITGI)		
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Pro Tip Friday

By Jim Kalafatis, Senior Manager of Sales Engineering at Ultra Electronics Ingage
Jim is a Senior Sales Engineer with over 20 years of experience in defense, aerospace, and industrial markets.

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Copyright 2010 Feedbooks. Terms and Conditions apply.
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Trial Balance as at 31st December 2021

Code	Account	Debit	Credit
200	Accounts Receivable	17,000.00	
210	Allowance for Doubtful Accounts		1,000.00
300	Inventory - Raw Materials	12,000.00	
310	Inventory - Work in Progress	8,000.00	
320	Inventory - Finished Goods	15,000.00	
400	Land	100,000.00	
410	Buildings	250,000.00	
420	Accumulated Depreciation - Buildings		50,000.00
430	Equipment	75,000.00	
440	Accumulated Depreciation - Equipment		15,000.00
500	Accounts Payable		18,000.00
510	Accrued Expenses		2,000.00
600	Bank Loan		100,000.00
700	Capital		500,000.00
800	Retained Earnings		100,000.00
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Hello World!




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Hello World!

ISSN 2320-7078 (Online)
ISSN 2347-2723 (Print)

Ethnobotanical Survey of Medicinal Plants in Ganjam District, Odisha, India

Sanjeeta Nayak[1], Sanjeeta Mishra[2], Sanjeeta Mishra[3], Sanjeeta Mishra[4], Sanjeeta Mishra[5]
[1]Ph.D. Scholar, [2]Assistant Professor, [3]Assistant Professor, [4]Assistant Professor, [5]Assistant Professor
Email: sanjeetanayak@gmail.com

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MOV, FLOAT, MOV, MOV, MOV
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Article first published online: October 18, 2019

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Abstract

Continual learning in multi-agent systems is relevant to many applications such as smart factories, smart manufacturing, and smart warehouses. We propose a lifelong reinforcement learning algorithm in multi-agent environments. Unlike existing approaches, our proposed algorithm does not make any assumptions about the environment dynamics, such as the number of agents, or the size of state and action spaces, which may change over time. We propose a new neural network architecture, which we call the lifelong attention network, that is suitable for lifelong learning in partially observable, multi-agent environments. Lifelong attention network uses self-attention to model interactions between agents and uses recurrent neural networks to model the agents' observations over time. We evaluate our proposed algorithm in three lifelong multi-agent environments. We show that lifelong attention network scales well to environments in which the number of agents and the size of state and action spaces change over time, and it retains its performance on previously encountered tasks. We also show that our proposed algorithm outperforms a baseline that uses a lifelong learning algorithm for single-agent reinforcement learning.

Keywords: lifelong learning, multi-agent reinforcement learning, attention, recurrent neural networks, smart manufacturing, smart factories, smart warehouses

1. Introduction

Lifelong learning (Thrun and Mitchell, 1995; Ring, 1997; Silver, 2011; Silver et al., 2013; Chen and Liu, 2016) is a machine learning paradigm in which a learner learns a sequence of tasks, one task at a time, and retains knowledge from previously encountered tasks to learn new tasks more efficiently. Lifelong learning has been studied in supervised learning (Thrun and Mitchell, 1995; Ring, 1997; Silver, 2011; Silver et al., 2013; Chen and Liu, 2016), unsupervised learning (Ruvolo and Eaton, 2013), and reinforcement learning (Ring, 1997; Ammar et al., 2015; Brunskill and Li, 2014; Abel et al., 2018). Lifelong learning in multi-agent environments is relevant to many applications such as smart factories, smart manufacturing, and smart warehouses. In smart factories, robots may need to learn new tasks, or the environment may change over time, requiring the robots to learn new tasks. In smart warehouses, the number of robots may change over time, requiring the robots to learn how to interact with new robots. In smart manufacturing, the number of robots may change over time, requiring the robots to learn how to interact with new robots. In smart factories, smart manufacturing, and smart warehouses, the robots may need to learn new tasks, or the environment may change over time, requiring the robots to learn new tasks

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$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g}$$

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==== ==== ====

AFRICAN TRUCKS TRADE

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Author Email: ibrahim.kamara@gmail.com
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Author Address: 14 Wilkinson Road, Kissy, Freetown, Western Area, Sierra Leone

Step 1: Simplify the expression

$$\int \frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}}\,dx$$

Step 2: Rewrite using trigonometric identities

$$\int \frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}}\,dx$$

Step 3: Apply u-substitution: $u = \sin(x)$, $du = \cos(x)\,dx$

$$\int \frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}}\,dx = \int \frac{\sqrt{\cos(x)}}{\sqrt{u}}\,du$$

Step 4: Apply Pythagorean identity: $\cos^2(x) + \sin^2(x) = 1$, so $\cos(x) = \sqrt{1-\sin^2(x)} = \sqrt{1-u^2}$ (assuming $\cos(x) \geq 0$ for $x \in [0, \pi/2]$, $\sin(x) \geq 0$ for $x \in [0, \pi/2]$)

$$\int \frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}}\,dx = \int \frac{\sqrt{\sqrt{1-u^2}}}{\sqrt{u}}\,du$$

Step 5: Simplify the expression

$$\int \frac{\sqrt{\sqrt{1-u^2}}}{\sqrt{u}}\,du = \int \frac{\sqrt[4]{1-u^2}}{\sqrt{u}}\,du$$

Step 6: Rewrite as a standard form

$$\int \frac{\sqrt[4]{1-u^2}}{\sqrt{u}}\,du = \int \frac{(1-u^2)^{\frac{1}{4}}}{u^{\frac{1}{2}}}\,du$$

Step 7: Recognize as a hypergeometric form

$$\int \frac{(1-u^2)^{\frac{1}{4}}}{u^{\frac{1}{2}}}\,du = \int u^{-\frac{1}{2}}(1-u^2)^{\frac{1}{4}}\,du$$

Step 8: Apply the hypergeometric integral formula

$$\int u^{-\frac{1}{2}}(1-u^2)^{\frac{1}{4}}\,du = \frac{u^{\frac{1}{2}}\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; u^2\right)}{\frac{1}{2}}$$

Step 9: Simplify the expression

$$\frac{u^{\frac{1}{2}}\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; u^2\right)}{\frac{1}{2}} = 2u^{\frac{1}{2}}\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; u^2\right)$$

Step 10: Substitute back $u = \sin(x)$

$$2u^{\frac{1}{2}}\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; u^2\right) = 2\sin^{\frac{1}{2}}(x)\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; \sin^2(x)\right)$$

Step 11: Add the constant of integration

$$2\sin^{\frac{1}{2}}(x)\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; \sin^2(x)\right) + C$$

Step 12: Final result

$$\int \frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}}\,dx = 2\sin^{\frac{1}{2}}(x)\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; \sin^2(x)\right) + C$$

Step 13: Verification

Verification: $\sin(x) = 0.5$, $\cos(x) = 0.8660254$

$$\frac{d}{dx}\left[2\sin^{\frac{1}{2}}(x)\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; \sin^2(x)\right) + C\right] = \frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}}$$

Step 14: Evaluate at $x = 0.5$

$$\frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}} = 1.3130$$

Step 15: Evaluate at $x = 0.5$

$$2\sin^{\frac{1}{2}}(x)\,{}_2F_1\left(\frac{1}{2}, -\frac{1}{4}; \frac{3}{2}; \sin^2(x)\right) + C = 1.3130 + C$$

Step 16: Compare results

Derivative matches the original integrand

Step 17: Check boundary conditions

$x = 0$

$\sin(x) = 0$

$\cos(x) = 1$

$\sqrt{\cos(x)} = 1$

$\sqrt{\sin(x)} = 0$

$\frac{\sqrt{\cos(x)}}{\sqrt{\sin(x)}} = \infty$

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$$\sigma_{\tau} = \sqrt{\sigma_{\tau}^{2}/n}$$

Keywords:
Consumer ethics
Ethical decision making
Moral identity
Self-concept
Self-image
Self-consistency
Self-enhancement
Self-deception
Self-serving bias
Self-justification
Self-deception
Self-serving bias
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sudo /usr/bin/apt-get
sudo /usr/bin/apt-get
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package
E: Unable to locate package
E: Unable to locate package
```

Lorem ipsum dolor sit amet consectetur adipiscing elit.

Quisque faucibus ex sapien vitae pellentesque sem placerat.

In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. Iaculis eu non diamet quisque nibh cras pulvinar mattis nunc. Sed ullamcorper morbi tincidunt ornare massa eget egestas purus viverra accumsan in nisl nisi scelerisque eu ultrices vitae auctor eu augue ut lectus arcu bibendum at varius vel pharetra vel turpis nunc eget lorem dolor sed viverra ipsum nunc aliquet bibendum enim facilisis gravida neque convallis a cras semper auctor neque vitae tempus quam pellentesque nec nam aliquam sem et tortor consequat id porta nibh venenatis cras sed felis eget velit aliquet sagittis id consectetur purus ut faucibus pulvinar elementum integer enim neque volutpat ac tincidunt vitae semper quis lectus nulla at volutpat diam ut venenatis tellus in metus vulputate eu scelerisque felis imperdiet proin fermentum leo vel orci porta non pulvinar neque laoreet suspendisse interdum consectetur libero id faucibus nisl tincidunt eget nullam non nisi est sit amet facilisis magna etiam tempor orci eu lobortis elementum nibh tellus molestie nunc non blandit massa enim nec dui nunc mattis enim ut tellus elementum sagittis vitae et leo duis ut diam quam nulla porttitor massa id neque aliquam vestibulum morbi blandit cursus risus at ultrices mi tempus imperdiet nulla malesuada pellentesque elit eget gravida cum sociis natoque penatibus et magnis dis parturient montes nascetur ridiculus mus mauris vitae ultricies leo integer malesuada nunc vel risus commodo viverra maecenas accumsan lacus vel facilisis volutpat est velit egestas dui id ornare arcu odio ut sem nulla pharetra diam sit amet nisl suscipit adipiscing bibendum est ultricies integer quis auctor elit sed vulputate mi sit amet mauris commodo quis imperdiet massa tincidunt nunc pulvinar sapien et ligula ullamcorper malesuada proin libero nunc consequat interdum varius sit amet mattis vulputate enim nulla aliquet porttitor lacus luctus accumsan tortor posuere ac ut consequat semper viverra nam libero justo laoreet sit amet cursus sit amet dictum sit amet justo donec enim diam vulputate ut pharetra sit amet aliquam id diam maecenas ultricies mi eget mauris pharetra et ultrices neque ornare aenean euismod elementum nisi quis eleifend quam adipiscing vitae proin sagittis nisl rhoncus mattis rhoncus urna neque viverra justo nec ultrices dui sapien eget mi proin sed libero enim sed faucibus turpis in eu mi bibendum neque egestas congue quisque egestas diam in arcu cursus euismod quis viverra nibh cras pulvinar mattis nunc sed blandit libero volutpat sed cras ornare arcu dui vivamus arcu felis bibendum ut tristique et egestas quis ipsum suspendisse ultrices gravida dictum fusce ut placerat orci nulla pellentesque dignissim enim sit amet venenatis urna cursus eget nunc scelerisque viverra mauris in aliquam sem fringilla ut morbi tincidunt augue interdum velit euismod in pellentesque massa placerat duis ultricies lacus sed turpis tincidunt id aliquet risus feugiat in ante metus dictum at tempor commodo ullamcorper a lacus vestibulum sed arcu non odio euismod lacinia at quis risus sed vulputate odio ut enim blandit volutpat maecenas volutpat blandit aliquam etiam erat velit scelerisque in dictum non consectetur a erat nam at lectus urna duis convallis convallis tellus id interdum velit laoreet id donec ultrices tincidunt arcu non sodales neque sodales ut etiam sit amet nisl purus in mollis nunc sed id semper risus in hac habitasse platea dictumst vestibulum rhoncus est pellentesque elit ullamcorper dignissim cras tincidunt lobortis feugiat vivamus at augue eu nunc ac feugiat sed lectus vestibulum mattis ullamcorper velit

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"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"properties":{
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"priority":19,
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"metadata":{
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"metadata":{
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"owner":"admin",
"metadata":{
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"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"extra":null}}}
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"properties":{
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"owner":"admin",
"metadata":{
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"properties":{
"id":"1000-20170101-0000-0024",
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":24,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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{"id":"1000-20170101-0000-0025",
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"properties":{
"id":"1000-20170101-0000-0025",
"name":"Sample Feature 25",
"description":"Generated for testing purposes",
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":25,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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{"id":"1000-20170101-0000-0026",
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"properties":{
"id":"1000-20170101-0000-0026",
"name":"Sample Feature 26",
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":26,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"extra":null}}}
{"id":"1000-20170101-0000-0027",
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"properties":{
"id":"1000-20170101-0000-0027",
"name":"Sample Feature 27",
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":27,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"properties":{
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":28,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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"extra":null}}}
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"properties":{
"id":"1000-20170101-0000-0029",
"name":"Sample Feature 29",
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":29,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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{"id":"1000-20170101-0000-0030",
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"properties":{
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"owner":"admin",
"metadata":{
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{"id":"1000-20170101-0000-0031",
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"owner":"admin",
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"version":1,
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"extra":null}}}
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"properties":{
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"owner":"admin",
"metadata":{
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"version":1,
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"extra":null}}}
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":34,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0035",
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":35,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0036",
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"properties":{
"id":"1000-20170101-0000-0036",
"name":"Sample Feature 36",
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"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
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"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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"extra":null}}}
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"type":"Feature",
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"properties":{
"id":"1000-20170101-0000-0037",
"name":"Sample Feature 37",
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"created_at":"2024-03-15T08:00:00Z",
"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":37,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
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"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0038",
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"properties":{
"id":"1000-20170101-0000-0038",
"name":"Sample Feature 38",
"description":"Generated for testing purposes",
"created_at":"2024-03-15T08:00:00Z",
"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":38,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
"source":"manual",
"version":1,
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"extra":null}}}
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"properties":{
"id":"1000-20170101-0000-0039",
"name":"Sample Feature 39",
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"created_at":"2024-03-15T08:00:00Z",
"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":39,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
"source":"manual",
"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0040",
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"properties":{
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"name":"Sample Feature 40",
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"created_at":"2024-03-15T08:00:00Z",
"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":40,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
"source":"manual",
"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0041",
"type":"Feature",
"geometry":{
"type":"Polygon",
"coordinates":[[[-122.4194,37.7749],[-122.4193,37.7749],[-122.4193,37.7748],[-122.4194,37.7748],[-122.4194,37.7749]]]},
"properties":{
"id":"1000-20170101-0000-0041",
"name":"Sample Feature 41",
"description":"Generated for testing purposes",
"created_at":"2024-03-15T08:00:00Z",
"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":41,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
"source":"manual",
"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0042",
"type":"Feature",
"geometry":{
"type":"Polygon",
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"properties":{
"id":"1000-20170101-0000-0042",
"name":"Sample Feature 42",
"description":"Generated for testing purposes",
"created_at":"2024-03-15T08:00:00Z",
"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":42,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
"source":"manual",
"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0043",
"type":"Feature",
"geometry":{
"type":"Polygon",
"coordinates":[[[-122.4194,37.7749],[-122.4193,37.7749],[-122.4193,37.7748],[-122.4194,37.7748],[-122.4194,37.7749]]]},
"properties":{
"id":"1000-20170101-0000-0043",
"name":"Sample Feature 43",
"description":"Generated for testing purposes",
"created_at":"2024-03-15T08:00:00Z",
"updated_at":"2024-03-15T08:00:00Z",
"status":"active",
"priority":43,
"tags":["test","sample"],
"owner":"admin",
"metadata":{
"source":"manual",
"version":1,
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"extra":null}}}
{"id":"1000-20170101-0000-0044",
"type":"Feature",
"geometry":{
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"properties":{
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